

MightyXDDoS™ - DDoS Mitigator

- High-Performance Inline Security Bridge
- Based on FD.io VPP and eBPF XDP

MightyLink Lab.
(www.mightyconn.com)
Doc. Revision: 0.94

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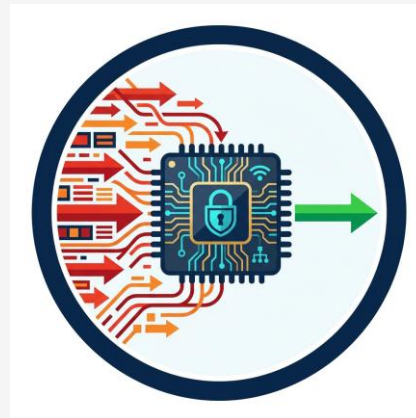
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3. MightyXDDoS
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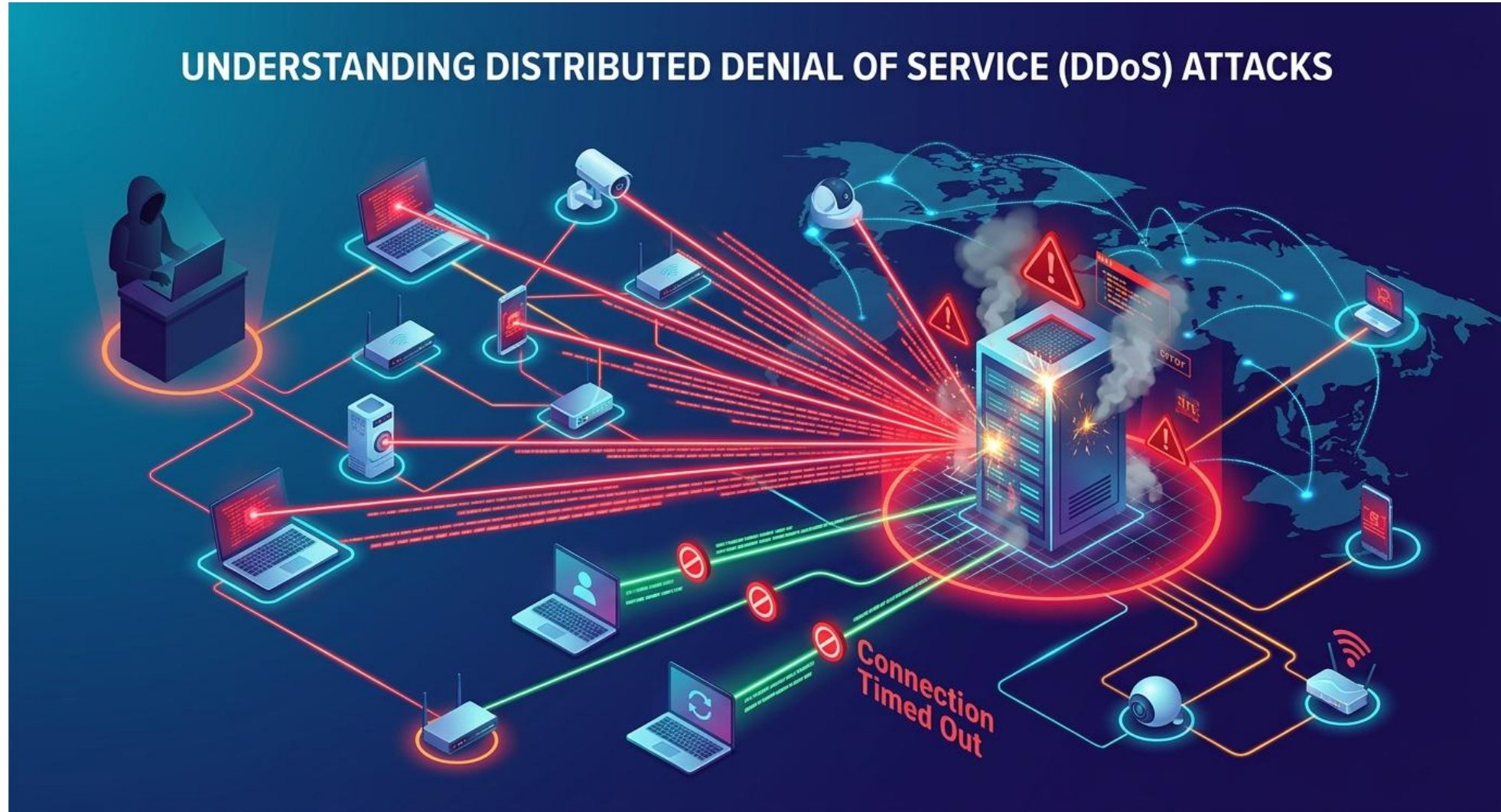


1. DDoS Attack & Mitigator

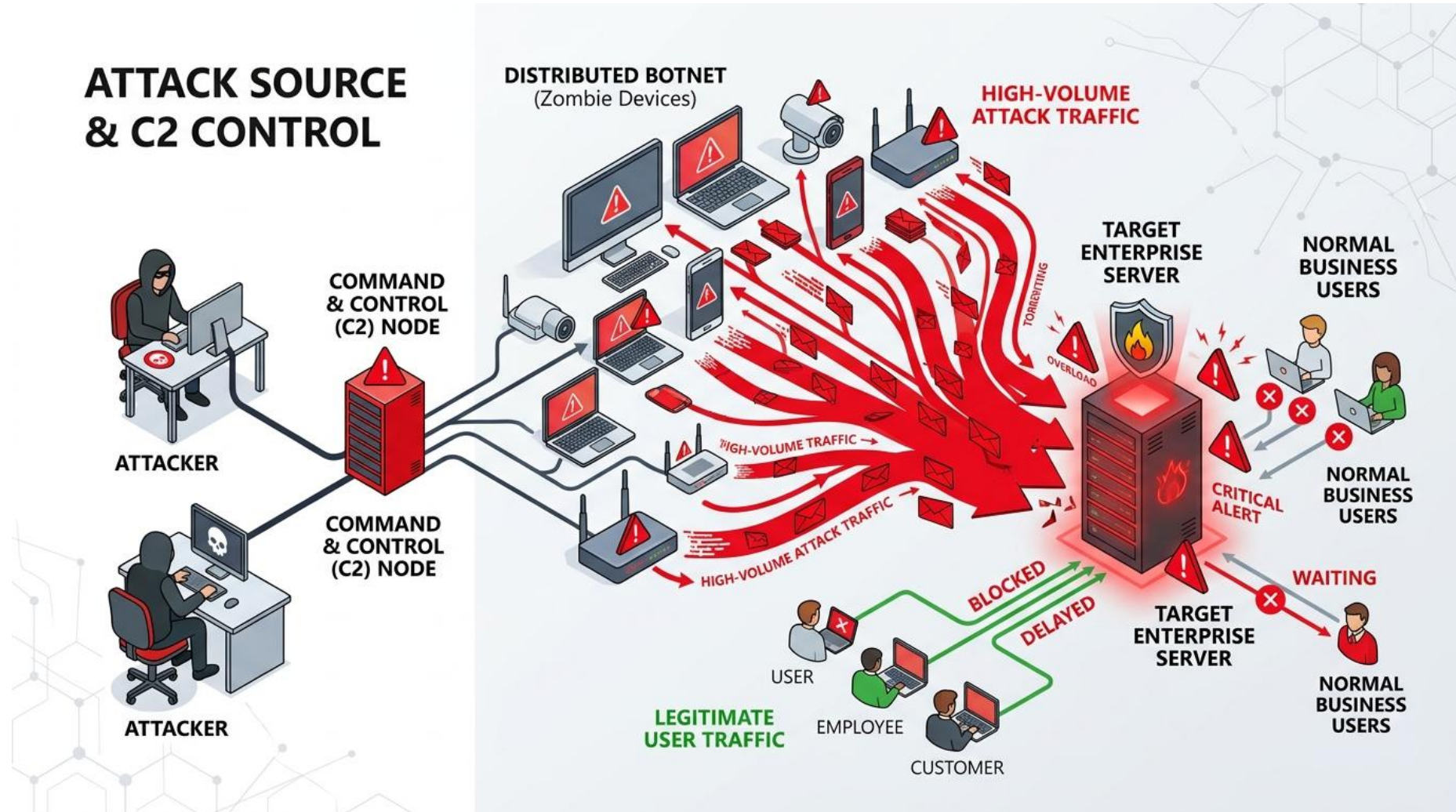
DDoS Attack and DDoS Mitigator



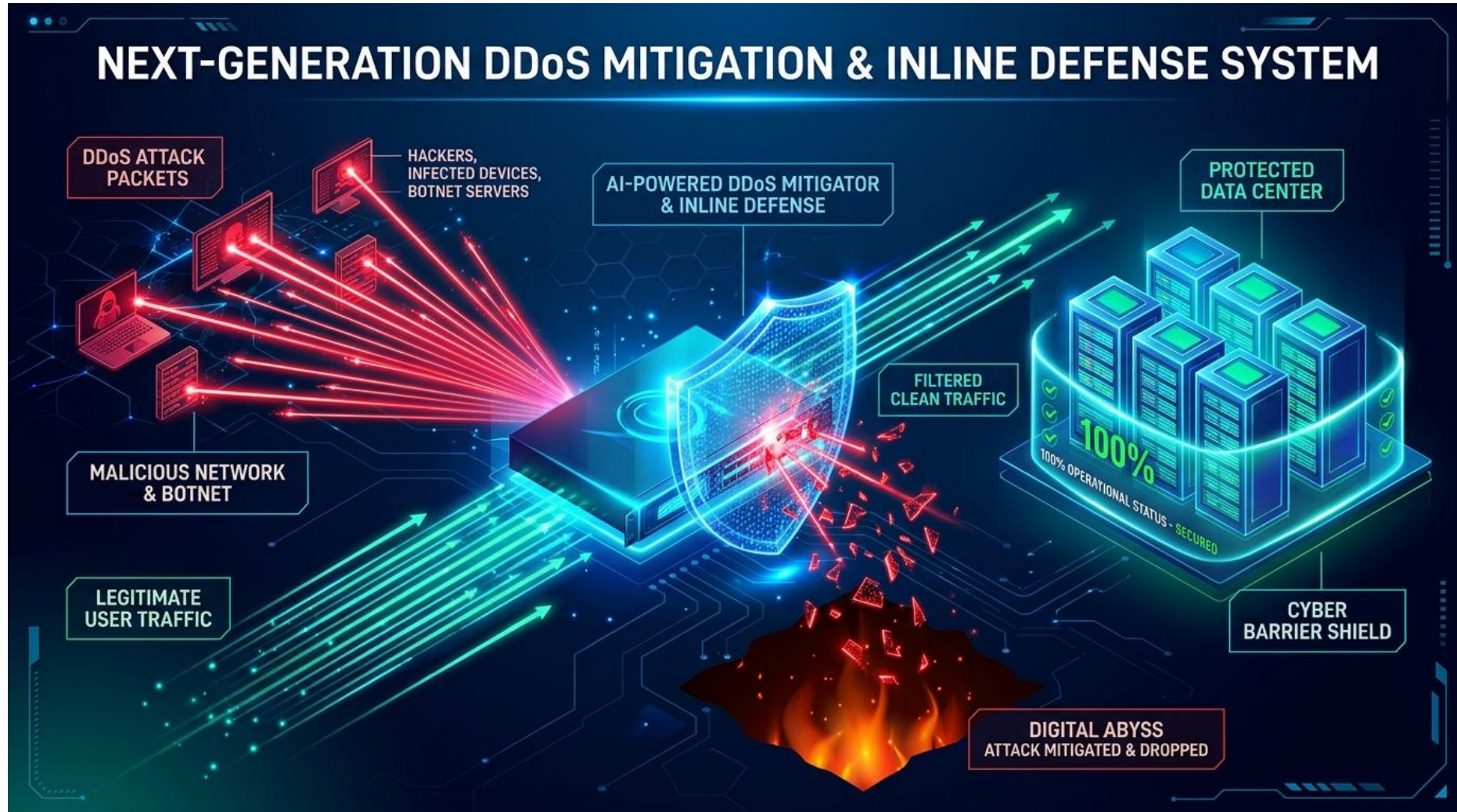
1. DDoS Attack & Mitigator(1) – Attack(1)



1. DDoS Attack & Mitigator(1) – Attack(2)

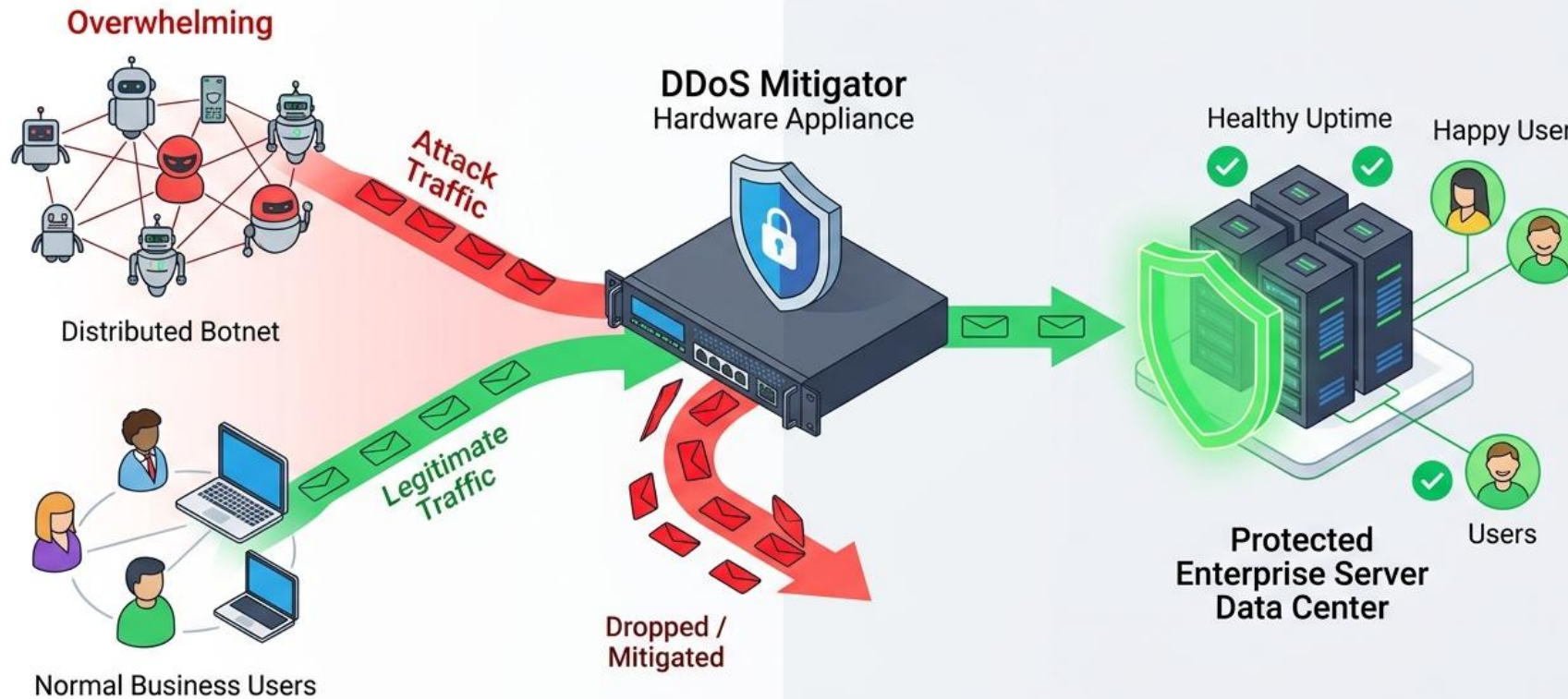


1. DDoS Attack & Mitigator(2) – Mitigator(1)

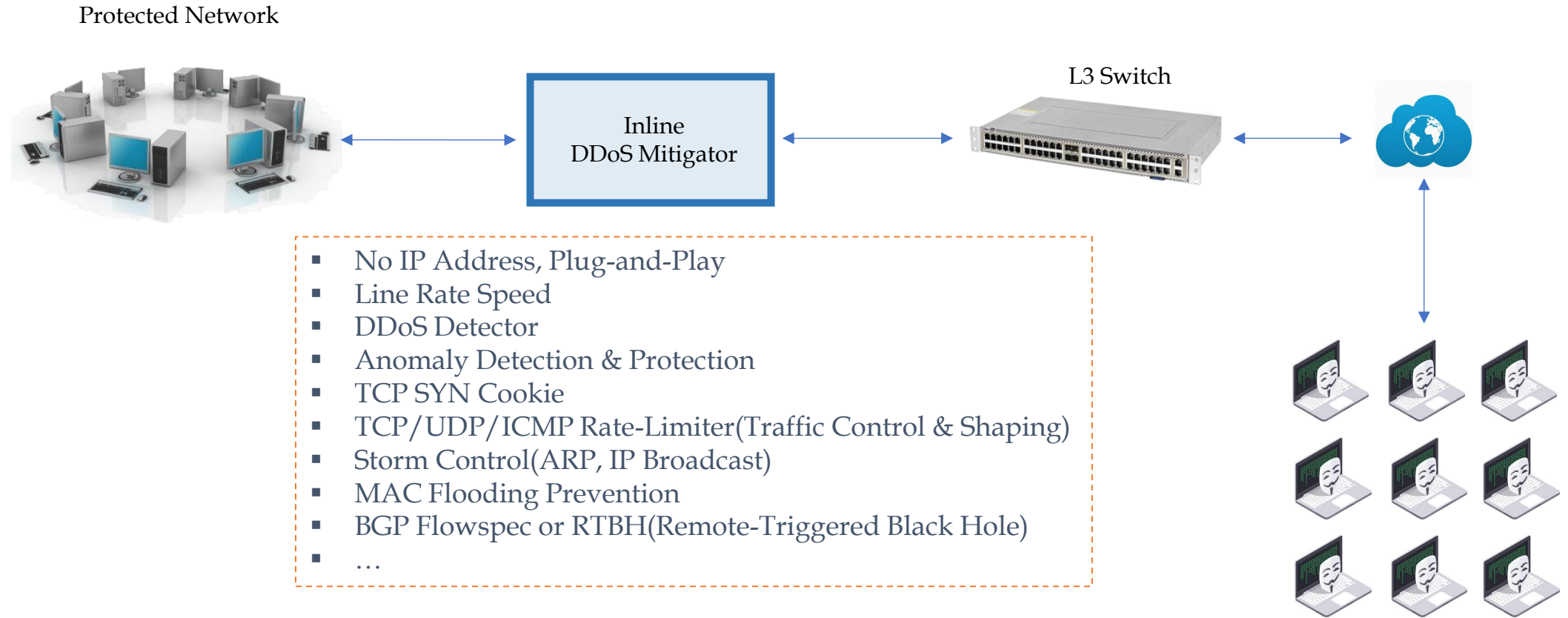


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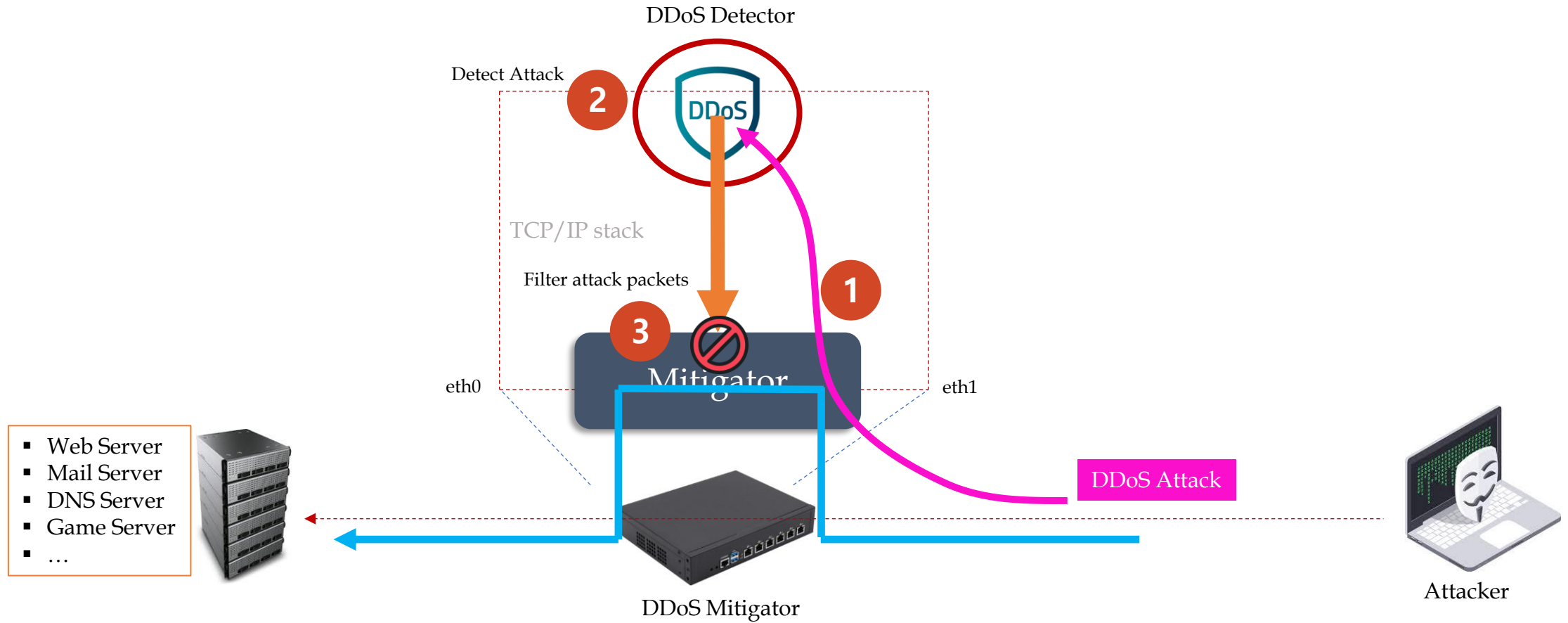
DDoS Mitigator and Inline Traffic Scrubber



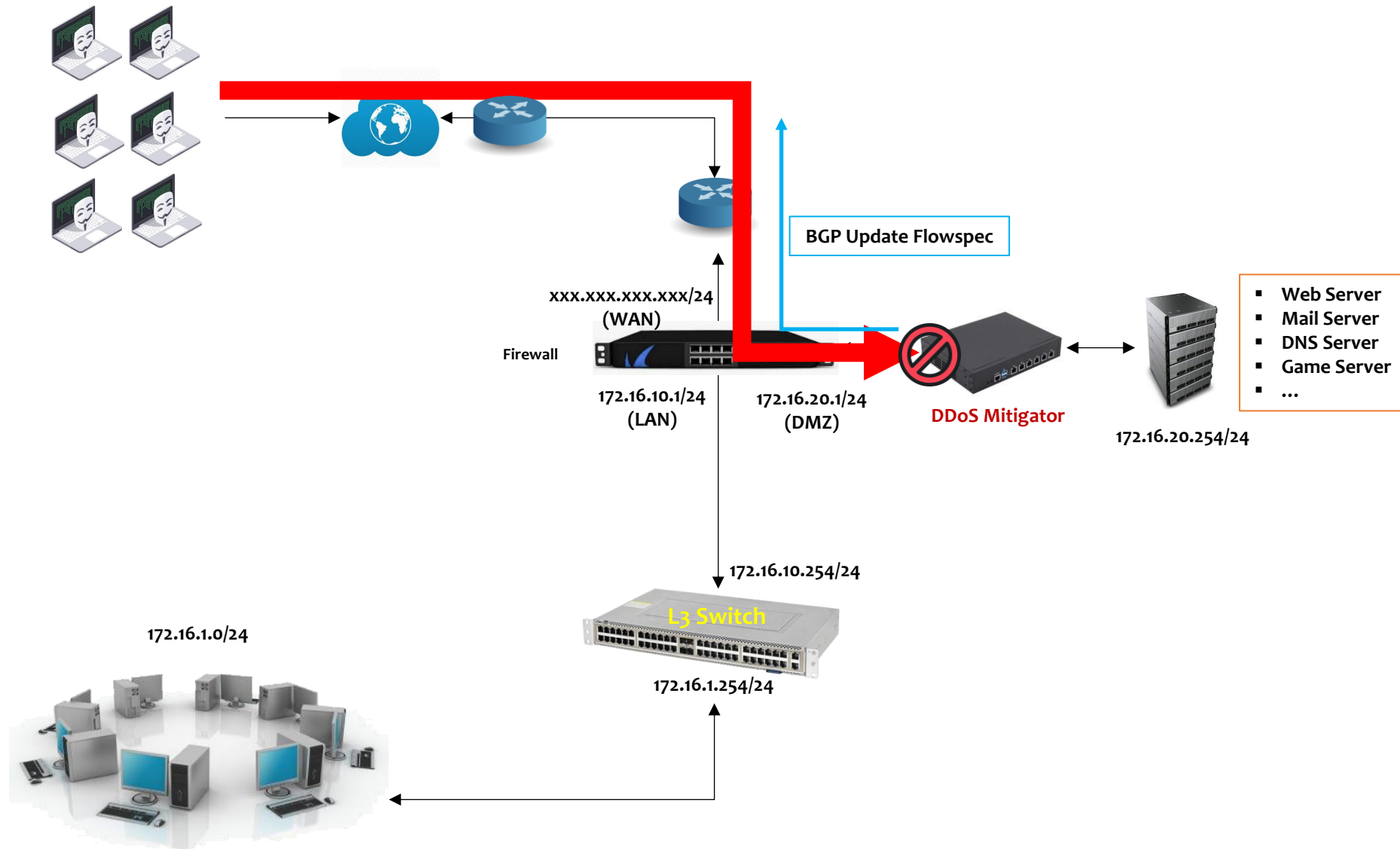
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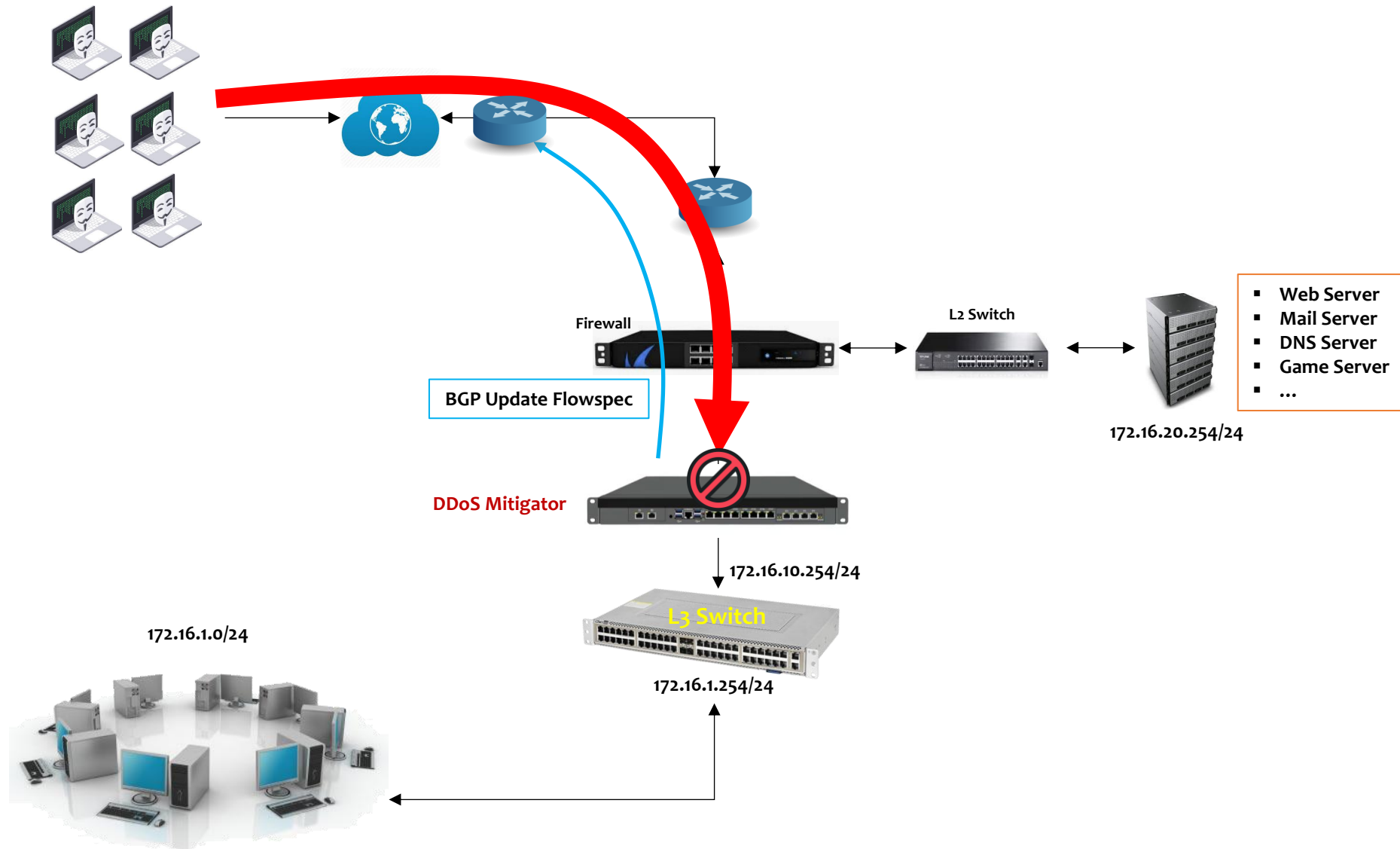
1. DDoS Attack & Mitigator(2) – Mitigator(4)



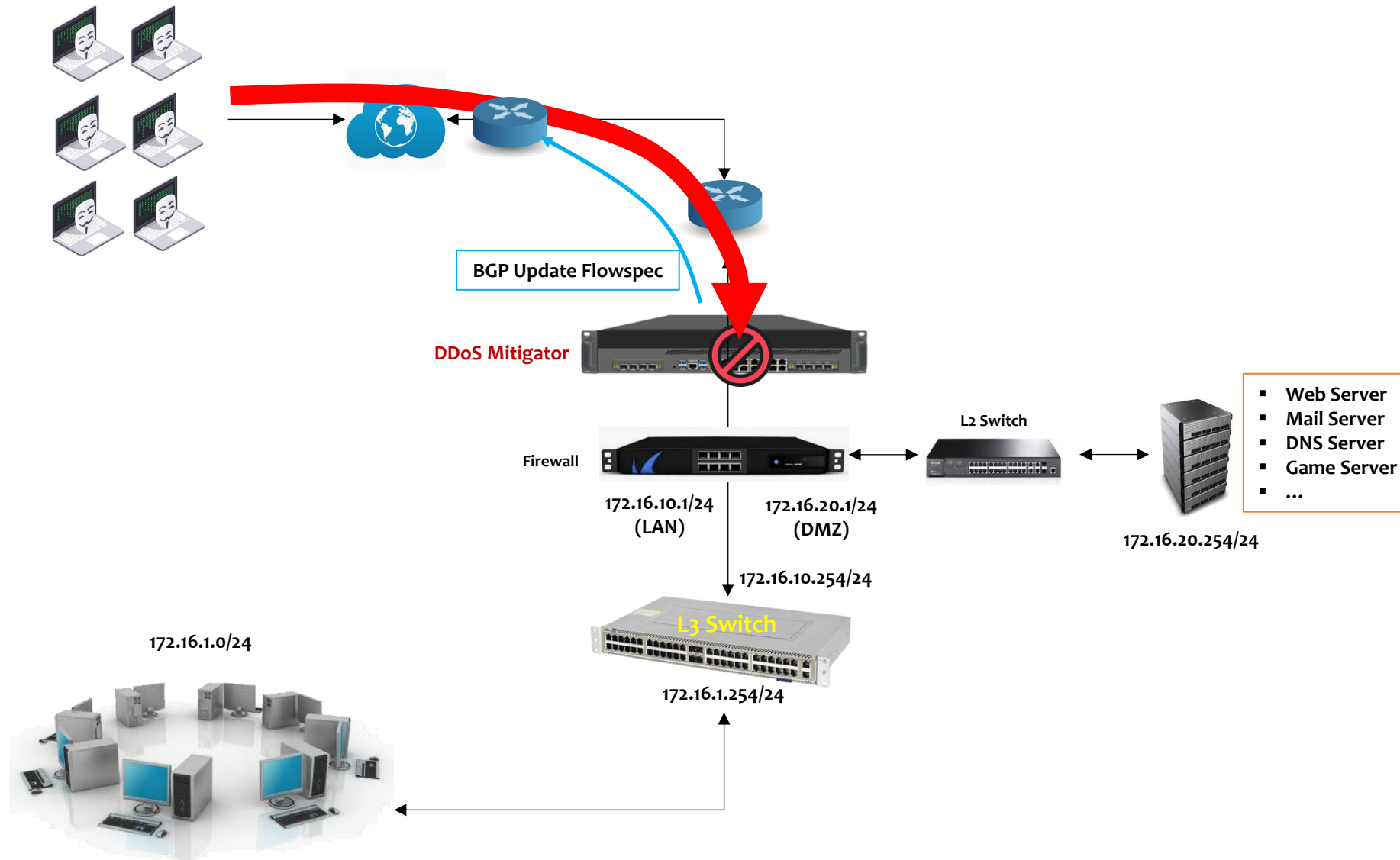
1. DDoS Attack & Mitigator(3) – Network 구성(1)



1. DDoS Attack & Mitigator(3) – Network 구성(2)



1. DDoS Attack & Mitigator(3) – Network 구성(3)



2. Base Technology

DPDK, FD.io VPP and XDP

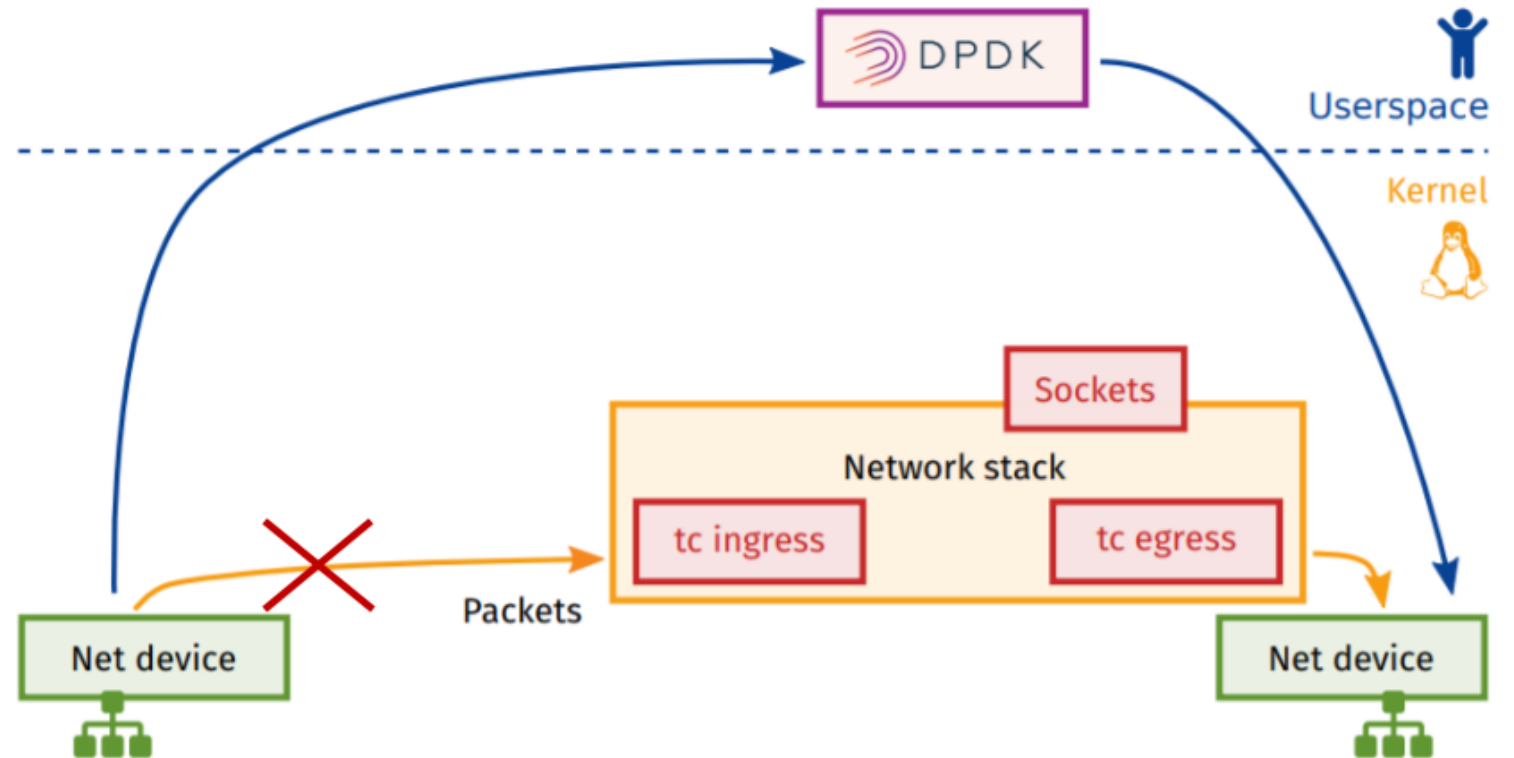
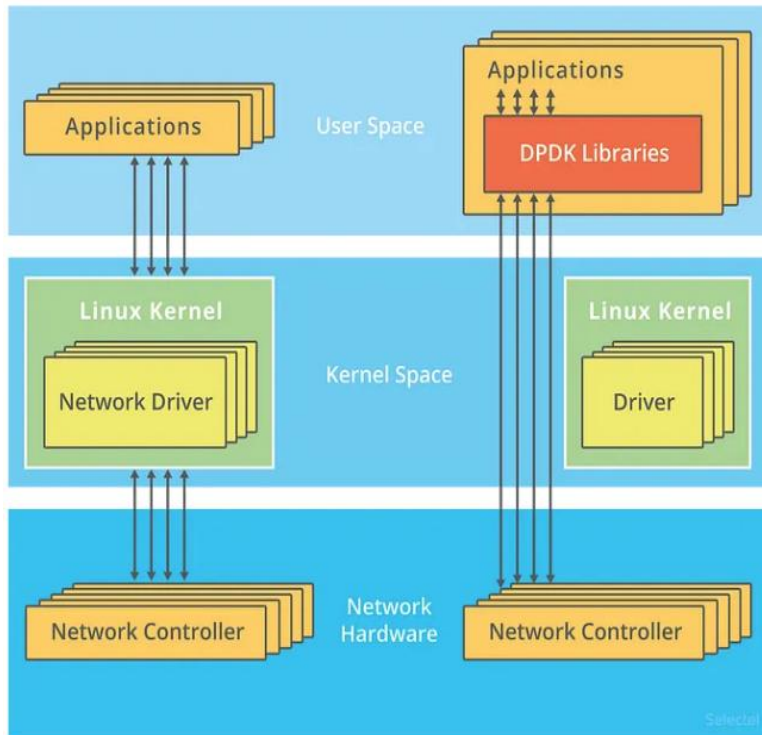
2.1 DPDK

Data Plane Development Kit



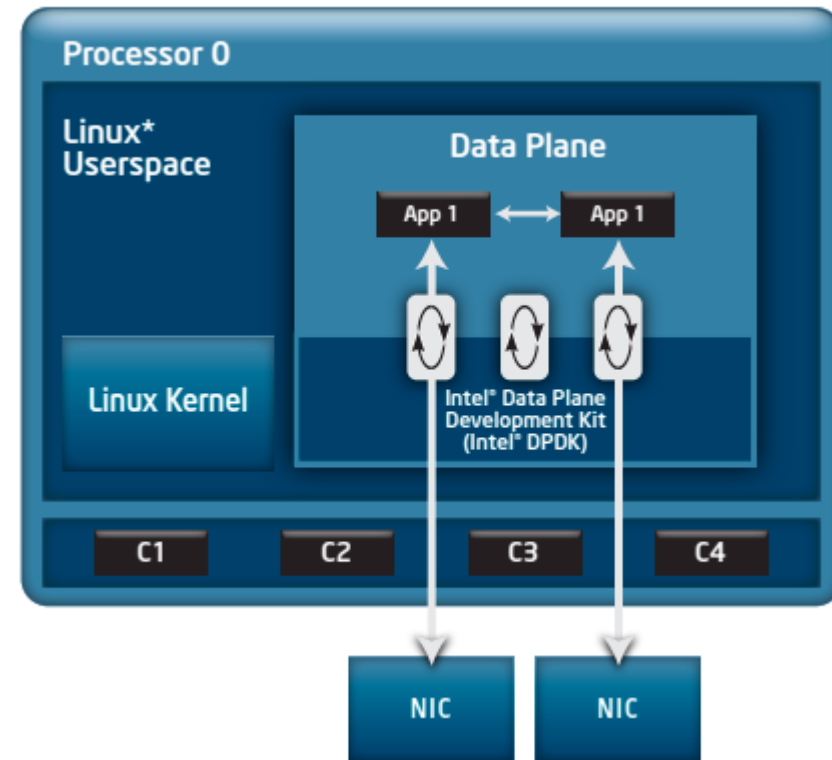
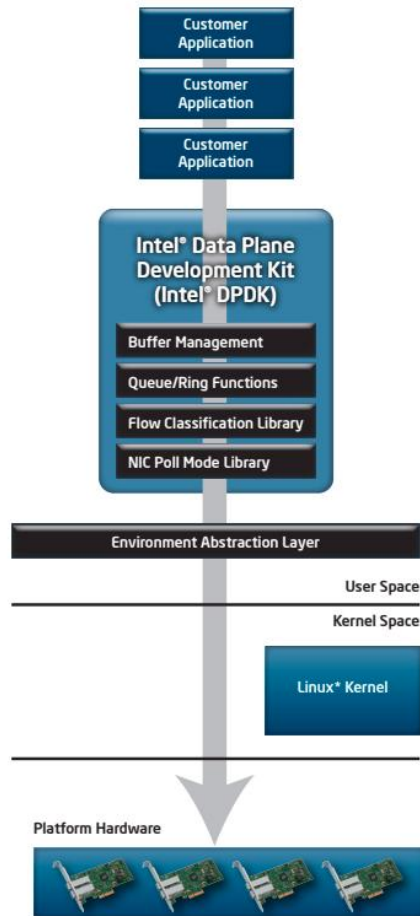
<https://www.dpdk.org/>

2.1 DPDK(1)



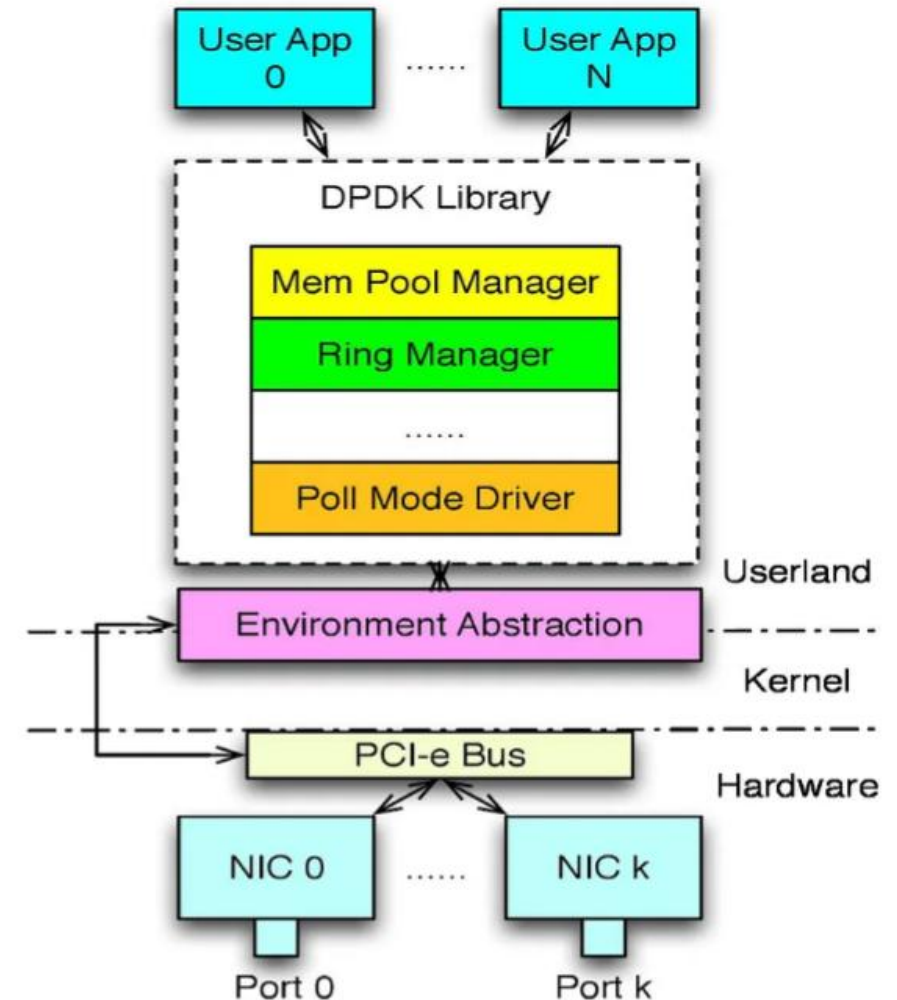
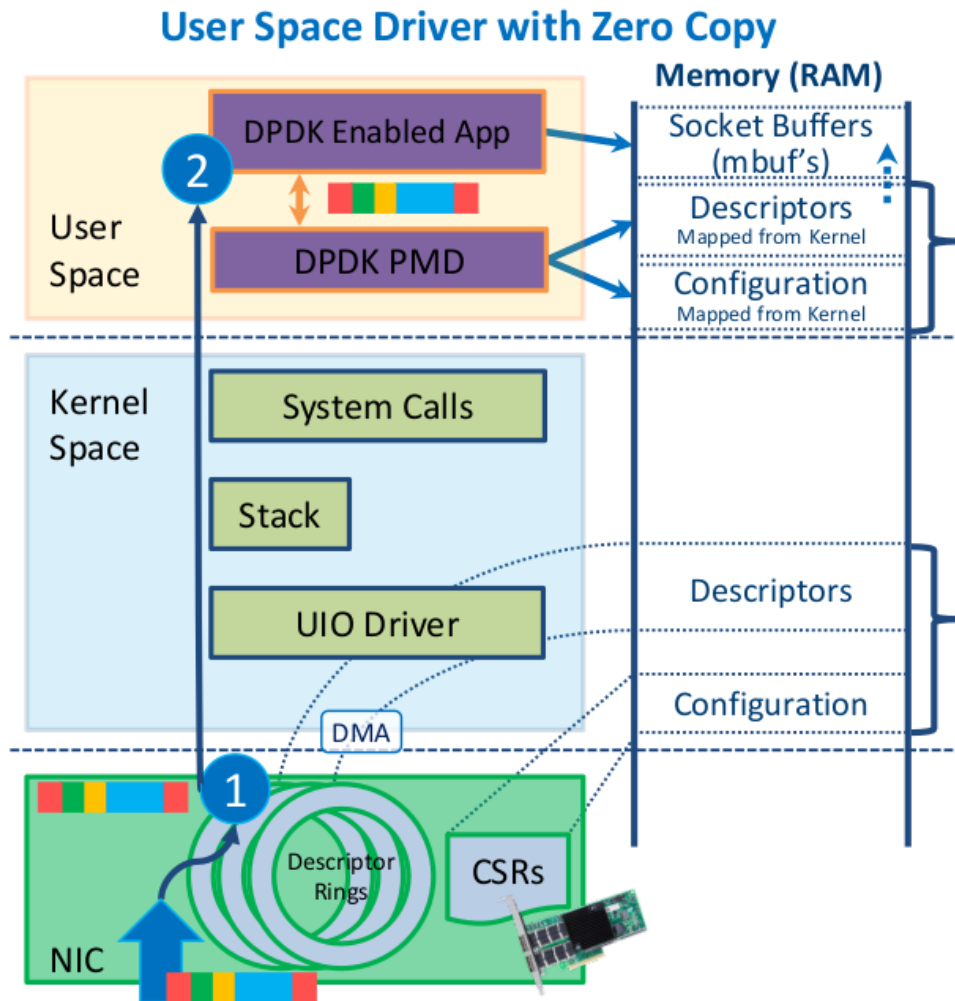
Ref) https://media.frnog.org/FRnOG_28/FRnOG_28-3.pdf

2.1 DPDK(2)

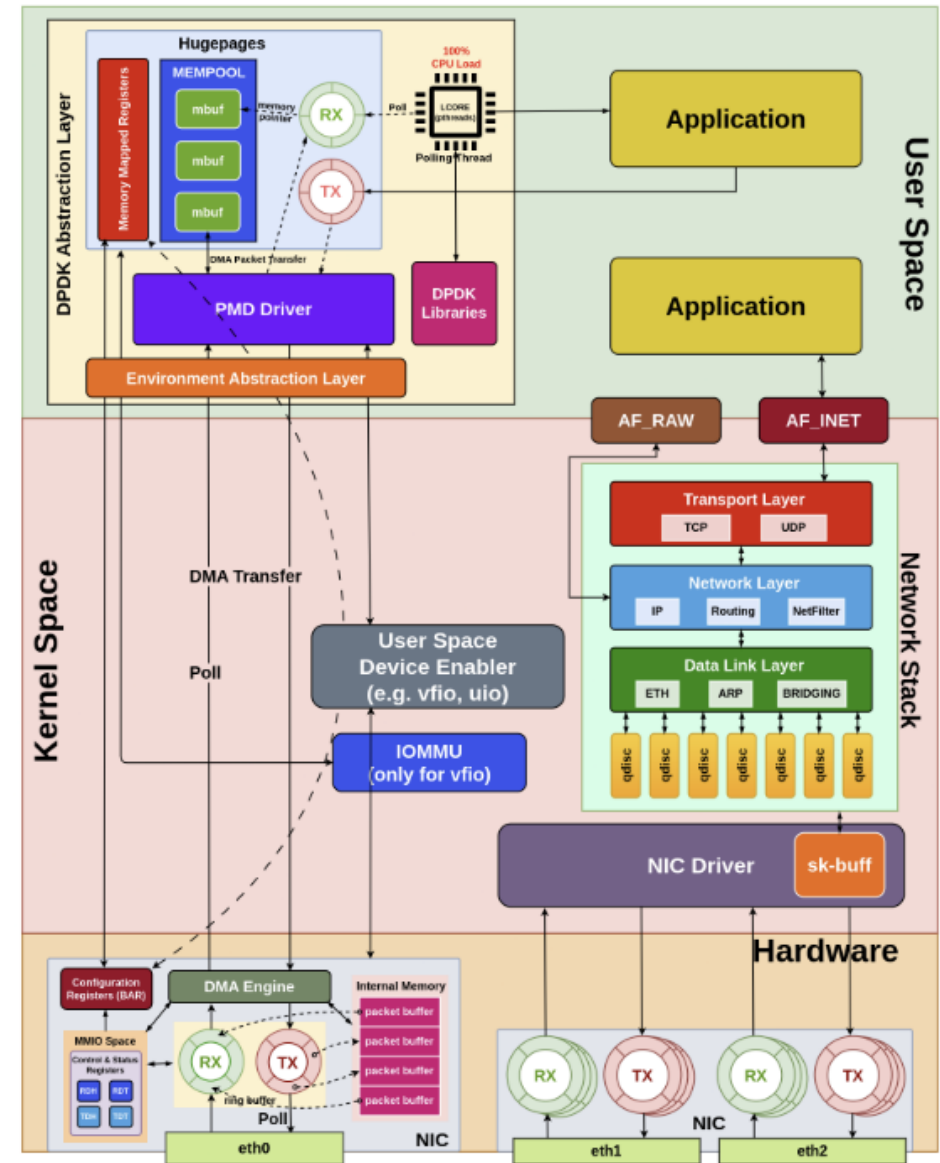
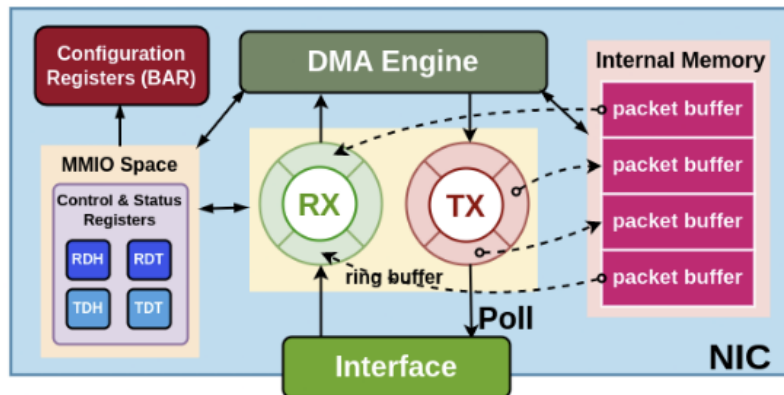
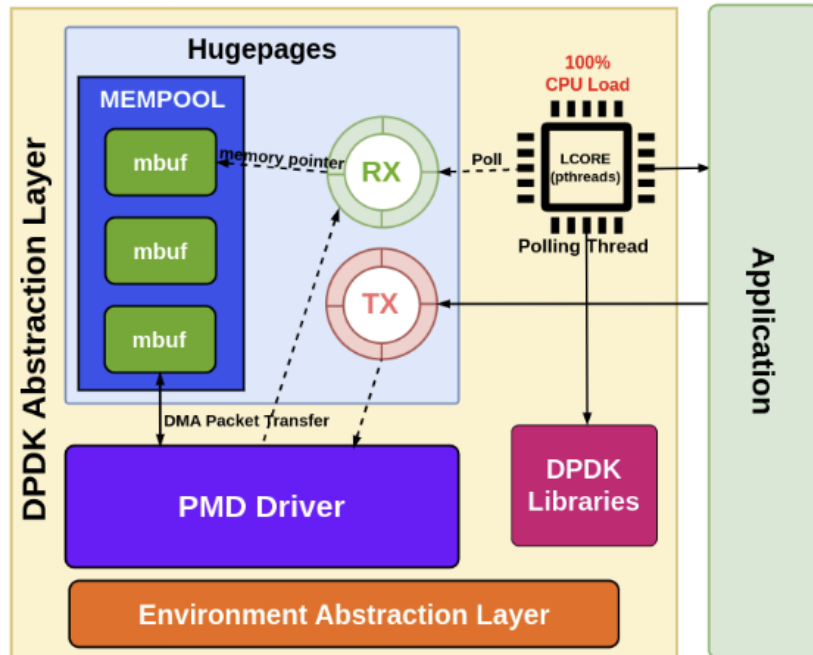


Ref) Impressive Packet Processing Performance Enables Greater Workload Consolidation, Intel

2.1 DPDK(3)



2.1 DPDK(4)



Ref) <https://docs.ngkore.com/blogs/dpdk/dpdk.html#hardware-requirements>

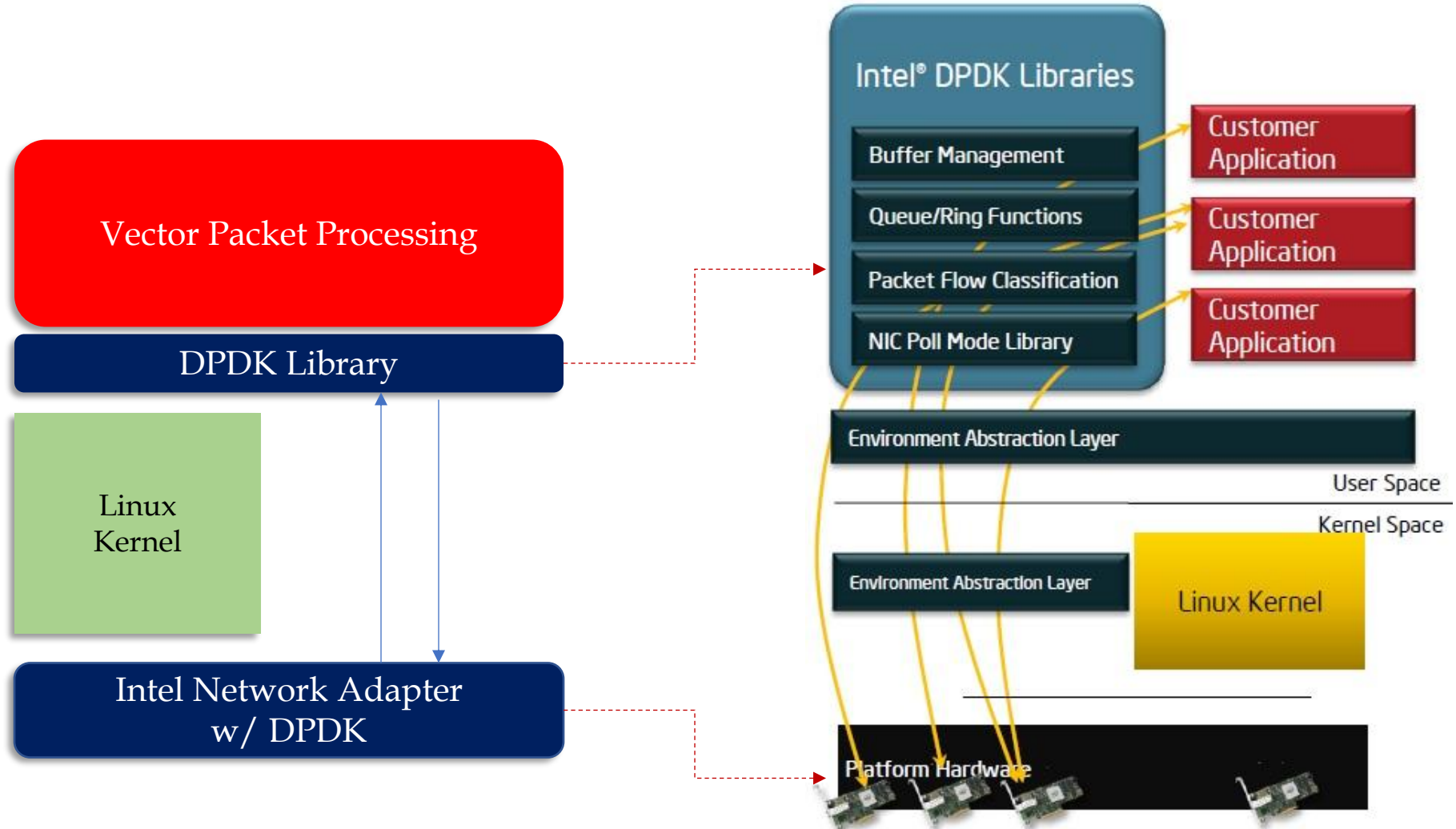
2.2 FD.io VPP

Vector Packet Processing

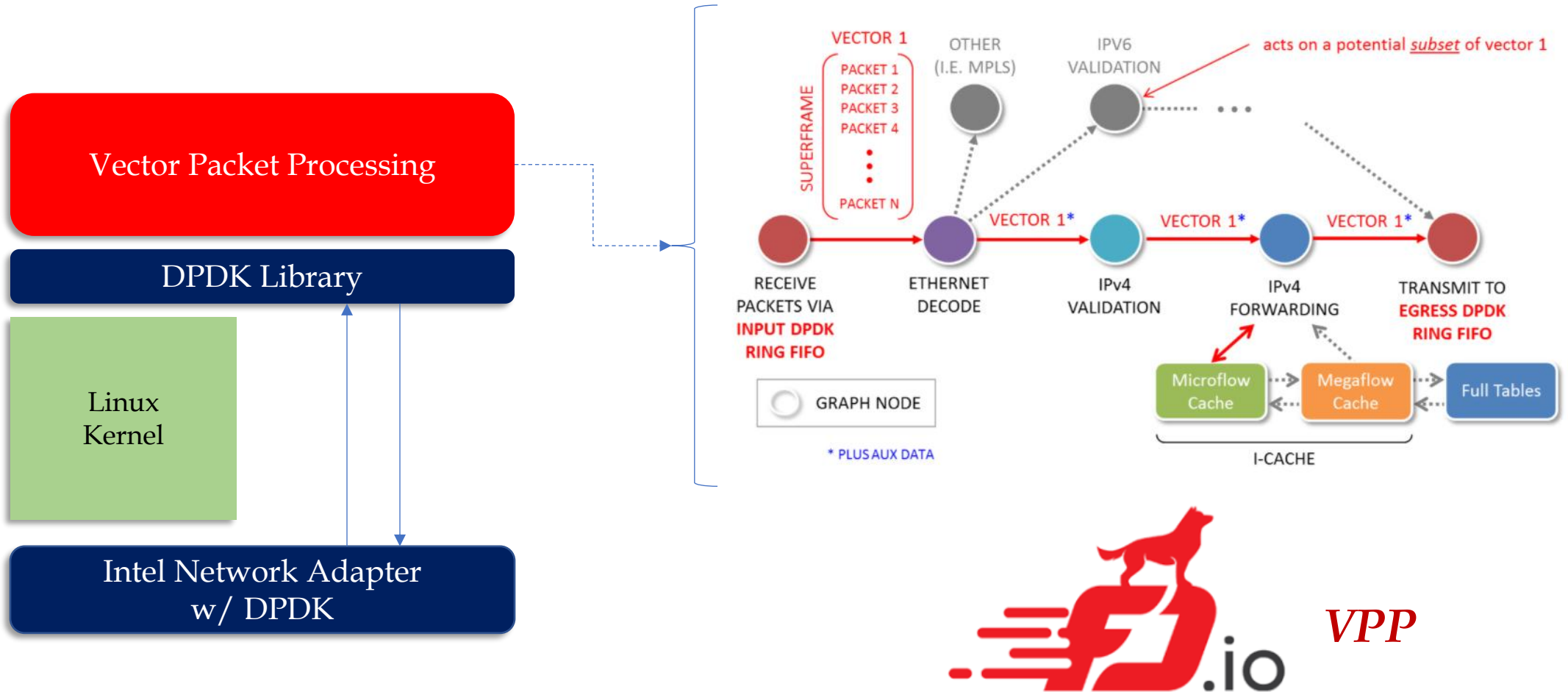


<https://fd.io/>

2.2 FD.io VPP(1)

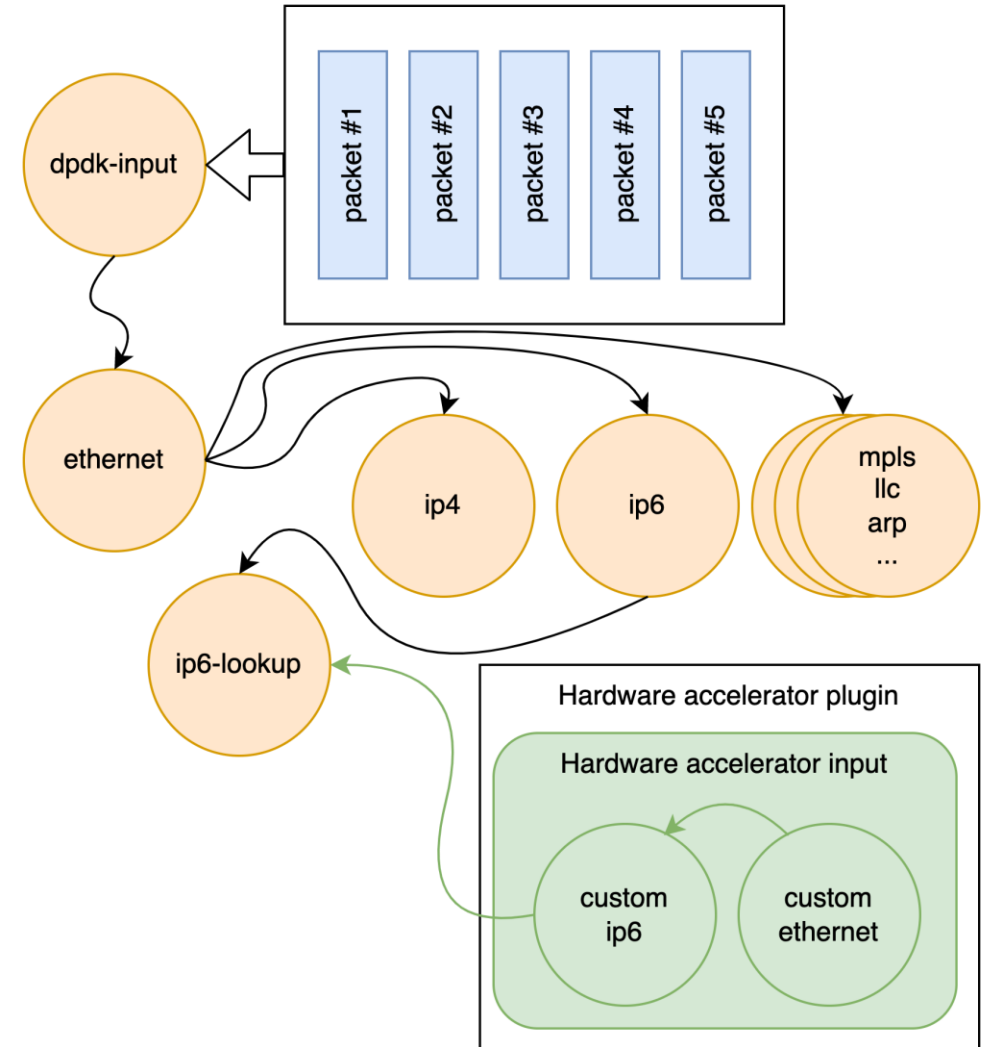
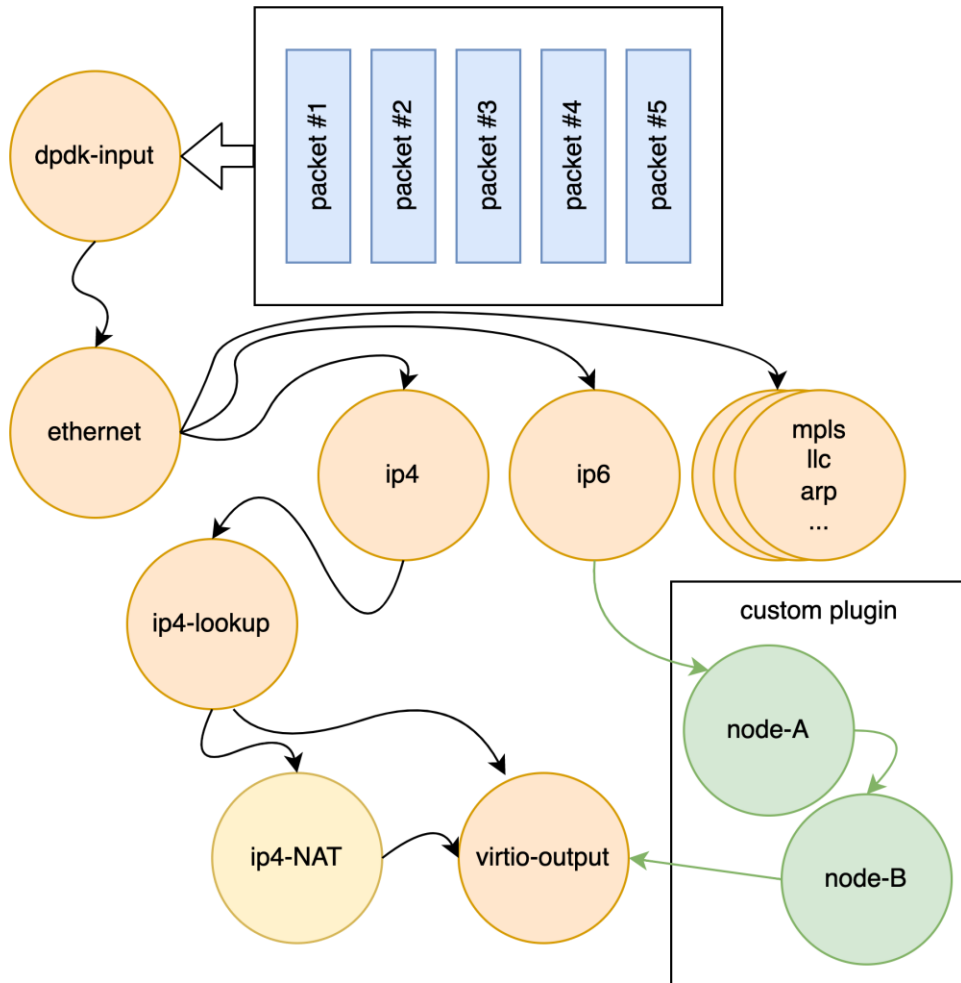


2.2 FD.io VPP(2)



2.2 FD.io VPP(3)

Ref <https://fd.io/technology/>



2.2 FD.io VPP(4) – Applications Area(1)



Routers/Universal CPE etc.

FD.io VPP supports entry hardware options from number of hardware vendors for building Customer Premise Equipmen FD.io based commercial options are available from vendors such as Netgate with TNSR, Cisco with the ASR 9000 and Carri Grade Services Engine, PANTHEON.tech with StoneWork Enterprise, and many more.

Commercial implementations integrate Strong Swan for IKE.

The implementations are accelerated with [DPDK](#) Cryptodev for whole platform crypto.

Featuring:

Data Plane	Control Plane
-----	-----
L2 Switch	Netconf/Yang
VLAN/ Q-inQ	BGP
L3 Router	IKEv2
NAT	
ACL (mac, ip, port)	



Broadband Network Gateway

FD.io VPP has a growing list of network traffic management and security features to support gateway uses cases such as Broadband Network Gateway.

Featuring:

Data Plane	Control Plane
-----	-----
L2 Switch	Netconf/Yang
L3 Router	BGP
Classification	
hQoS	
ACL	
TM (Policing, Metering)	



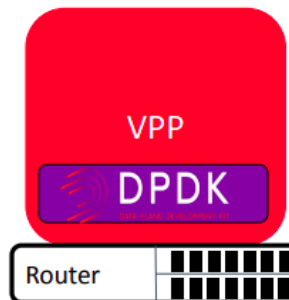
Cloud Load Balancer

FD.io VPP has a rich set of plugin's to enhance its capabilities. Cloud load-balancing is just one of number of feature enhancin plugins available to the end user.

For example: Google Maglev Implementation, Consistent Hashing, Stateful and stateless load balancing, Kube-proxy integrat

Featuring:

Data Plane	Control Plane
-----	-----
Bonding	K8s Kube-Proxy
VLAN / Q-in-Q	
NAT	
ACL (blacklist)	
TM (policing, metering)	
L4 Load Balancer	



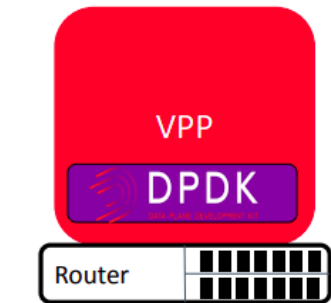
Intrusion Prevention System

FD.io VPP has four different Access Control List technologies; ranging from the simple IP-address whitelisting (called COP) to the sophisticated FD.io VPP Classifiers.

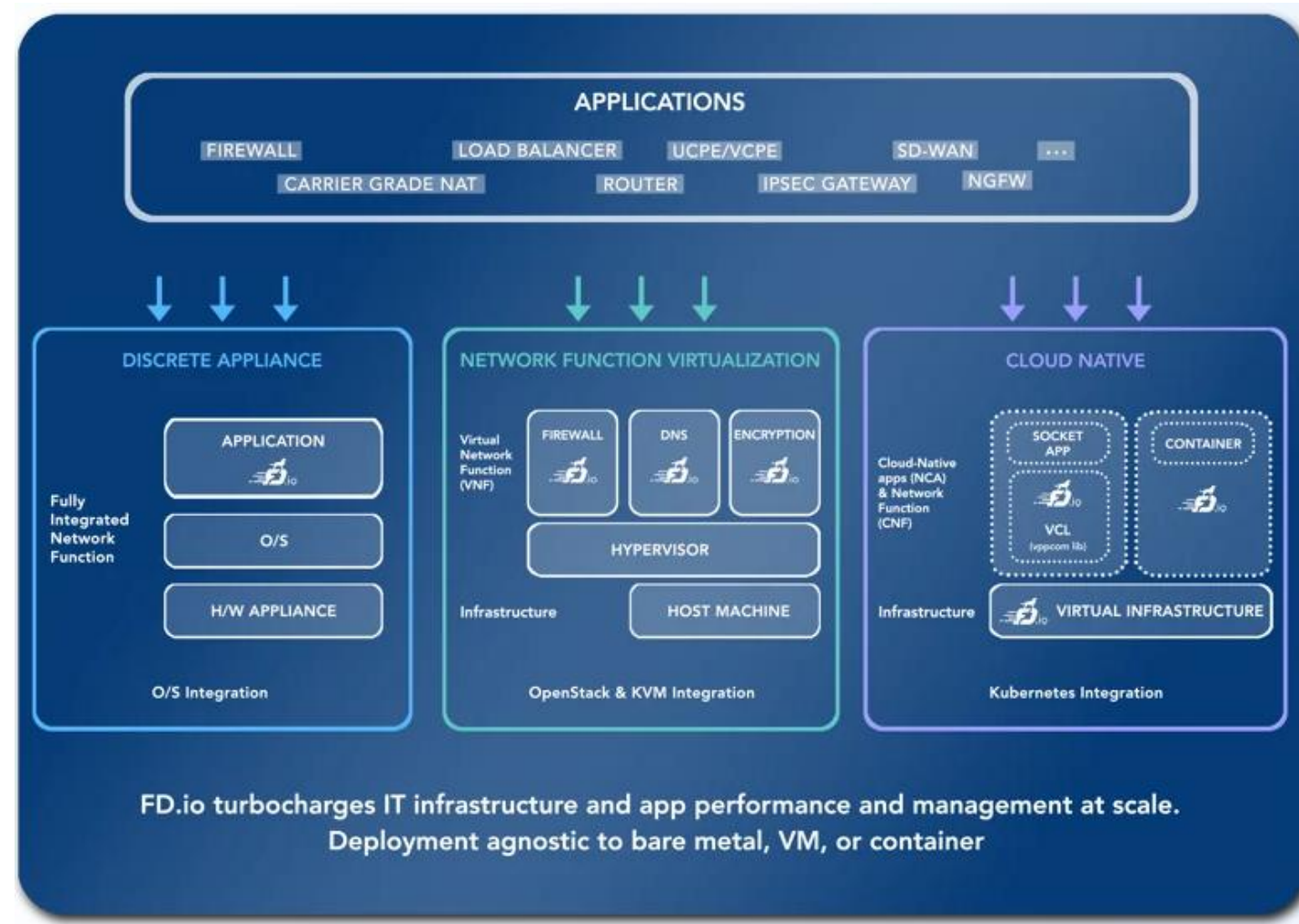
Featuring:

Data Plane	Control Plane
-----	-----
L2 Switch	Netconf/Yang
L3 Router	BGP
Classification	
NAT (mac, ip, port)	
ACL (mac, ip, port)	

2.2 FD.io VPP(4) – Applications Area(2)



- L2 Switch
- L3 Router
- CGNAT
- L4 Switch
- WireGuard VPN
- IPSec VPN
- IPS
- VxLAN Gateway
- ...



Ref) <https://www.netgate.com/resources/articles-vector-packet-processing>

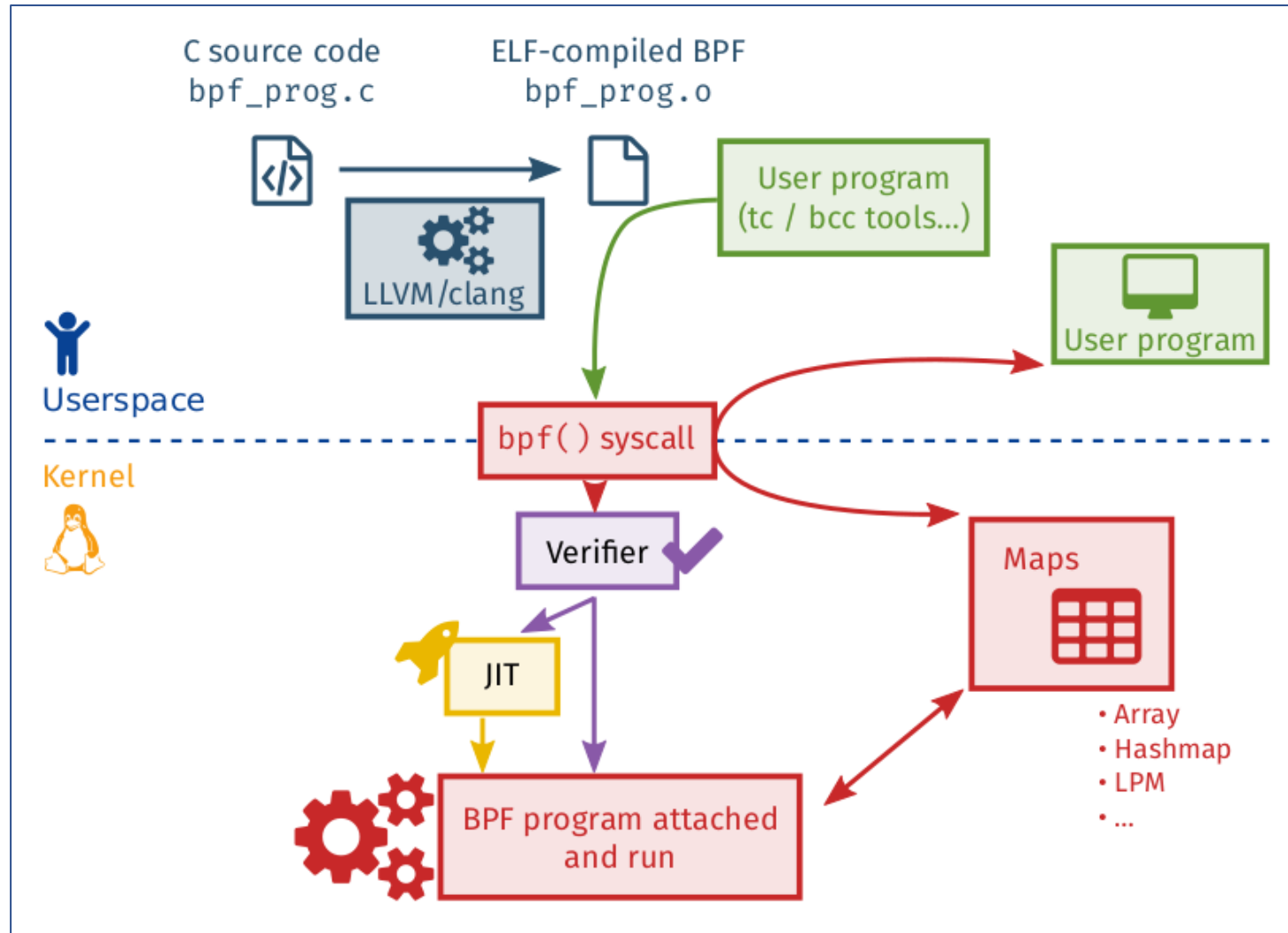
2.3 eBPF XDP

eBPF XDP



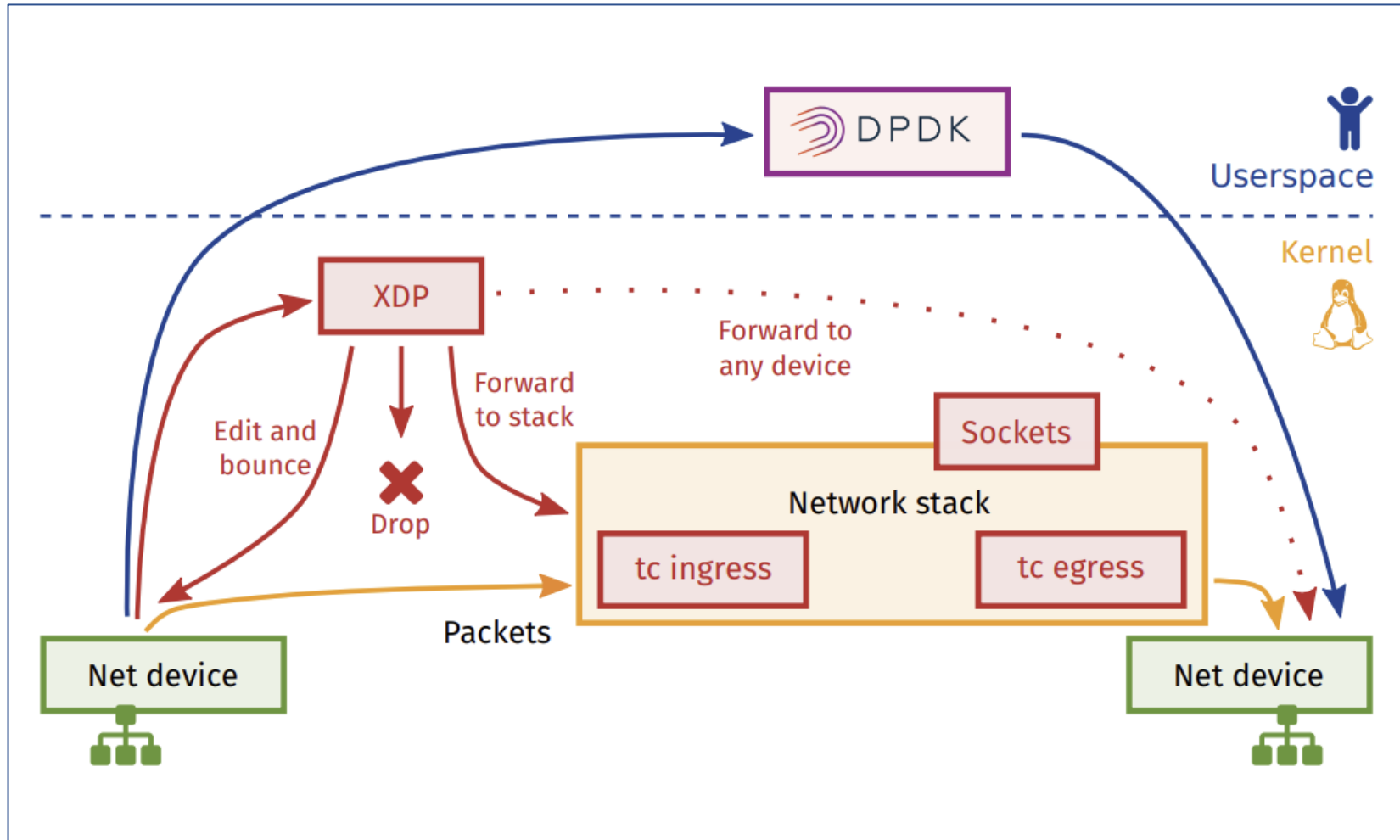
<https://ebpf.io/>

2.3 eBPF XDP(1) – eBPF Concept



Ref) https://media.frnog.org/FRnOG_28/FRnOG_28-3.pdf

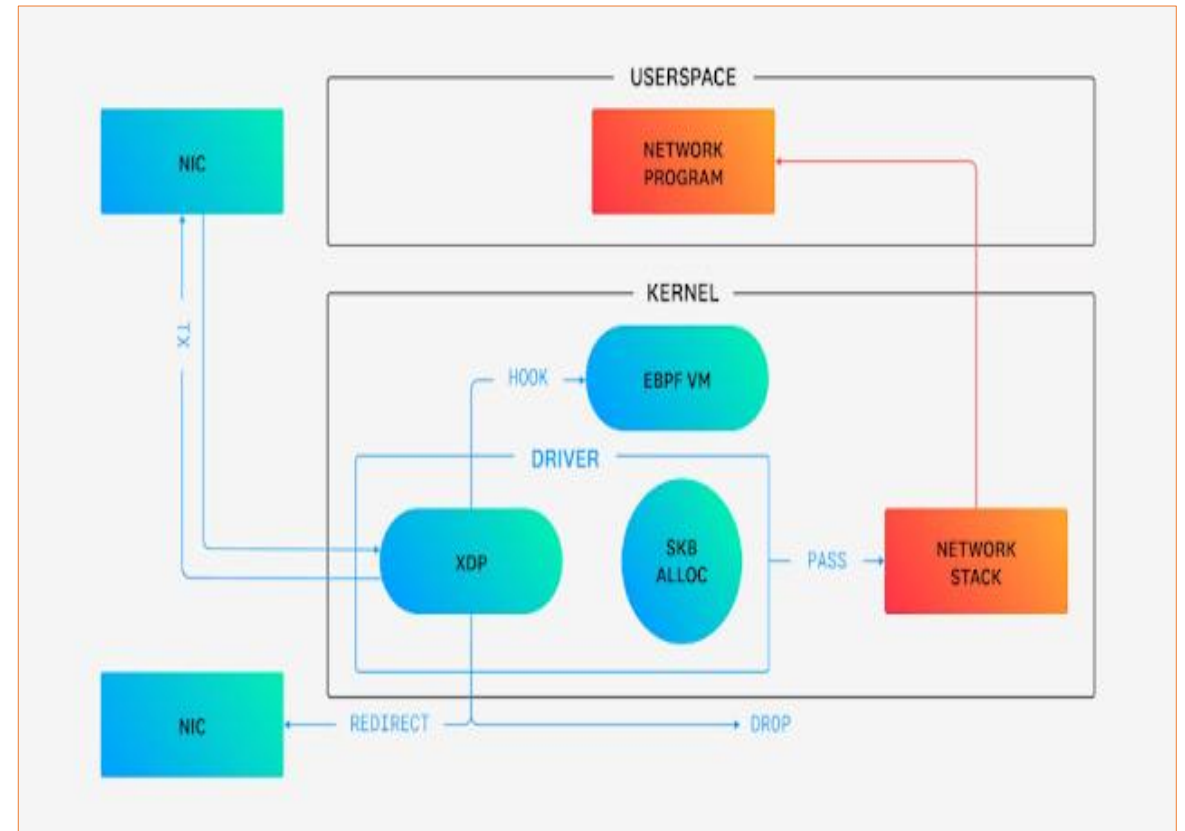
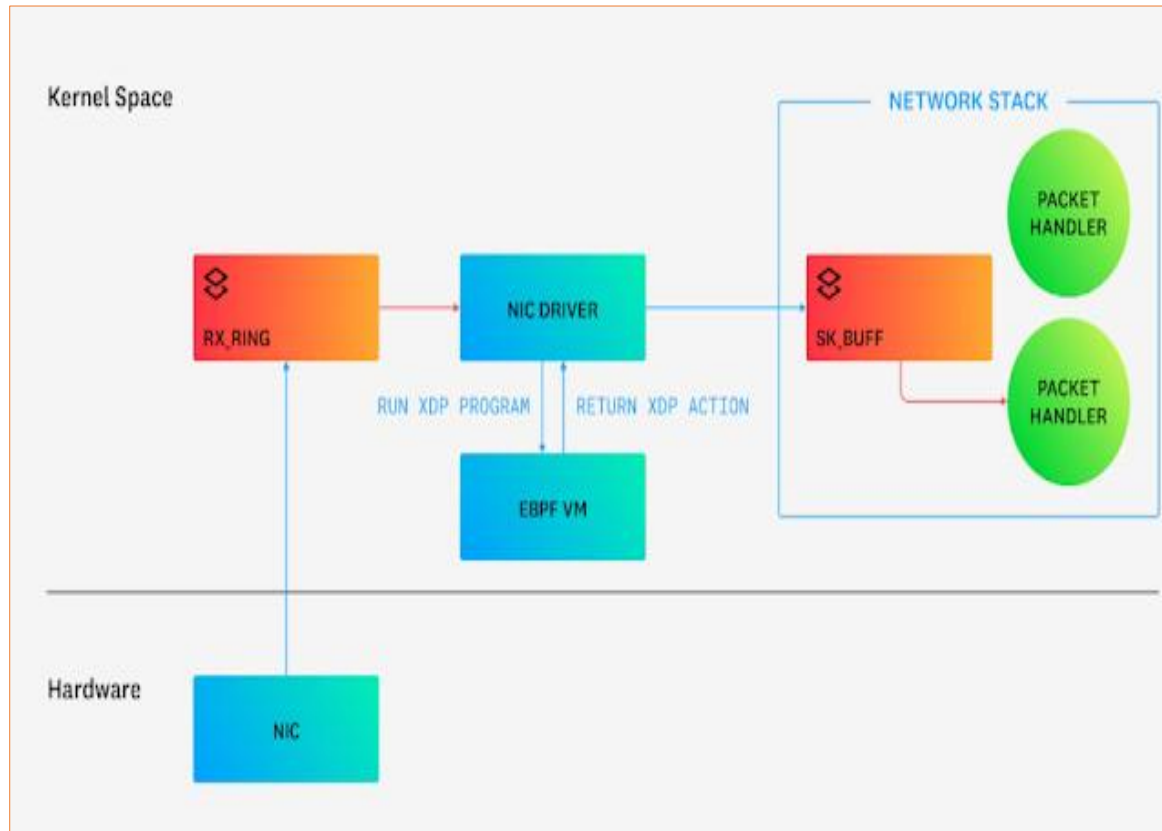
2.3 eBPF XDP(2) – XDP Concept(1)



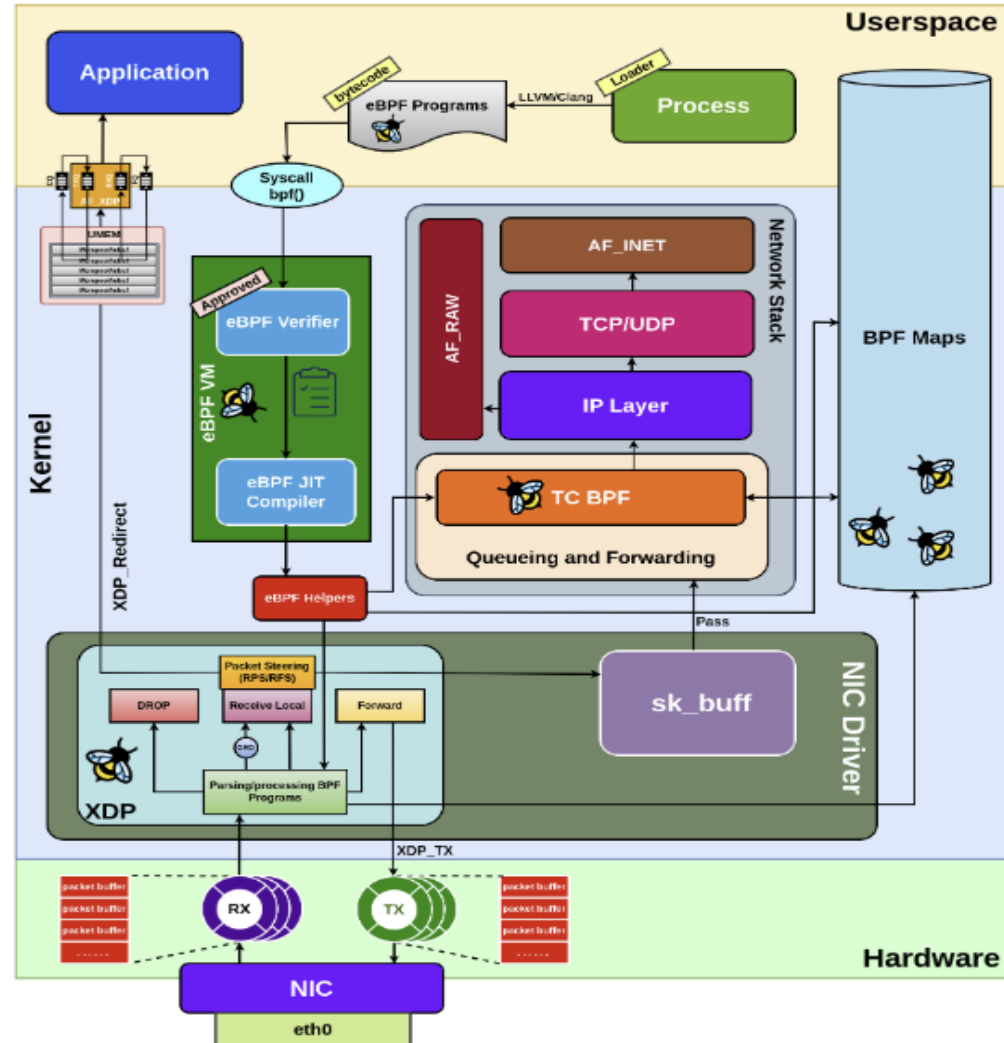
Ref) https://media.frnwg.org/FRnWG_28/FRnWG_28-3.pdf

2.3 eBPF XDP(2) – XDP Concept(2)

Ref) <https://www.datadoghq.com/blog/xdp-intro/>

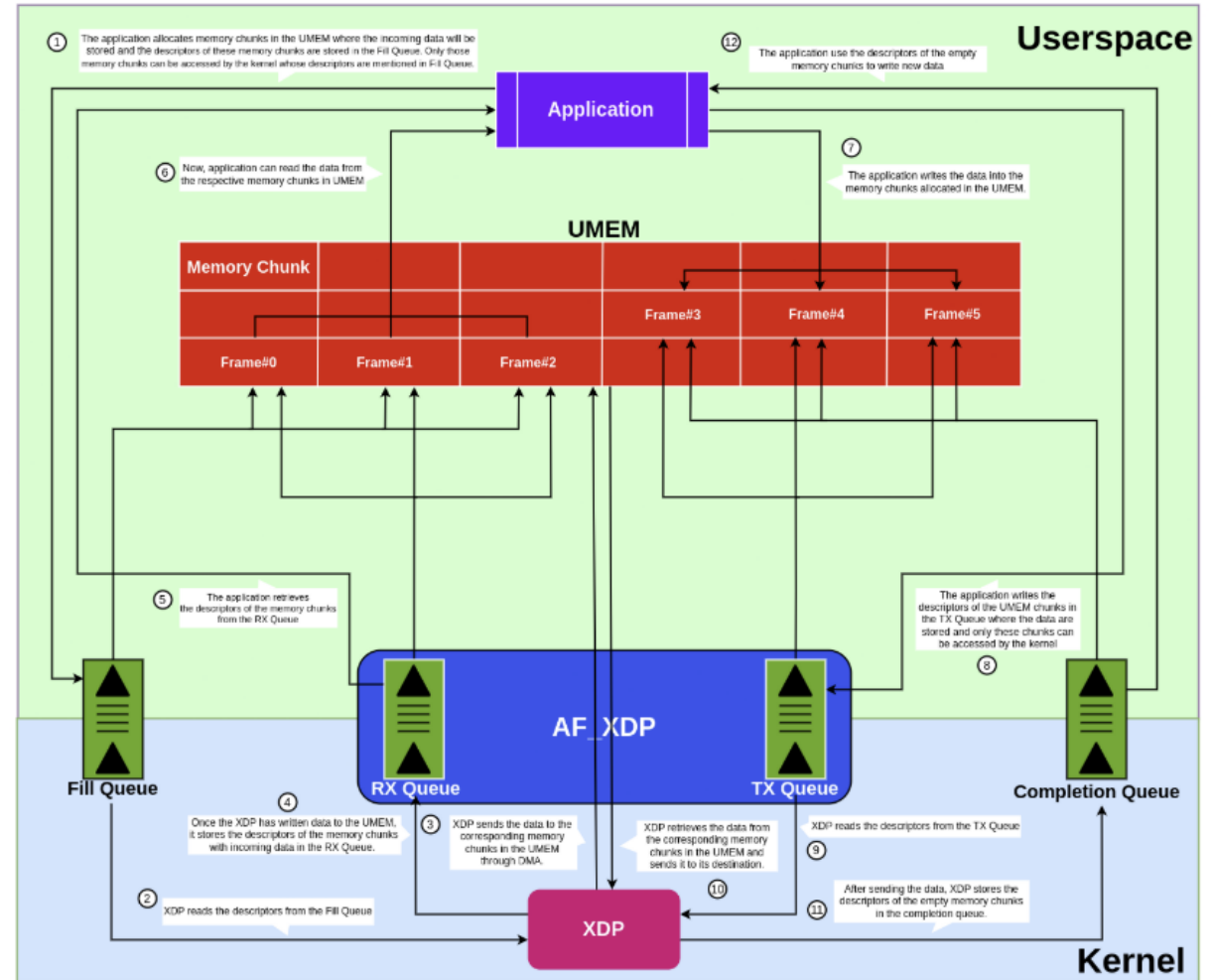
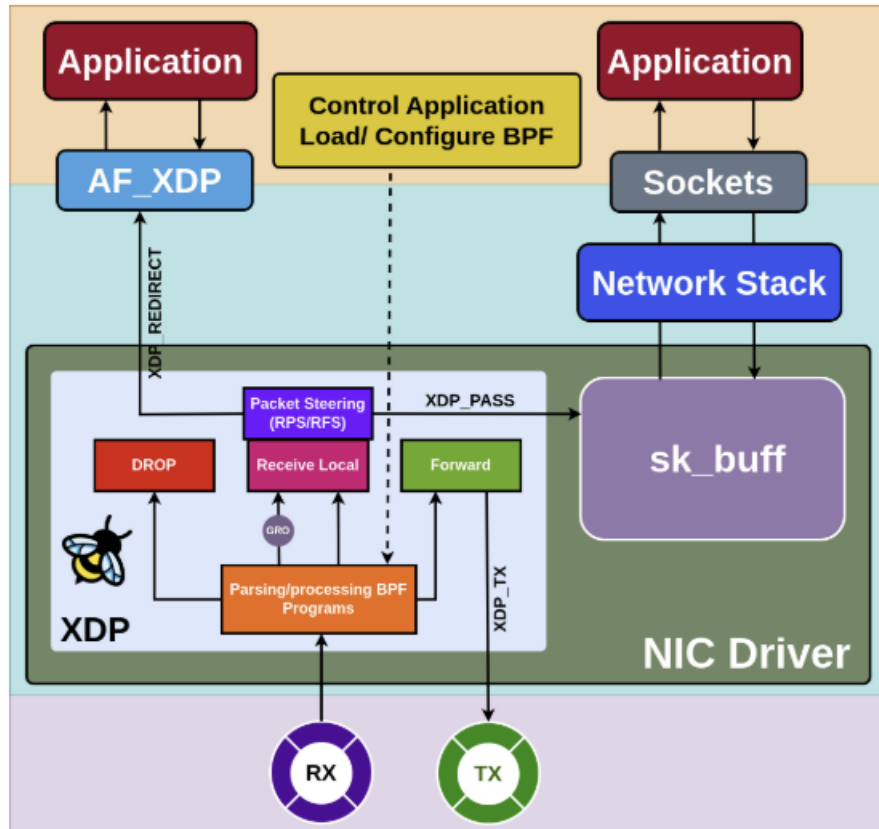


2.3 eBPF XDP(3) – XDP & AF_XDP(1)



Ref) <https://docs.ngkore.com/blogs/afxdp/afxdp.html#hardware-requirements>

2.3 eBPF XDP(3) - XDP & AF_XDP(2)

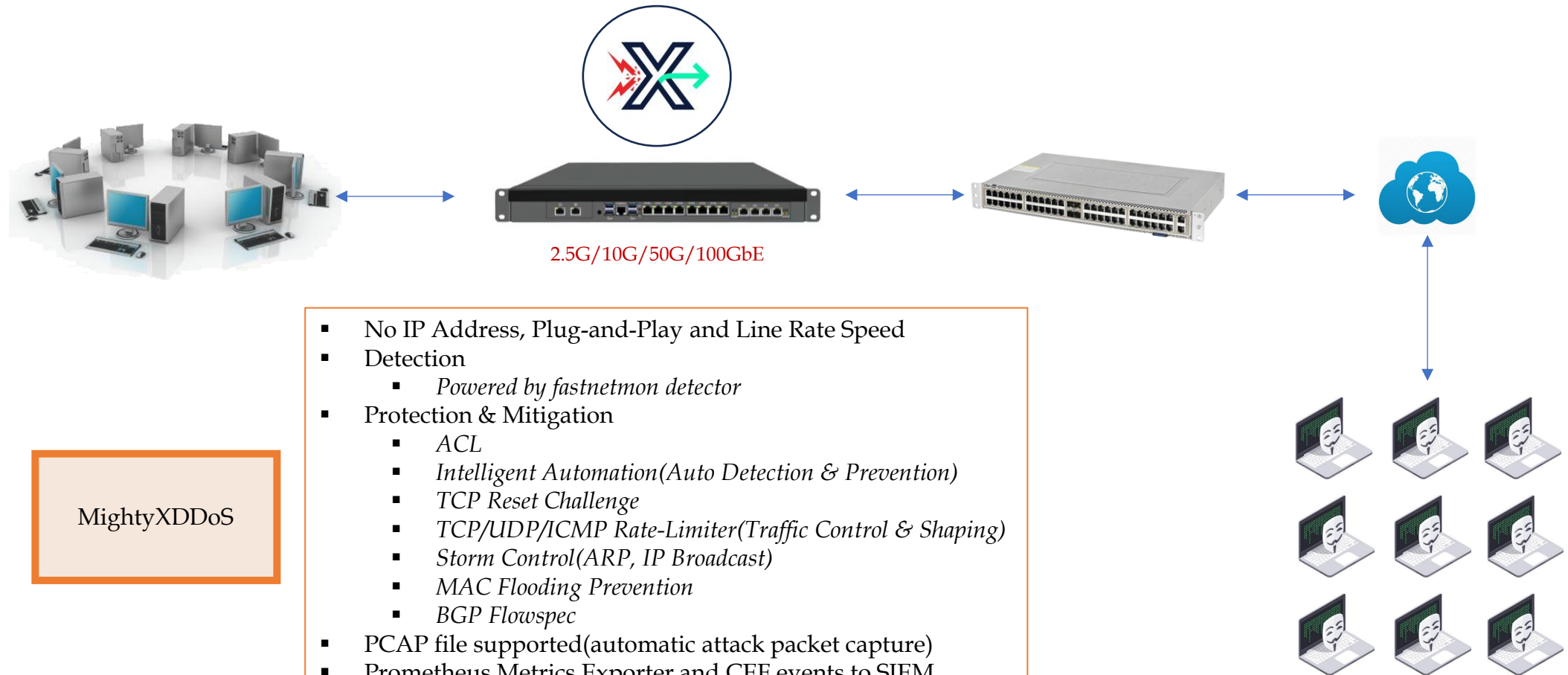


3. MightyXDDoS™

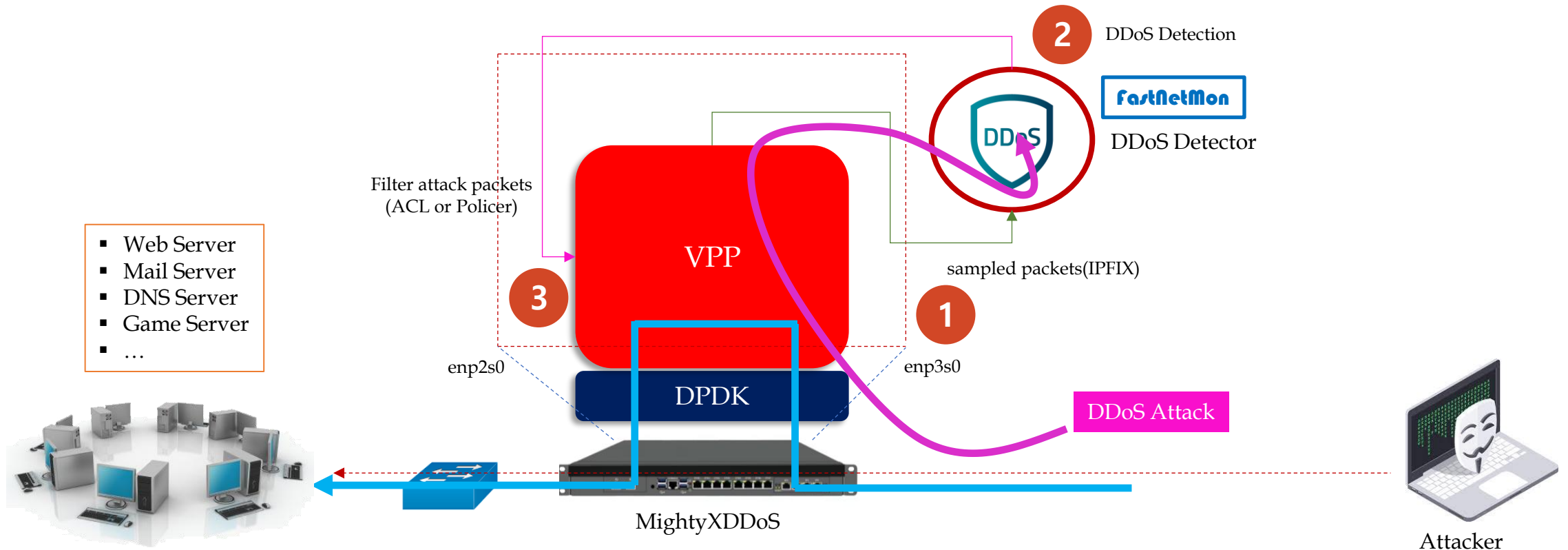


XDDoS Mitigator

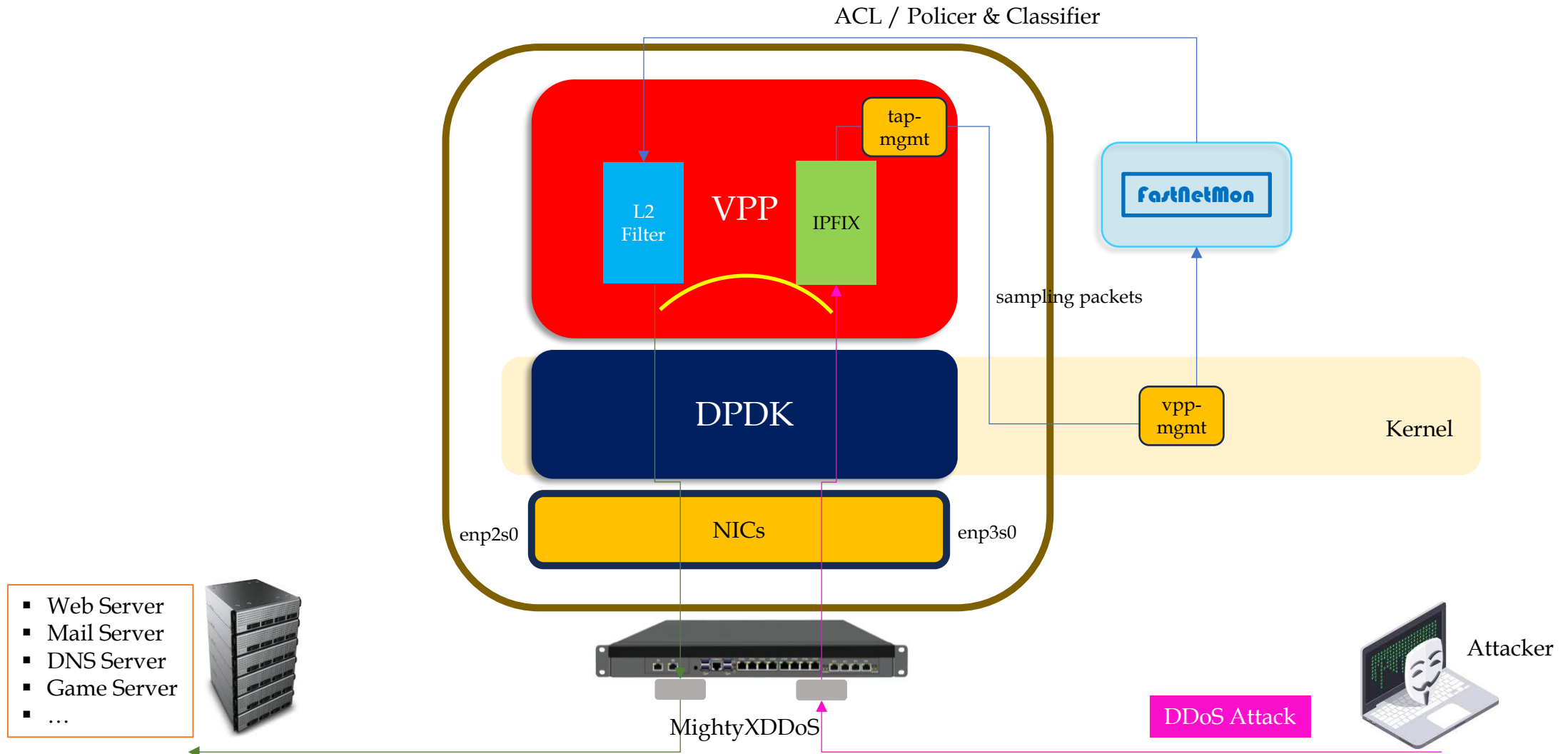
3. MightyXDDoS(1) – Inline Mitigator Box



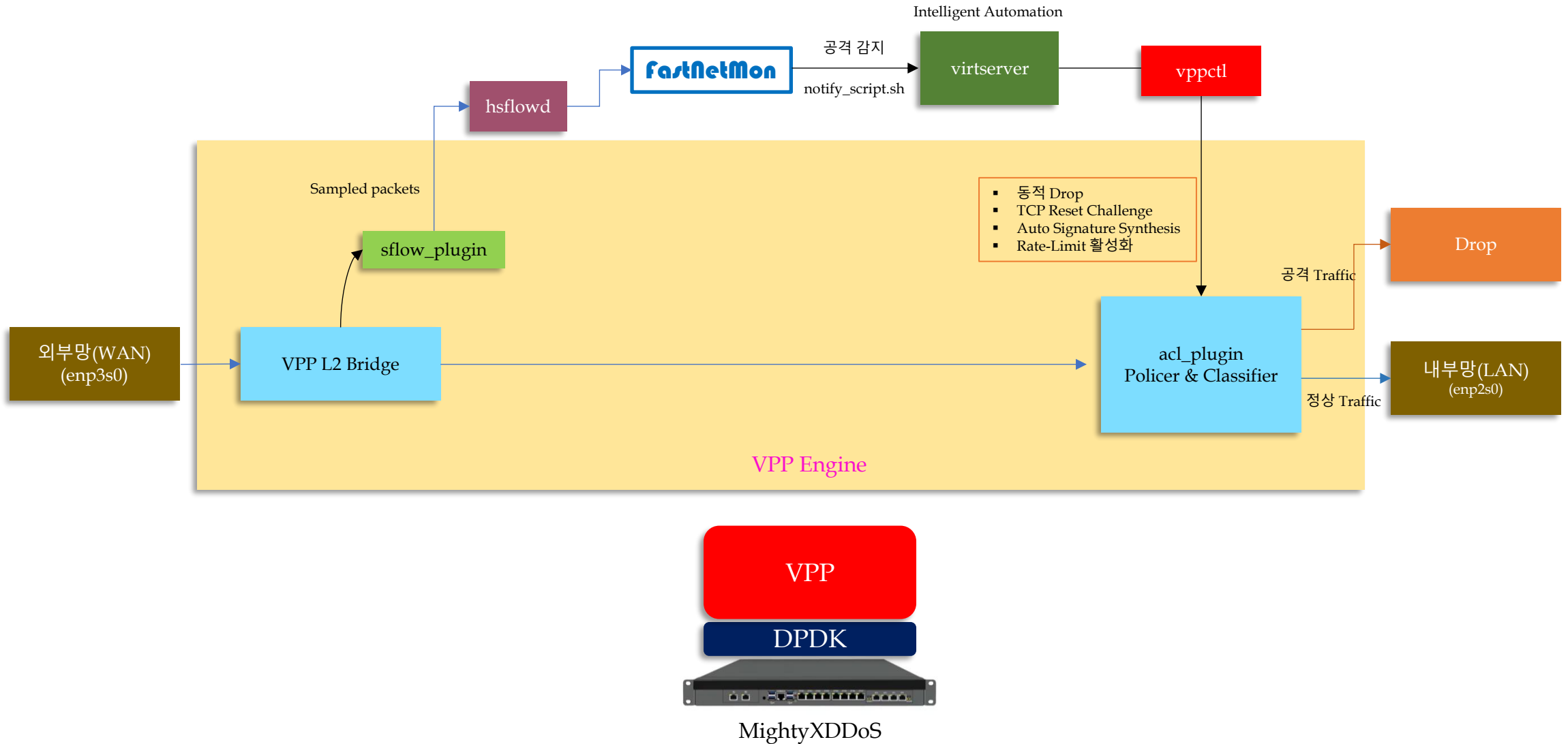
3. MightyXDDoS(2) – VPP Mitigator(1)



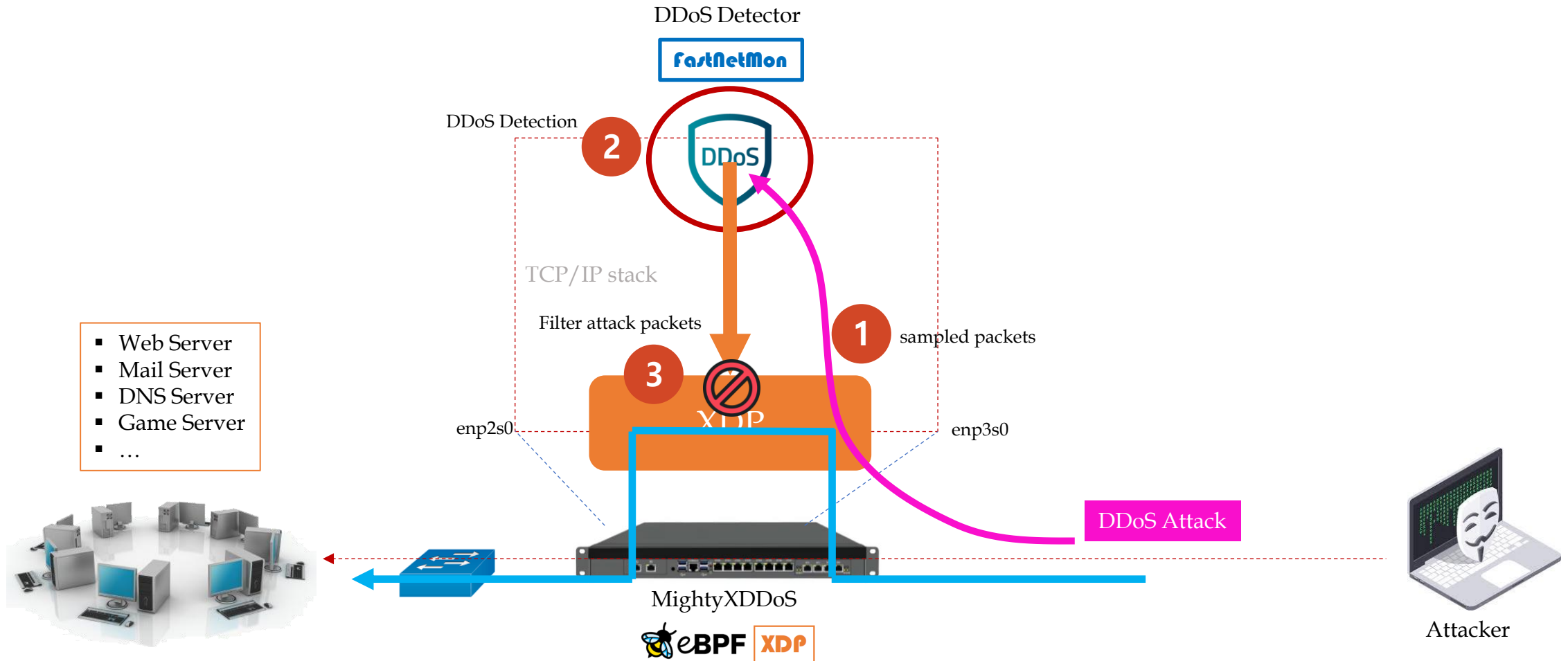
3. MightyXDDoS(2) – VPP Mitigator(2)



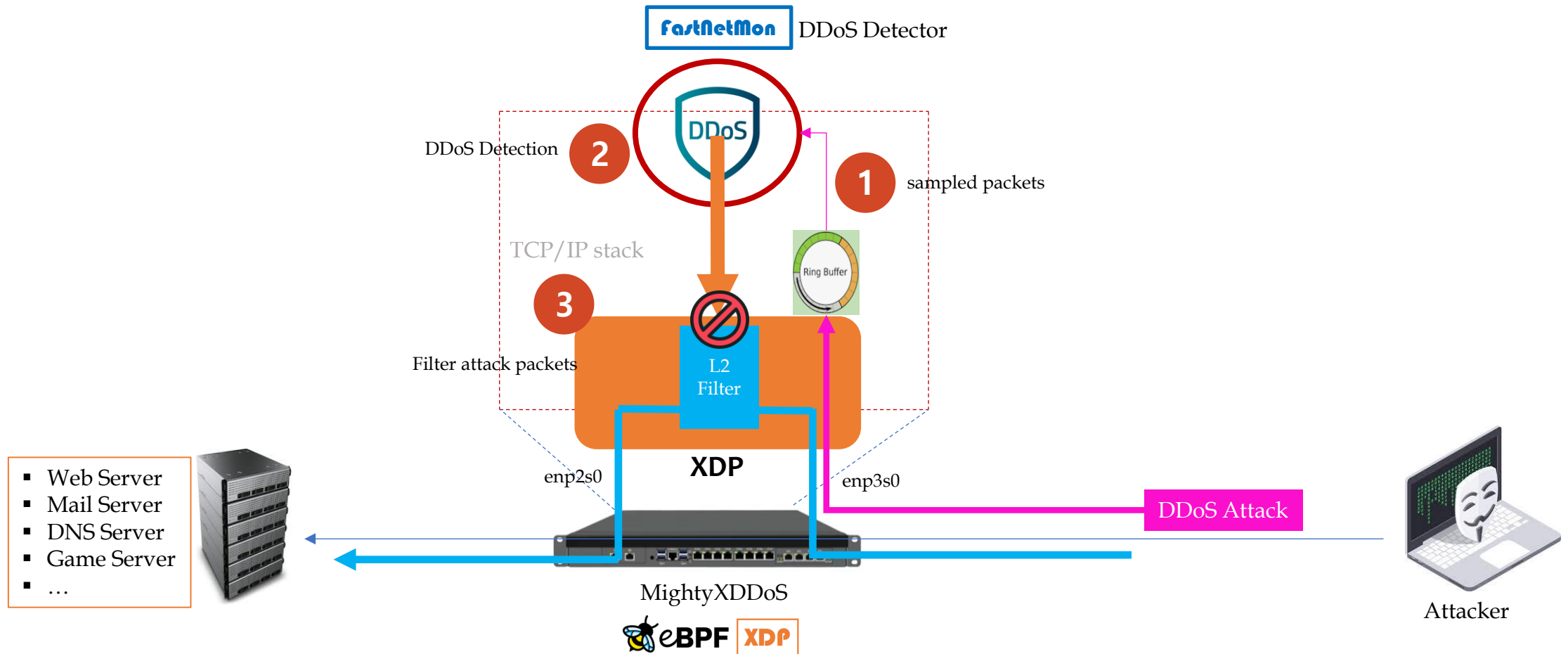
3. MightyXDDoS(2) – VPP Mitigator(3)



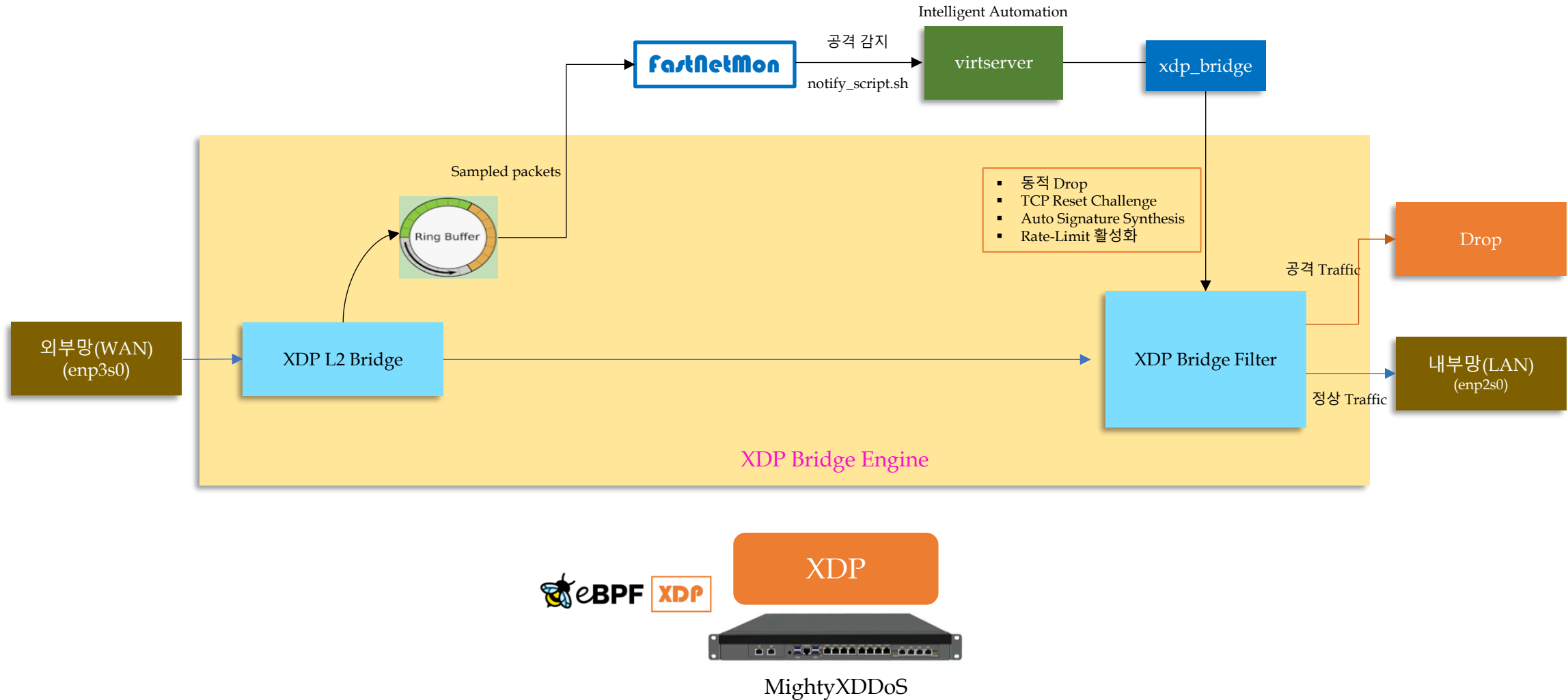
3. MightyXDDoS(3) – XDP Mitigator(1)



3. MightyXDDoS(3) – XDP Mitigator(2)

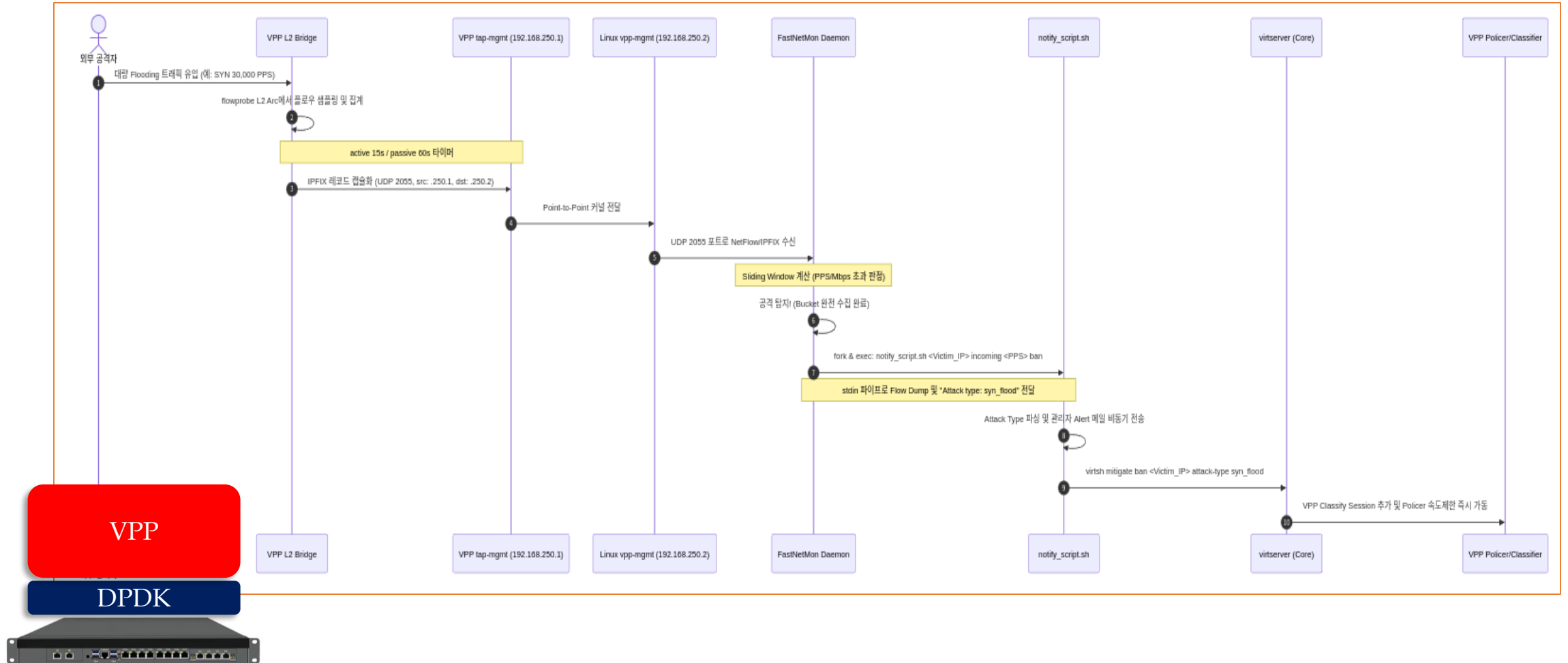


3. MightyXDDoS(3) – XDP Mitigator(3)



3. MightyXDDoS(4) – w/ FastNetMon Detector(1)

fastNetMon



3. MightyXDDoS(4) – w/ FastNetMon Detector(2)

The screenshot shows the MightyXDDoS Detector web interface. The left sidebar contains navigation links: admin (Security Administrator), Dashboard (System Monitor, Traffic Monitor), Detector (selected), View Attackers, and Mitigator (Policy, Flooding Attack, Broadcast Filter, Blocked Attacker, BGP Flowspec, Statistics). The main content area displays the Detector status with four key metrics: Detection Engine (ACTIVE), Live Traffic State (TRAFFIC DETECTED), Trigger Baselines (20000 pps / 800 Mbps), and Attack Incidents (20). Below these, the Detector Monitor (Live Telemetry) section shows a real-time stream of FastNetMon Engine data. The stream output is as follows:

```
FastNetMon 1.2.10 Try Advanced edition: https://fastnetmon.com/product-overview/
IPs ordered by: bytes
Incoming traffic
192.168.8.220      74475 pps    23 mbps    1893 flows
                  74475 pps    23 mbps    989 flows  *banned*

Outgoing traffic
                  0 pps     0 mbps     0 flows

Internal traffic
                  0 pps     0 mbps

Other traffic
                  0 pps     0 mbps

Screen updated in: 5.3e-05 sec
Traffic calculated in: 0 sec 67 microseconds
Not processed packets: 0 pps
```

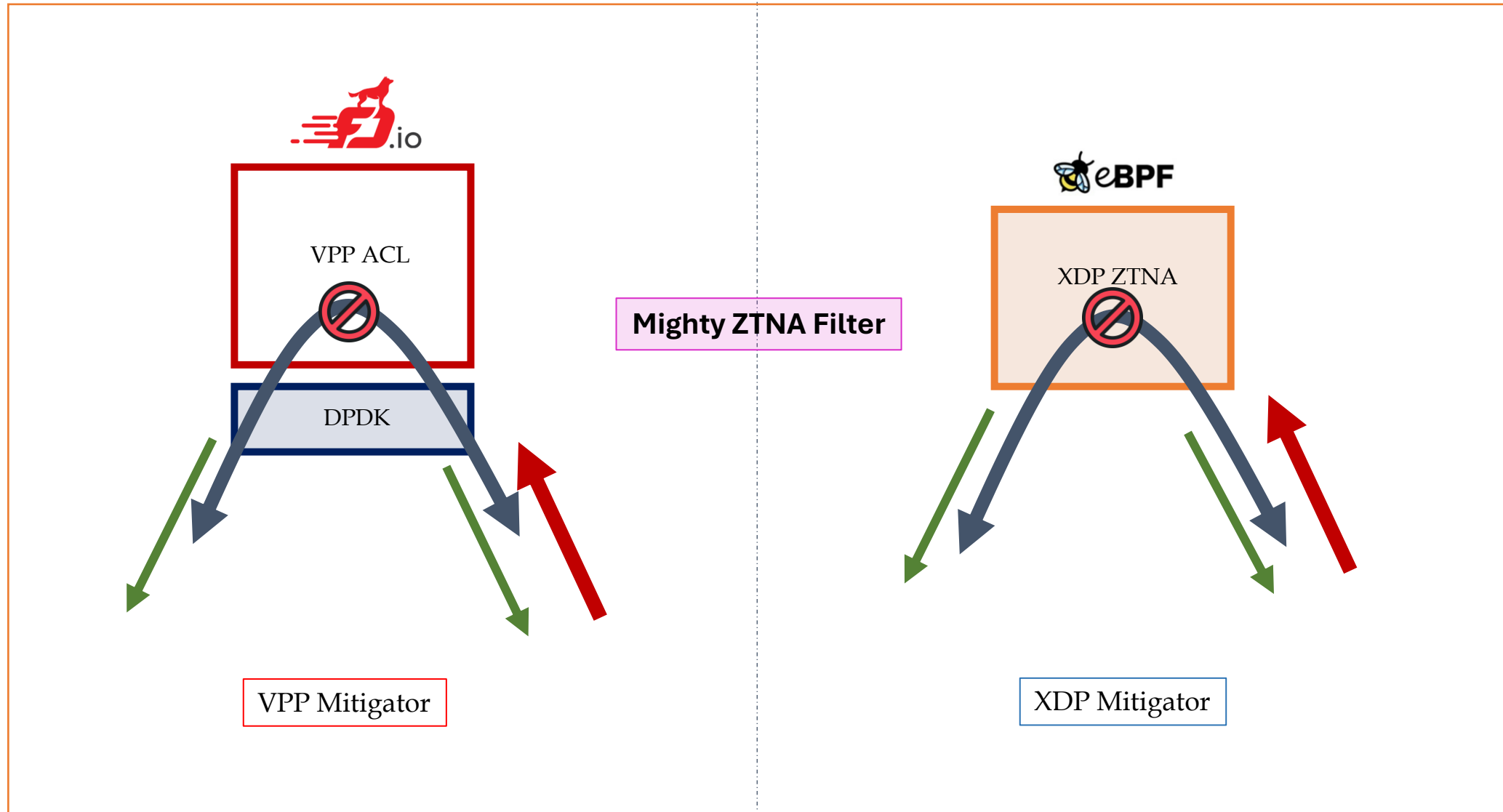
The screenshot shows the MightyXDDoS Detector web interface, specifically the Recent Attack Sessions section. The left sidebar is the same as the previous screenshot. The main content area displays a table of recent attack sessions, with a search filter for victim IP or file. The table has columns for Target Victim IP, Incident Timestamp, Forensics File, and Action. Below the table, the details for a specific attack session are shown.

TARGET VICTIM IP	INCIDENT TIMESTAMP	FORENSICS FILE	ACTION
192.168.8.220	17 09 26 11:52:36	192.168.8.220_17_09_26_11:52:36.txt	Inspect
192.168.8.220	17 09 26 11:52:26	192.168.8.220_17_09_26_11:52:26.txt	Inspect
192.168.8.220	17 09 26 11:46:56	192.168.8.220_17_09_26_11:46:56.txt	Inspect

ATTACK DETAILS 192.168.8.220_17_09_26_11:52:36.txt

IP: 192.168.8.220
Attack type: syn_flood
Initial attack power: 27875 packets per second
Peak attack power: 27875 packets per second
Attack direction: incoming
Attack protocol: tcp
Total incoming traffic: 8 mbps
Total outgoing traffic: 0 mbps
Total incoming pps: 27875 packets per second
Total outgoing pps: 0 packets per second
Total incoming flows: 366 flows per second
Total outgoing flows: 0 flows per second
Incoming ip fragmented traffic: 0 mbps

3. MightyXDDoS(5) – ZTNA ACL(1)



3. MightyXDDoS(5) – ZTNA ACL(2)

The screenshot shows the 'Policy' page in the MightyXDDoS interface. The left sidebar contains navigation links: admin, DASHBOARD, DETECTOR, MITIGATOR, and Policy (selected). The main content area is titled 'Policy' and shows the 'ACL / Mitigator Interface Binding Scope' section. This section contains two tabs: 'All (LAN & WAN)' (selected) and 'WAN Only'. The 'All (LAN & WAN)' tab is described as 'Default' and 'Binds to every bridge member interface, LAN included — a LAN-originated attack (e.g. a compromised internal host) is also detected and blocked, not just WAN-originated ones. Slightly higher per-interface ACL overhead.' The 'WAN Only' tab is described as 'Binds only to WAN-role interfaces — lower overhead, appropriate when attacks are only expected from the WAN side. LAN-originated attacks are still detected by Detector but not enforced against by this box. Requires every bridge member interface to have a LAN/WAN role assigned first.' Below these tabs is a 'Save' button. The 'WAN Service Port Policy' section is also visible, showing a toggle for 'Enable WAN Service Port Policy' set to 'ENABLED' and a table for 'ALLOWED PORTS'.

Policy

ACL / Mitigator Interface Binding Scope

Controls which bridge member interfaces get the enforcement layer bound to them: manual Firewall rules, the fastnetmon/SYN-flood automatic attacker-block ACL, and the SYN-flood classify-policer session. Detector's own traffic visibility is unaffected either way — it always watches every bridge member.

All (LAN & WAN) Default

Binds to every bridge member interface, LAN included — a LAN-originated attack (e.g. a compromised internal host) is also detected and blocked, not just WAN-originated ones. Slightly higher per-interface ACL overhead.

WAN Only

Binds only to WAN-role interfaces — lower overhead, appropriate when attacks are only expected from the WAN side. LAN-originated attacks are still detected by Detector but not enforced against by this box. Requires every bridge member interface to have a LAN/WAN role assigned first.

Applies immediately to VPP. **Save**

WAN Service Port Policy ZTNA-STYLE, AUTO-GENERATED

Enter only a protocol and port — no source/destination/interface needed. Once enabled, only these ports are reachable from WAN into the protected network; everything else is denied by default. Traffic from LAN to the Internet is never affected by this setting. For ad-hoc manual rules with full source/destination control, use the **Firewall** page instead — that layer is evaluated first regardless of this setting.

Enable WAN Service Port Policy **ENABLED**

ALLOWED PORTS **+ Add Port Rule**

The screenshot shows the 'Policy' page in the MightyXDDoS interface, specifically the 'WAN Service Port Policy' section. The 'Enable WAN Service Port Policy' toggle is set to 'ENABLED'. Below this is a table titled 'ALLOWED PORTS' with columns for Protocol, Port, Description, and Action. The table lists several rules: TCP 443 (https), TCP 8080 (https), UDP 6000 (test udp), ICMP Any (no ports) (icmp test), and TCP 80 (http). Each rule has a delete icon in the Action column. Below the table is a 'Save' button. At the bottom, there are two graphs: 'ACL Permit Packets (cumulative)' and 'ACL Deny Packets (cumulative)'. The 'ACL Permit Packets' graph shows a green line at approximately 100,000 packets. The 'ACL Deny Packets' graph shows a red line at approximately 140 packets. Both graphs are labeled 'Last updated: 오후 2:29:47'.

Policy

Enable WAN Service Port Policy **ENABLED**

ALLOWED PORTS **+ Add Port Rule**

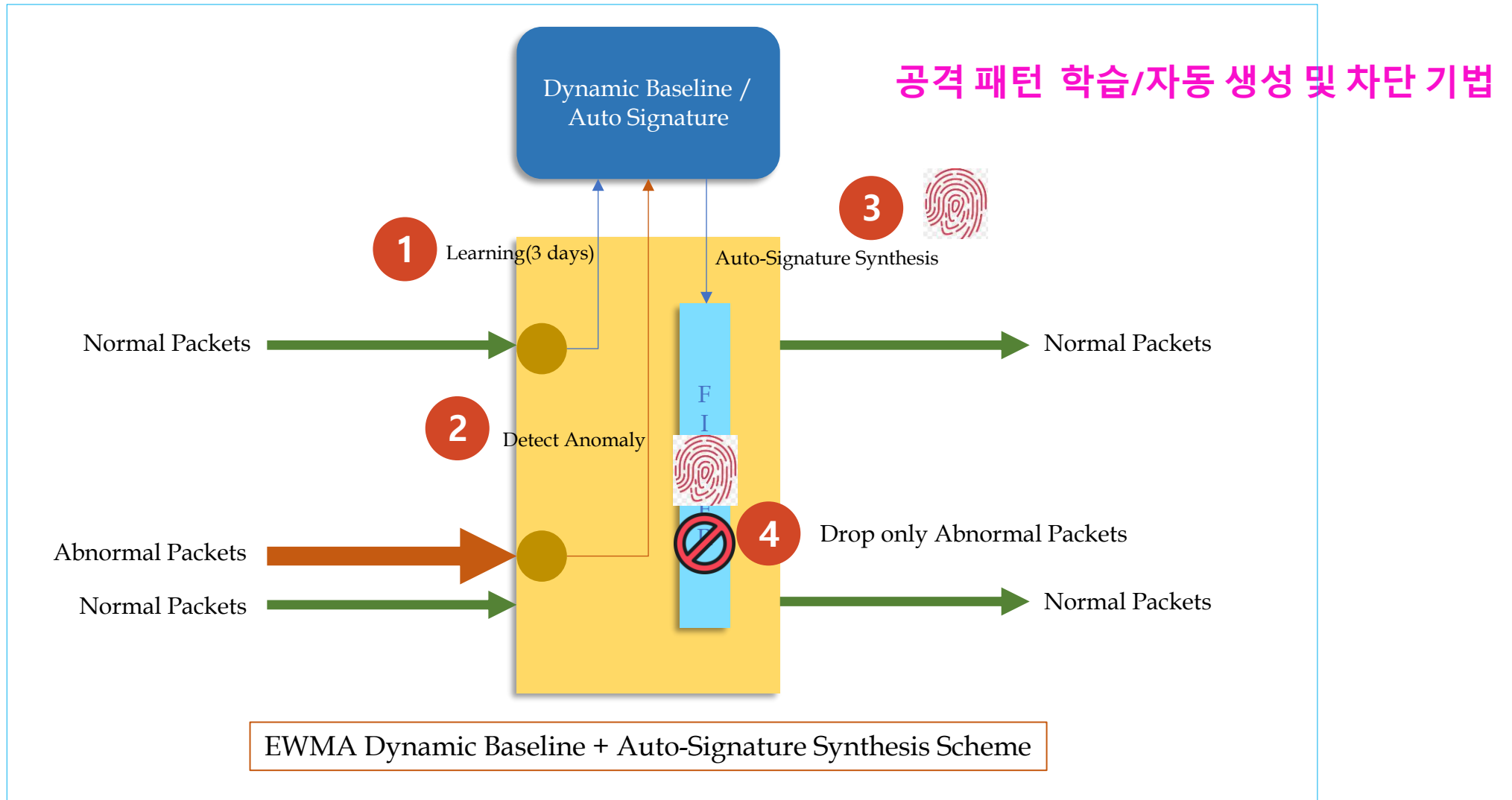
Protocol	Port	Description	Action
TCP	443	https	
TCP	8080	https	
UDP	6000	test udp	
ICMP	Any (no ports)	icmp test	
TCP	80	http	

Applies immediately. **Save**

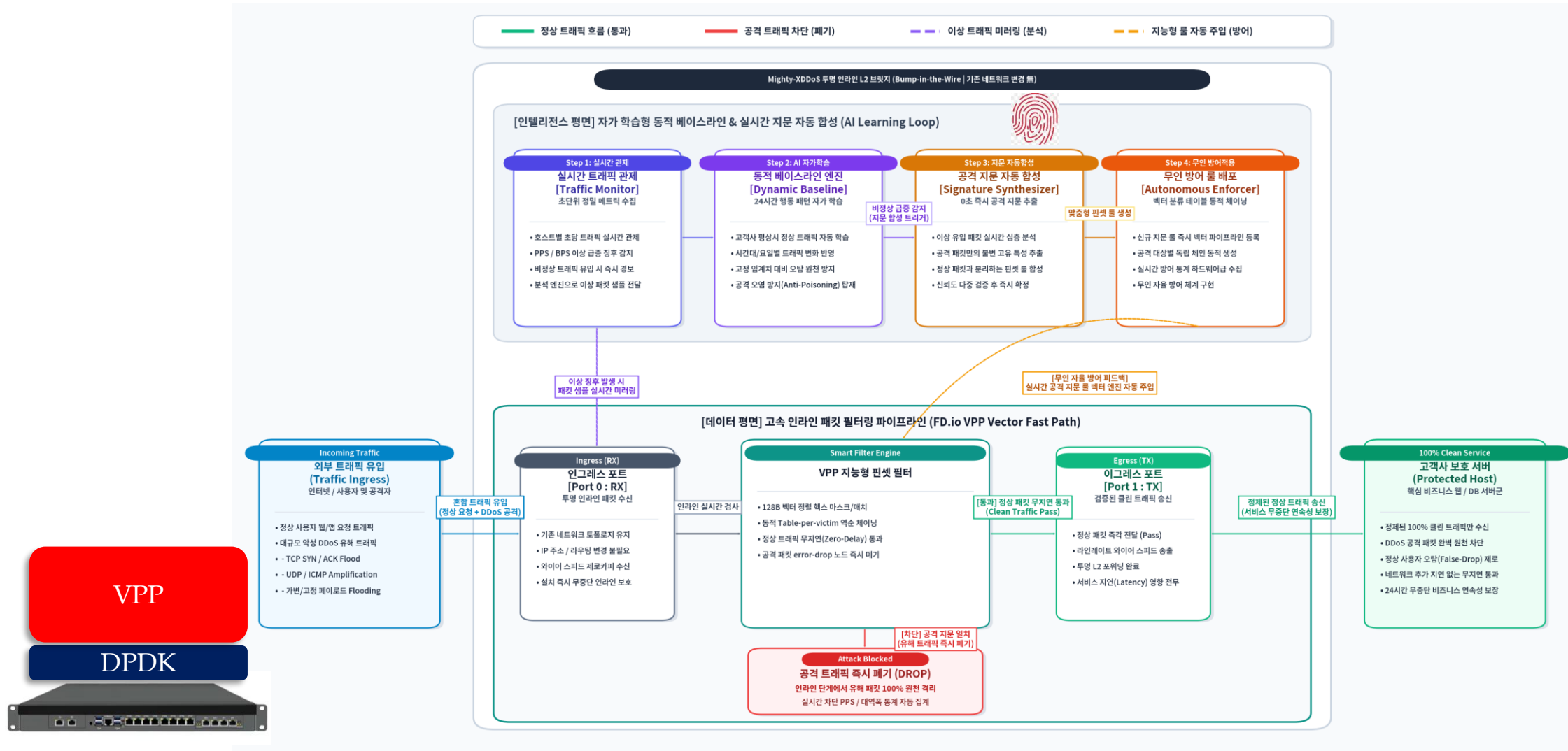
ACL Permit Packets (cumulative) Last updated: 오후 2:29:47

ACL Deny Packets (cumulative) Last updated: 오후 2:29:47

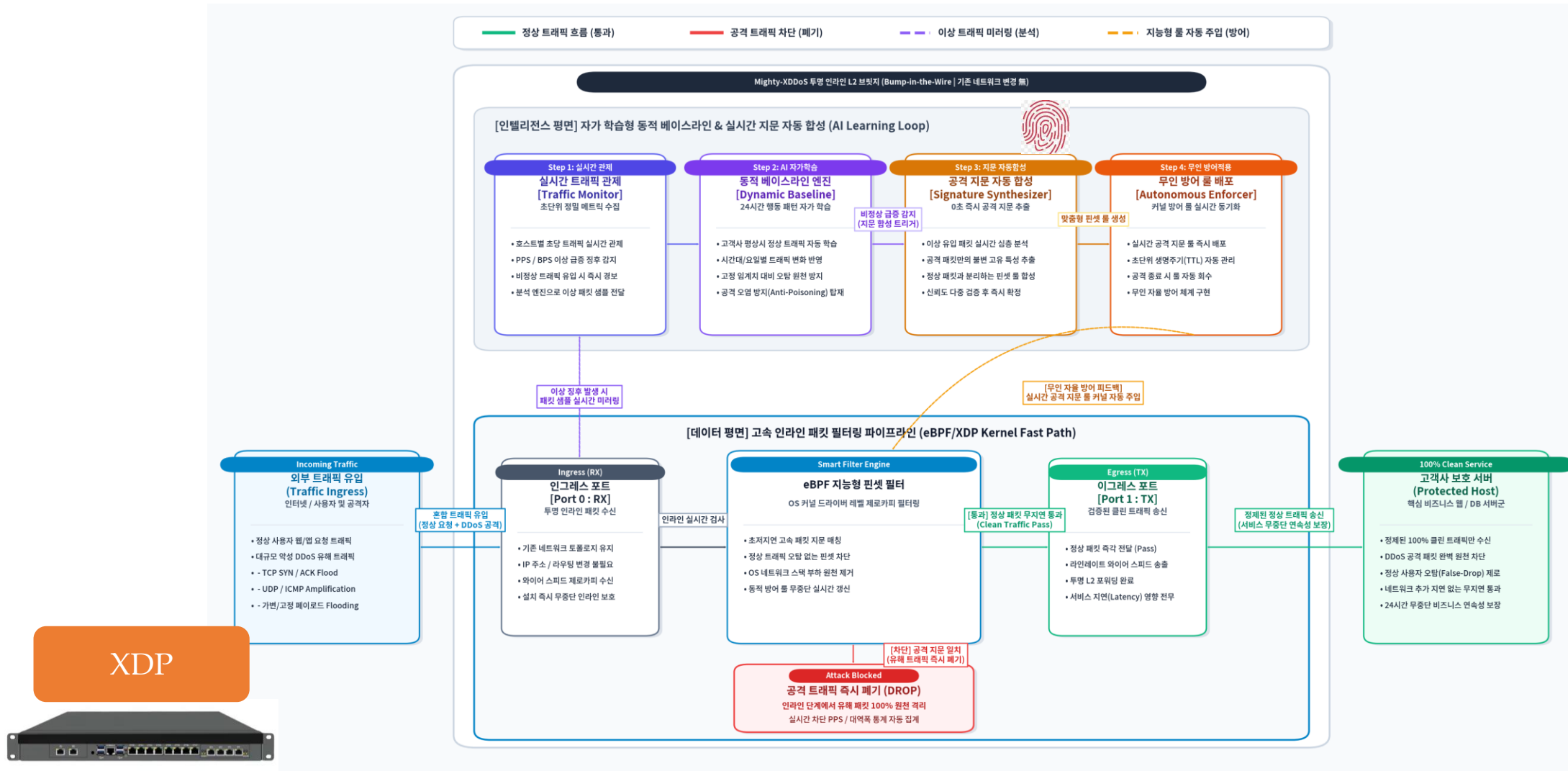
3. MightyXDDoS(6) – Intelligent Automation(1)



3. MightyXDDoS(6) – Intelligent Automation(2)

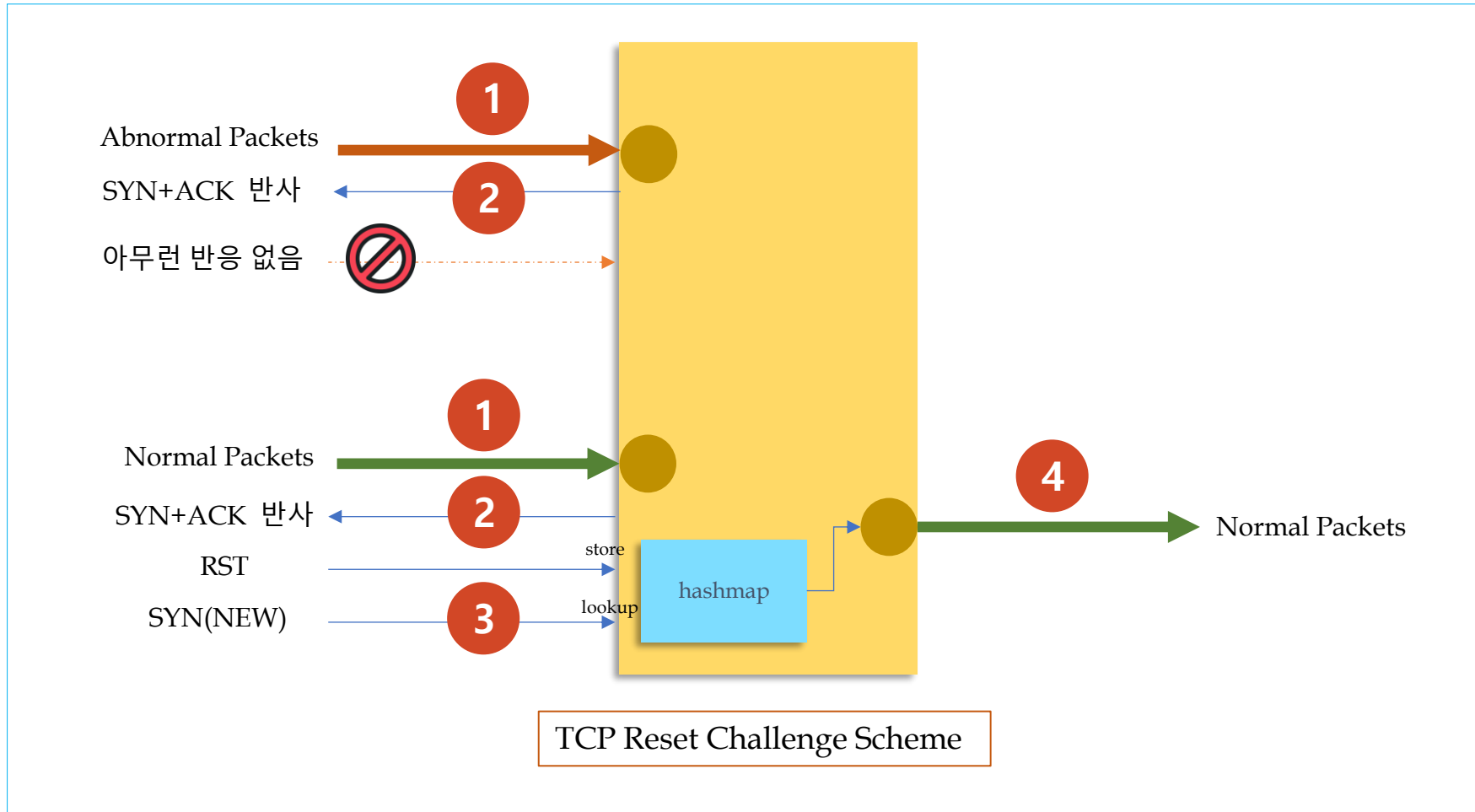


3. MightyXDDoS(6) – Intelligent Automation(3)

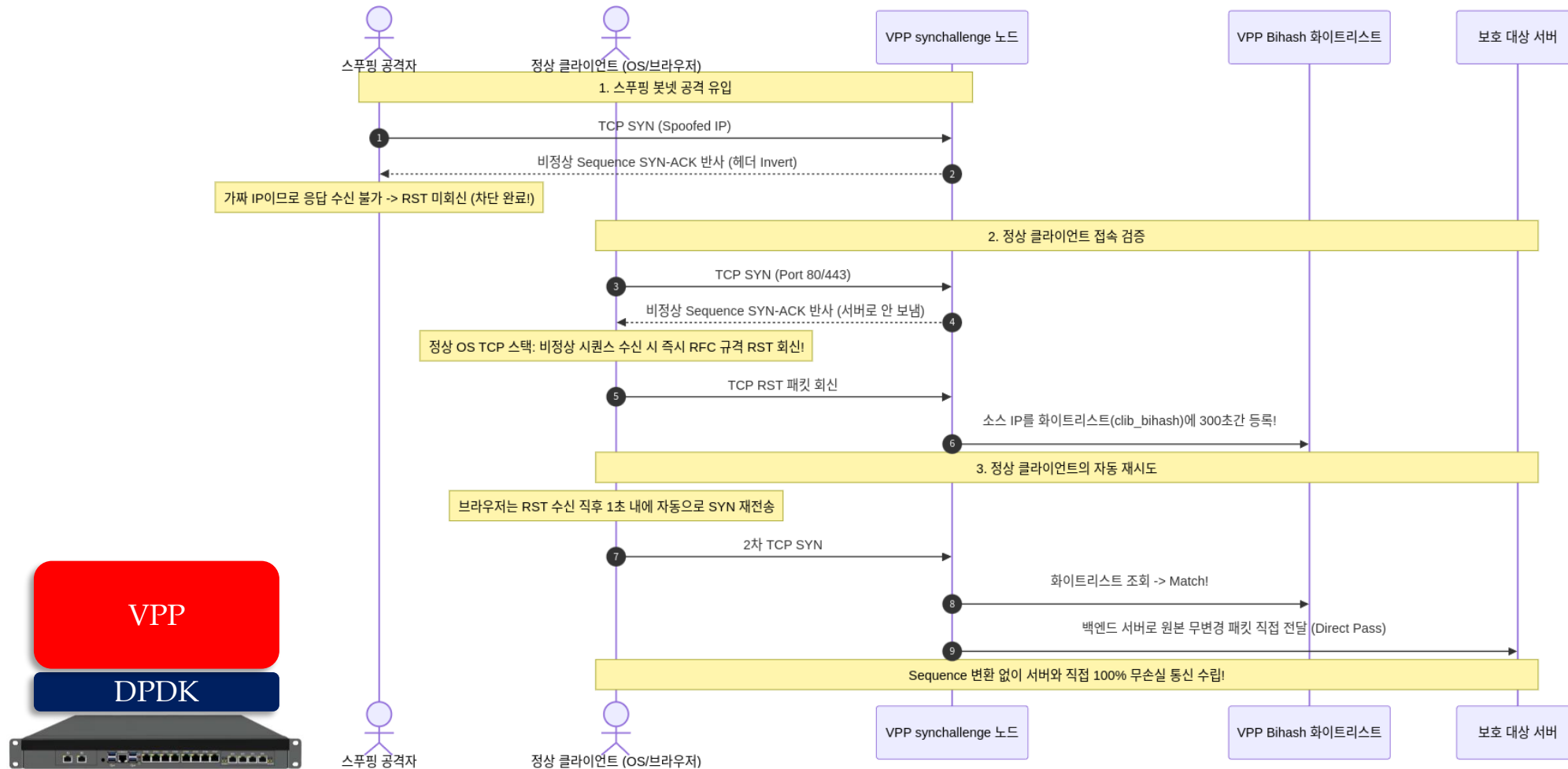


3. MightyXDDoS(7) – TCP Reset Challenge Scheme(1)

공격 패킷 차단, 정상 TCP 접속만 선별 후 통과 기법

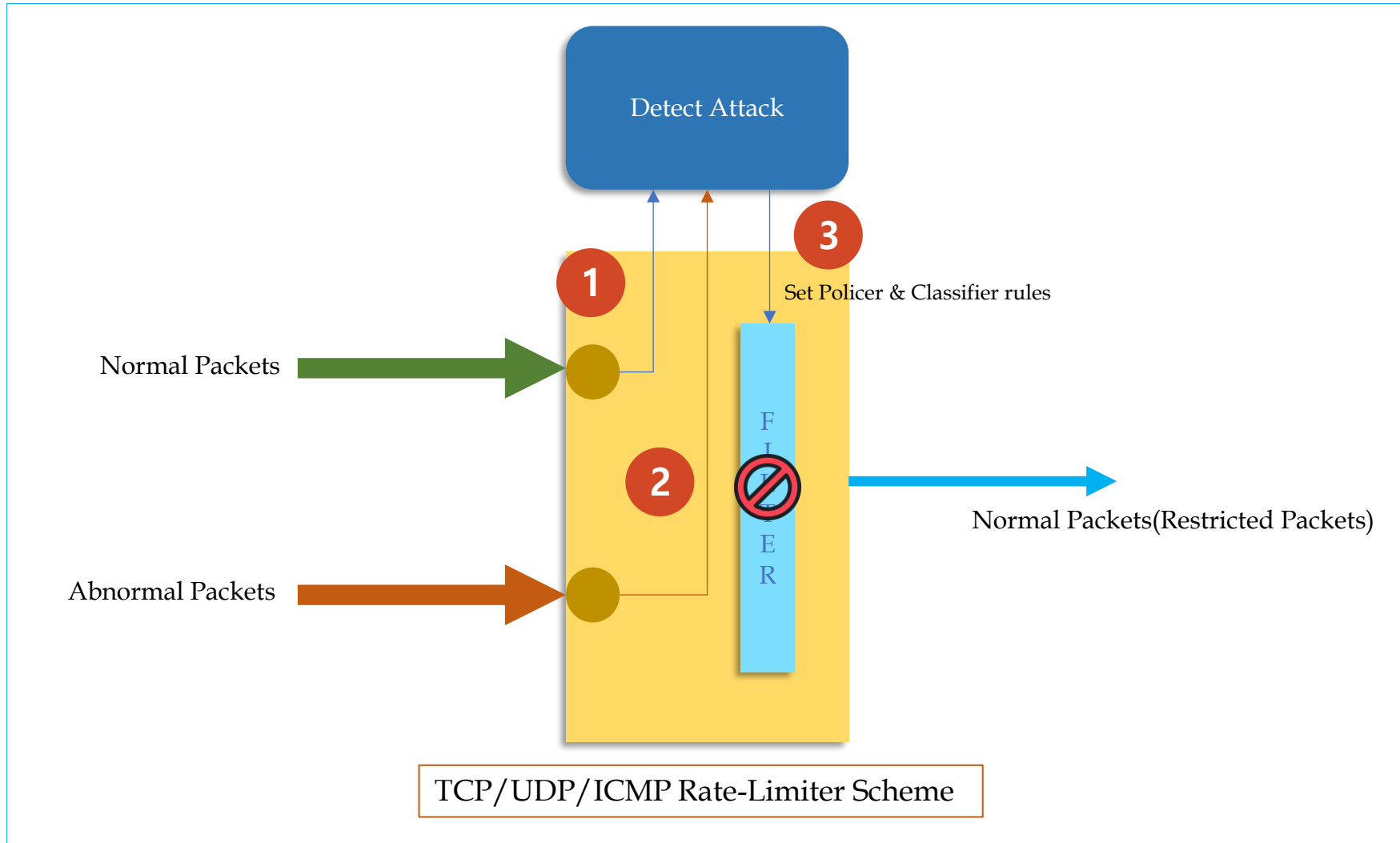


3. MightyXDDoS(7) – TCP Reset Challenge Scheme(2)

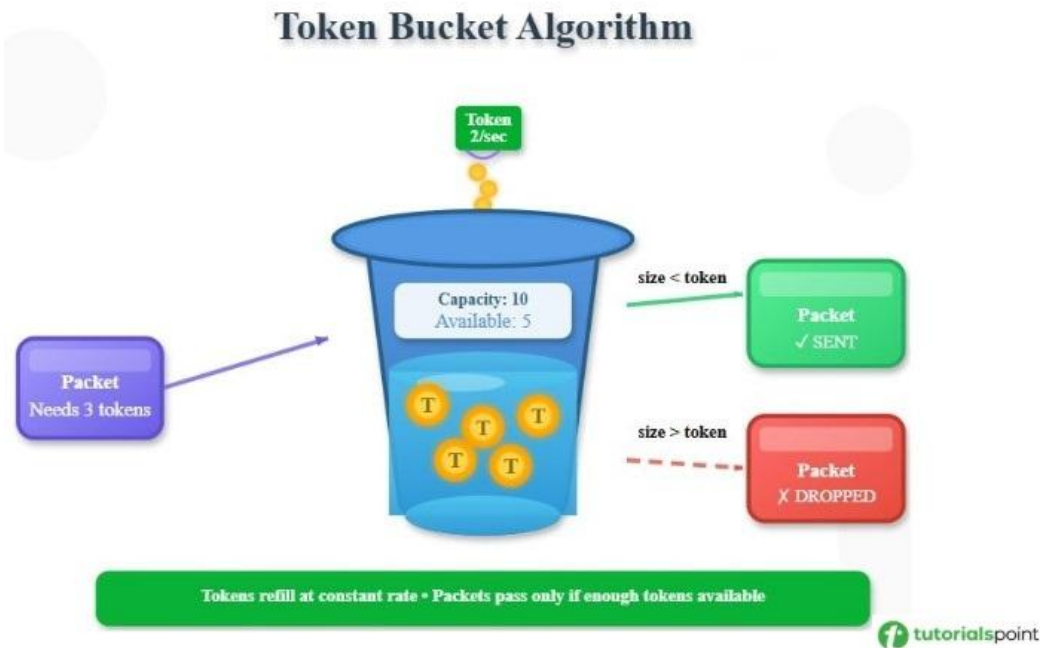


3. MightyXDDoS(8) – Rate-Limiter Scheme(1)

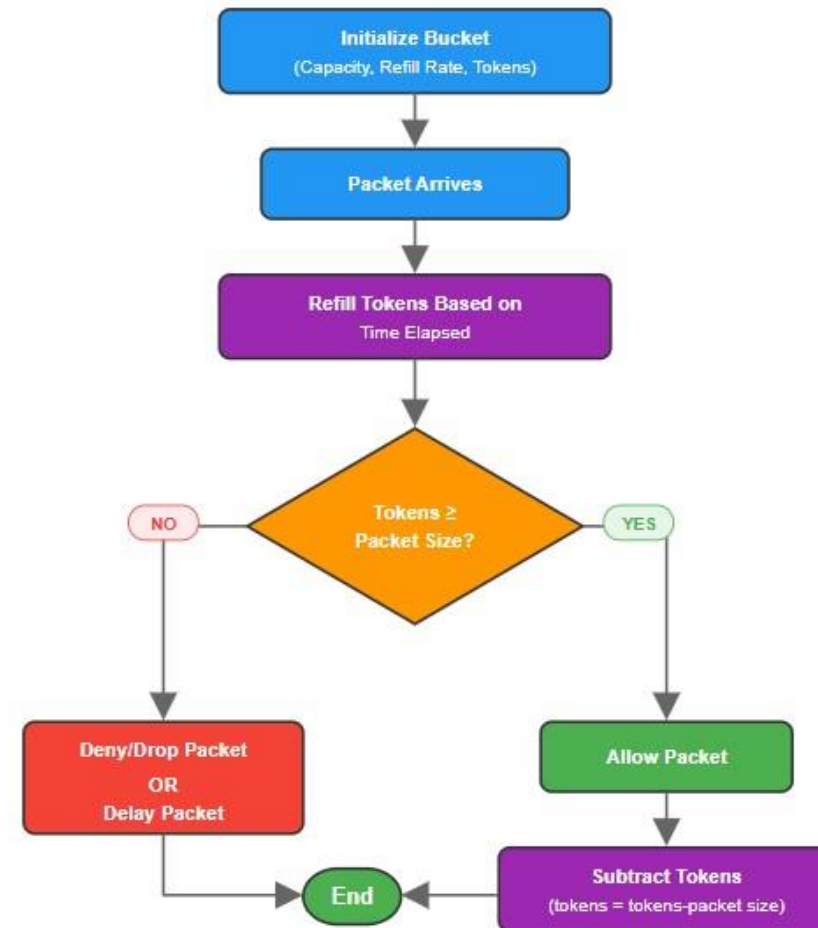
Traffic Control & Shaping(Policer & Classifier) 기법



3. MightyXDDoS(8) – Rate-Limiter Scheme(2)



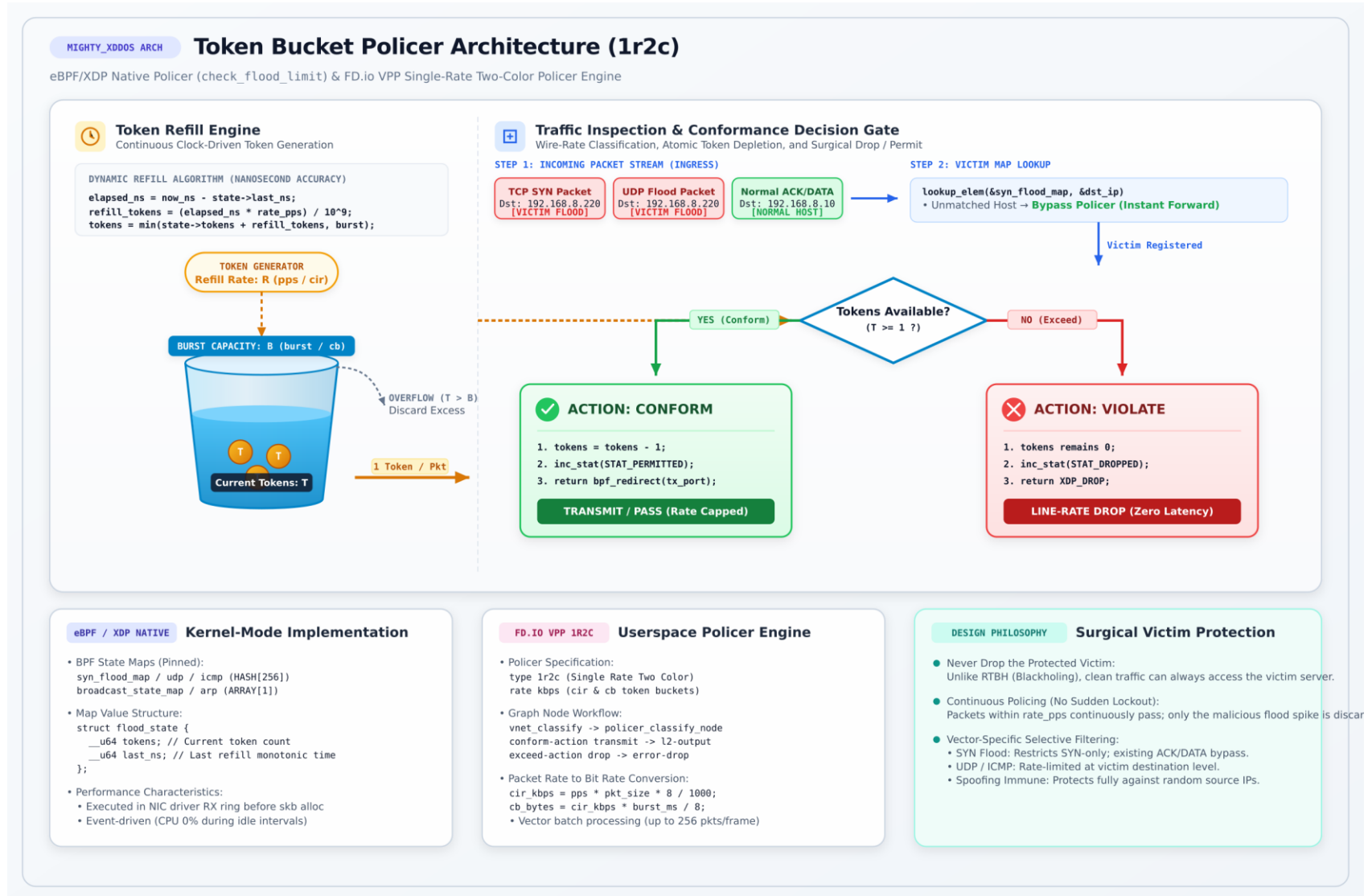
Token Bucket Algorithm Flowchart



3. MightyXDDoS(8) – Rate-Limiter Scheme(3)

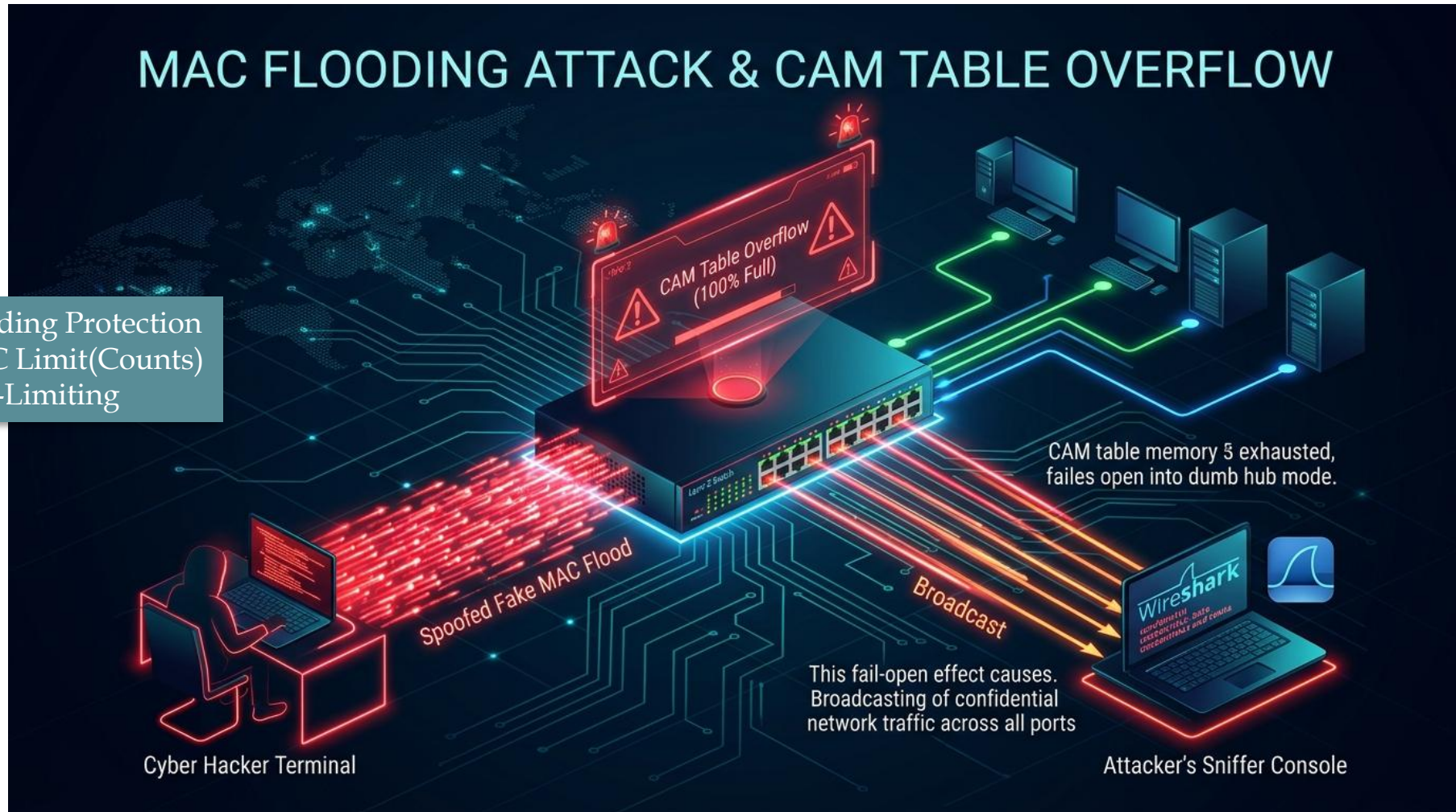
Rate-Limiter

- TCP Syn Flooding
- UDP Flooding
- ICMP Flooding
- ARP Broadcast Storm
- IP Broadcast Storm
- MAC Flooding
- ...

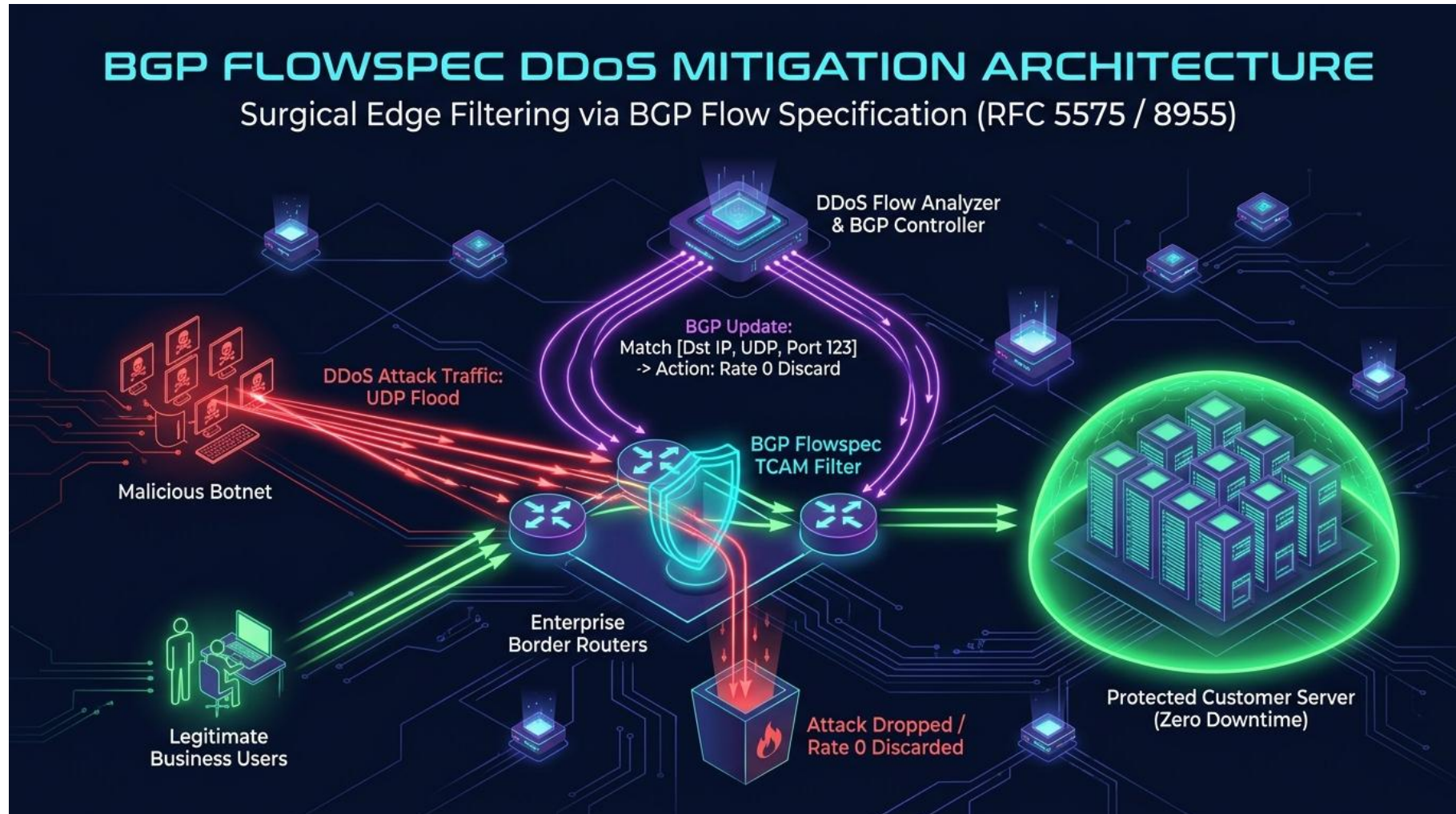


3. MightyXDDoS(9) – MAC Flooding Prevention

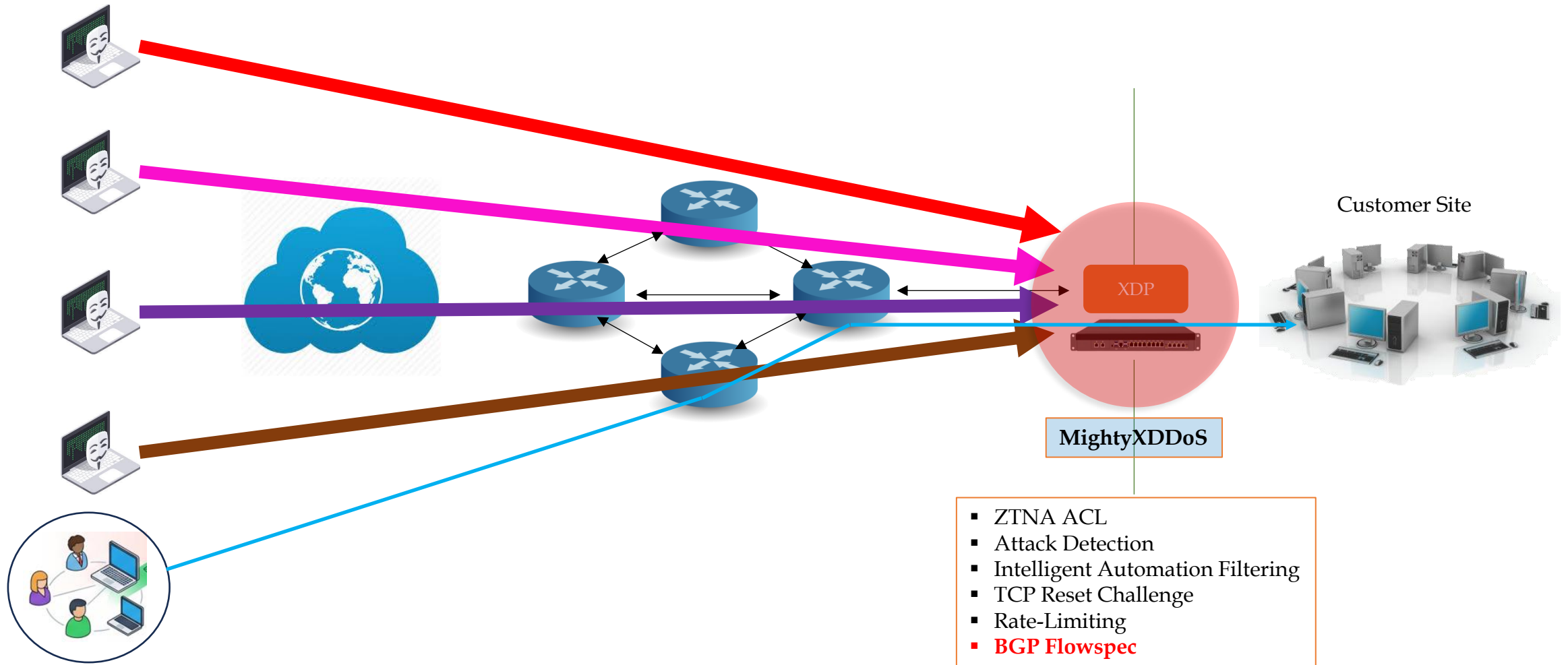
- MAC Flooding Protection
 - ✓ MAC Limit(Counts)
 - ✓ Rate-Limiting



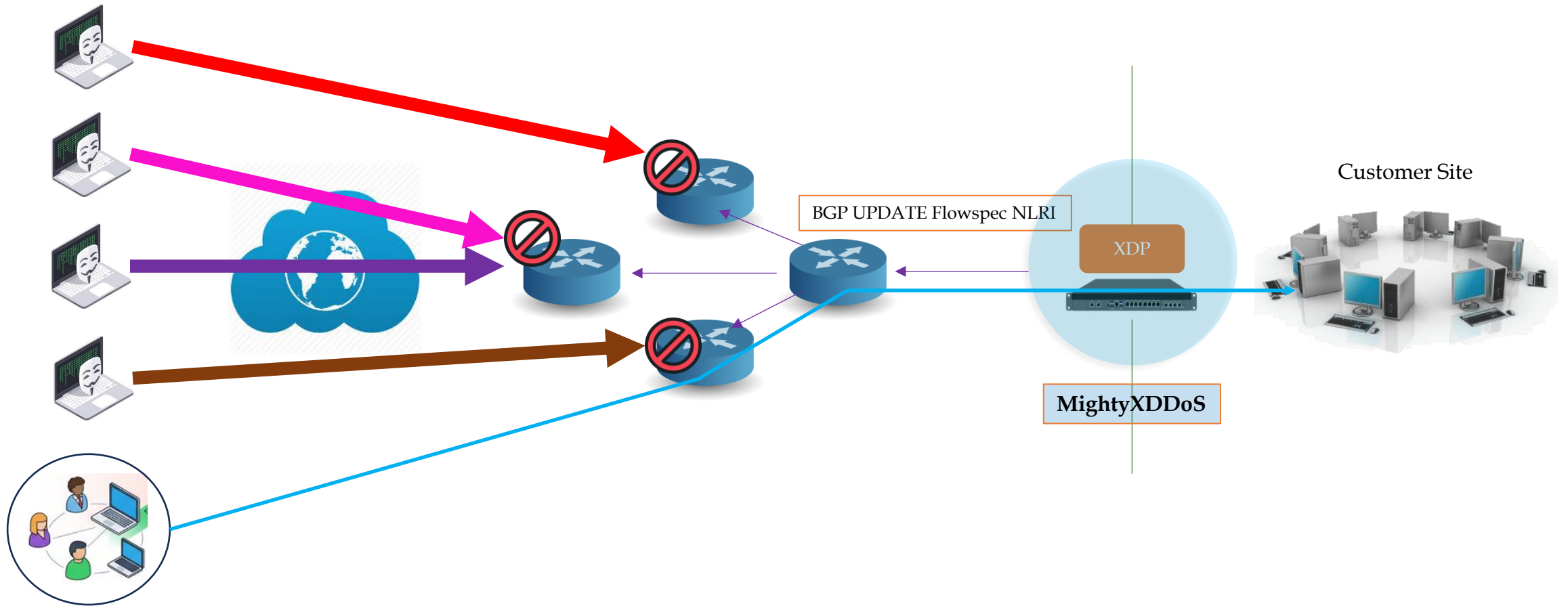
3. MightyXDDoS(10) – BGP Flowspec(1)



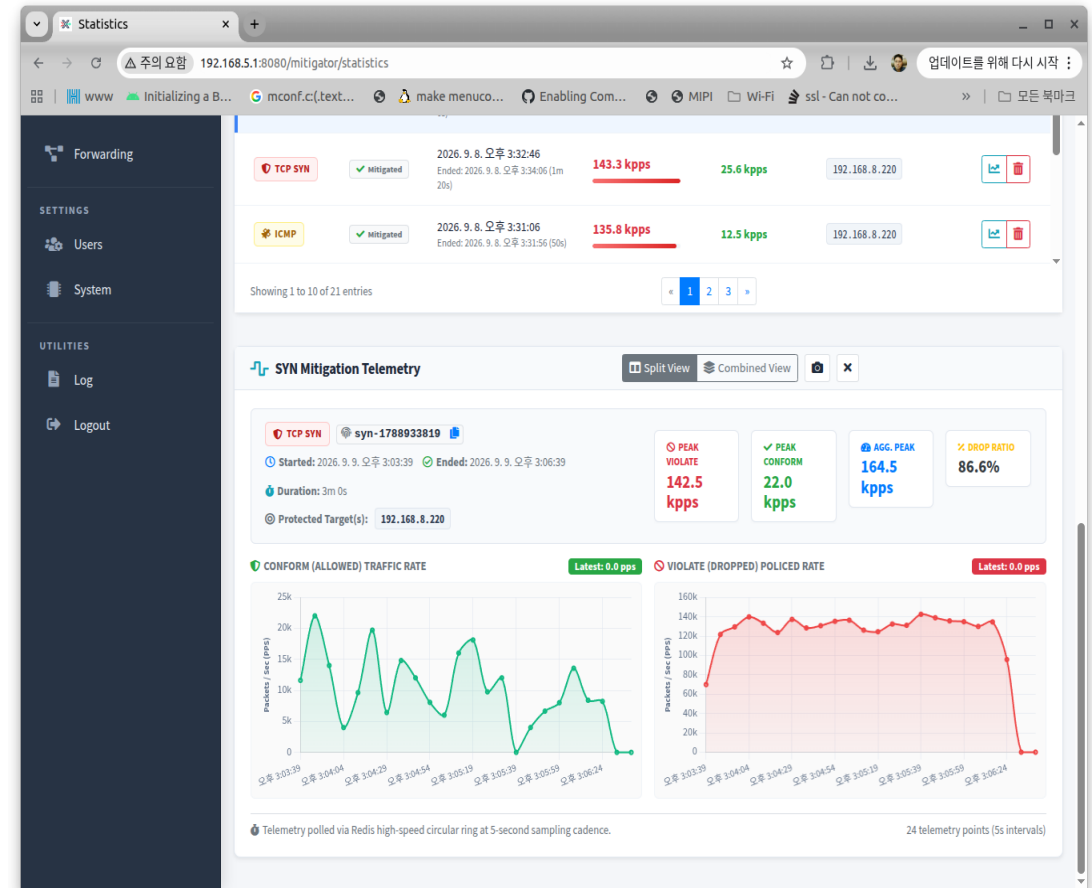
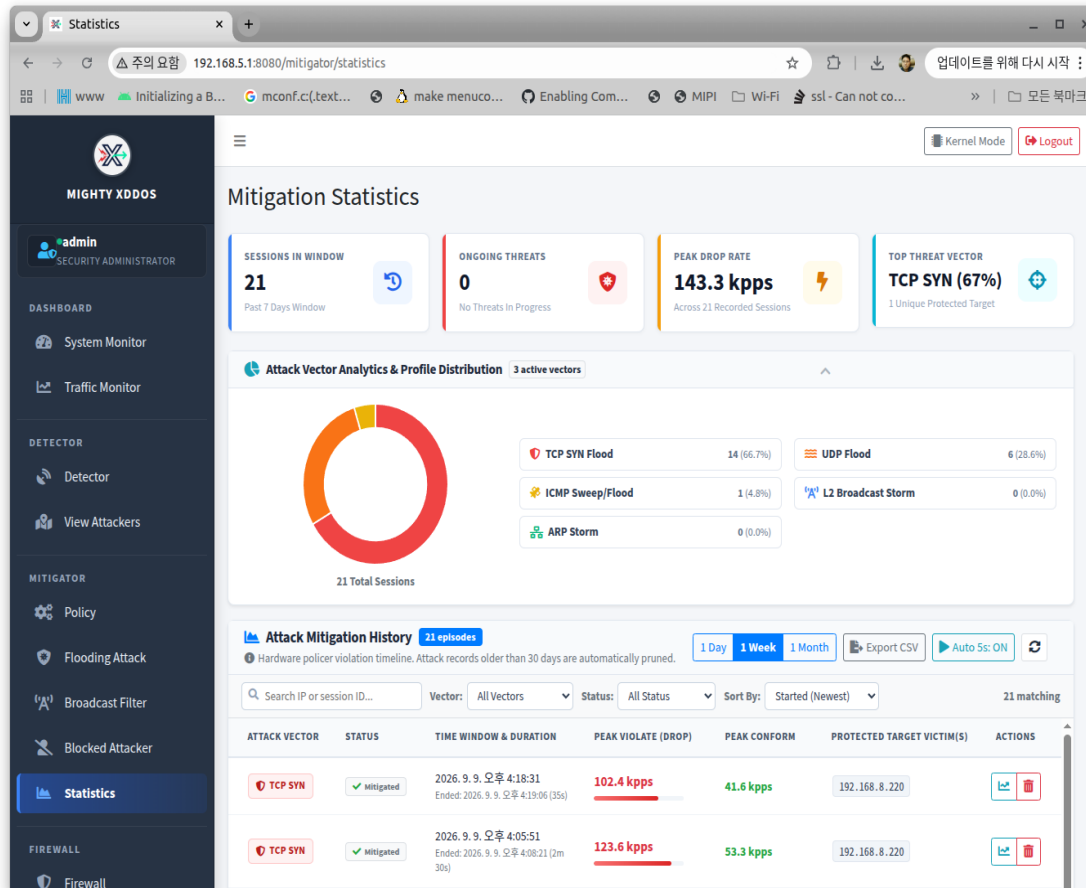
3. MightyXDDoS(10) – BGP Flowspec(2)



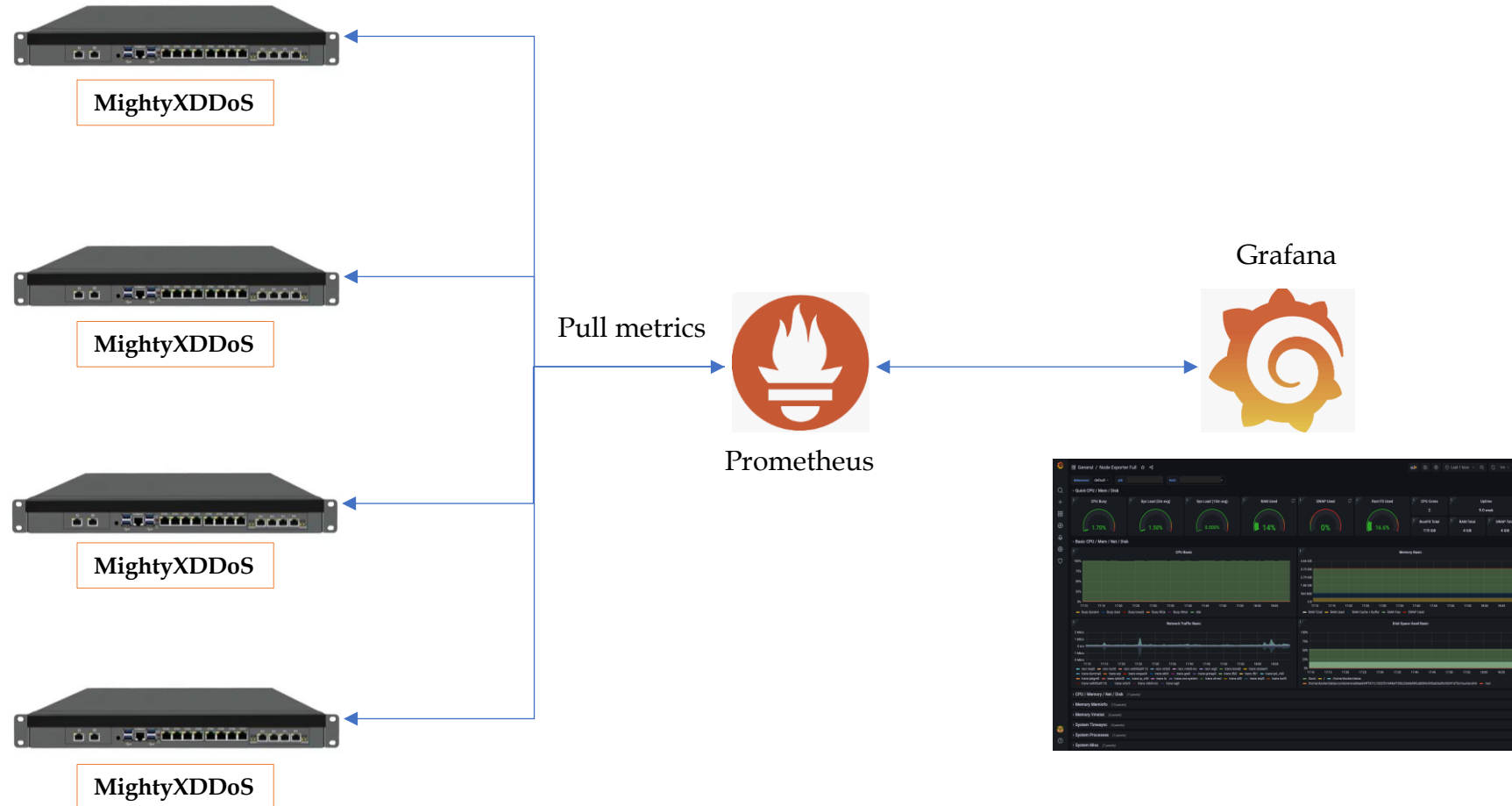
3. MightyXDDoS(10) – BGP Flowspec(3)



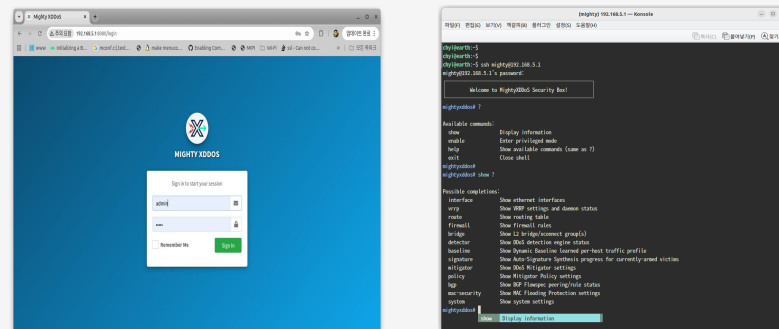
3. MightyXDDoS(11) – Statistics



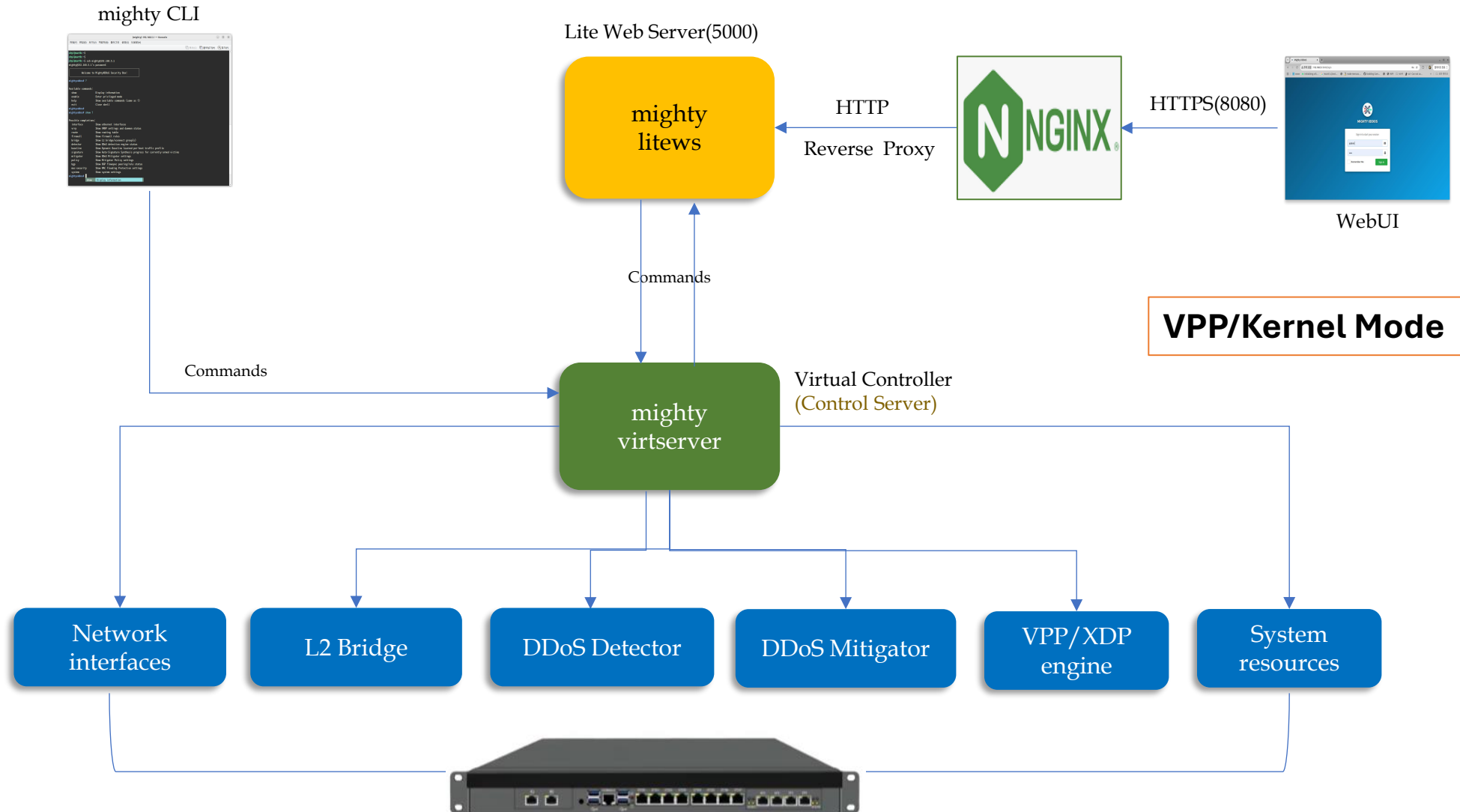
3. MightyXDDoS(12) – Prometheus Metrics Exporter



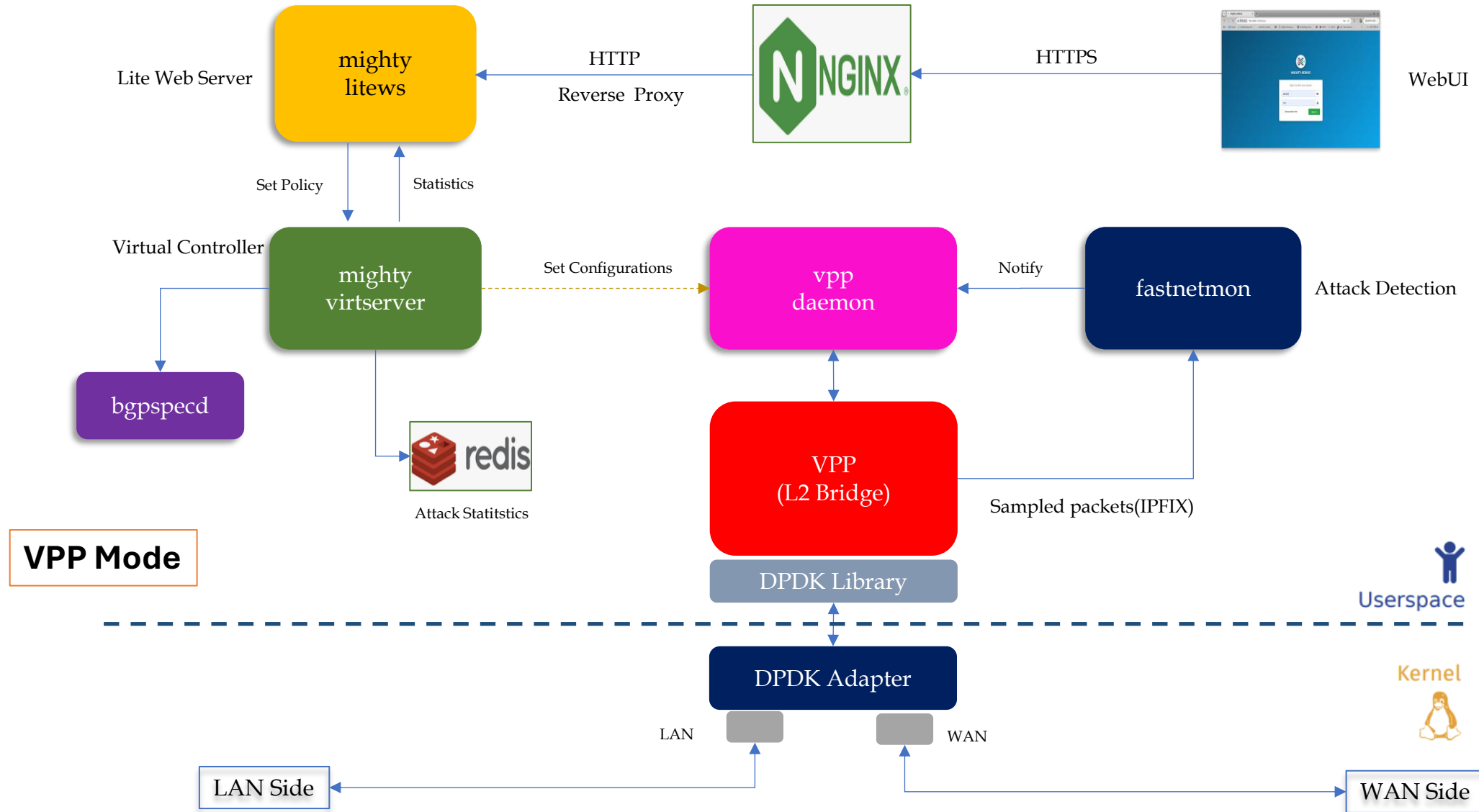
4. Mighty User Interface



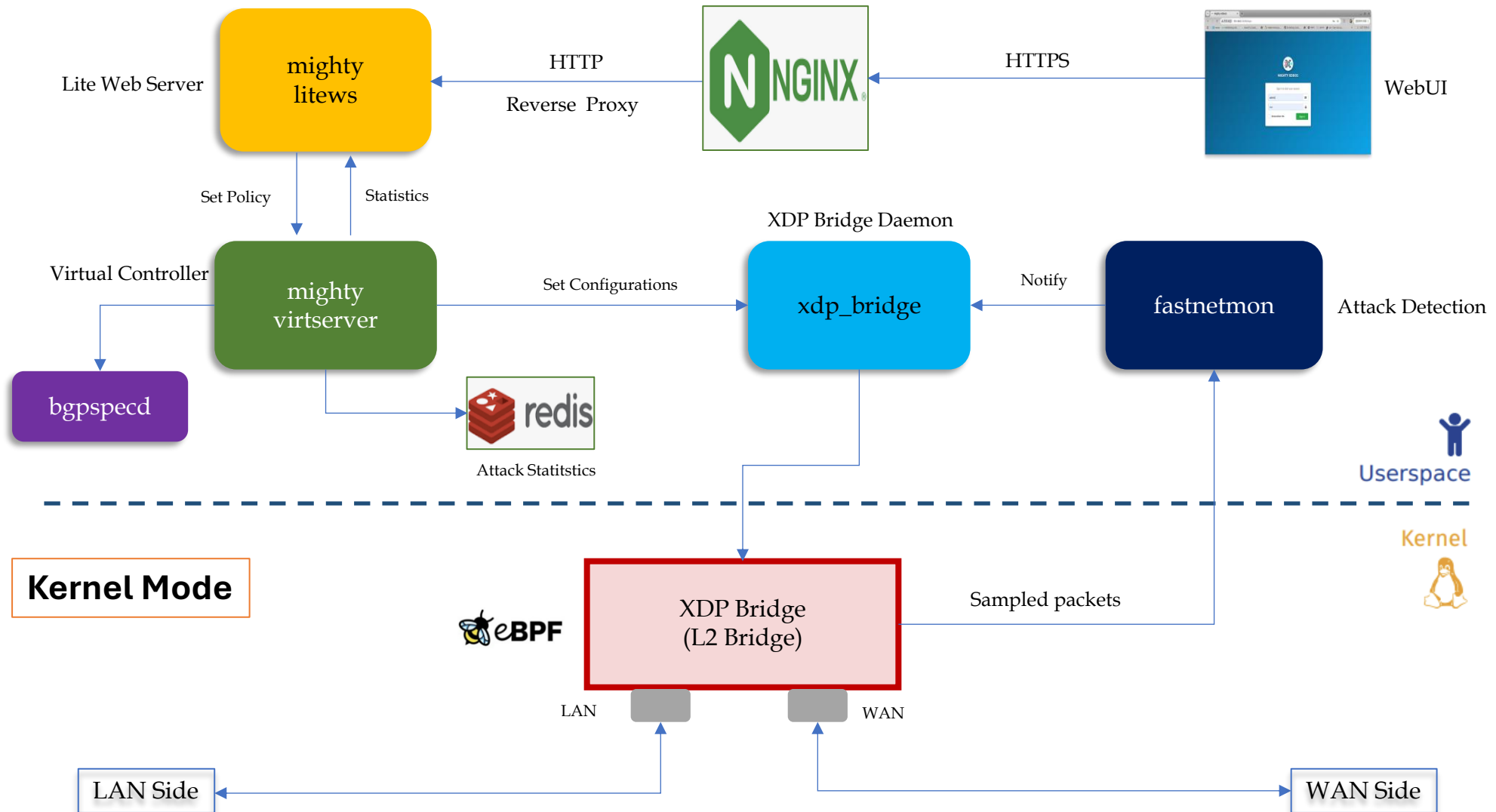
4. Mighty User Interface(1) – Mighty Engine(1)



4. Mighty User Interface(1) – Mighty Engine(2)

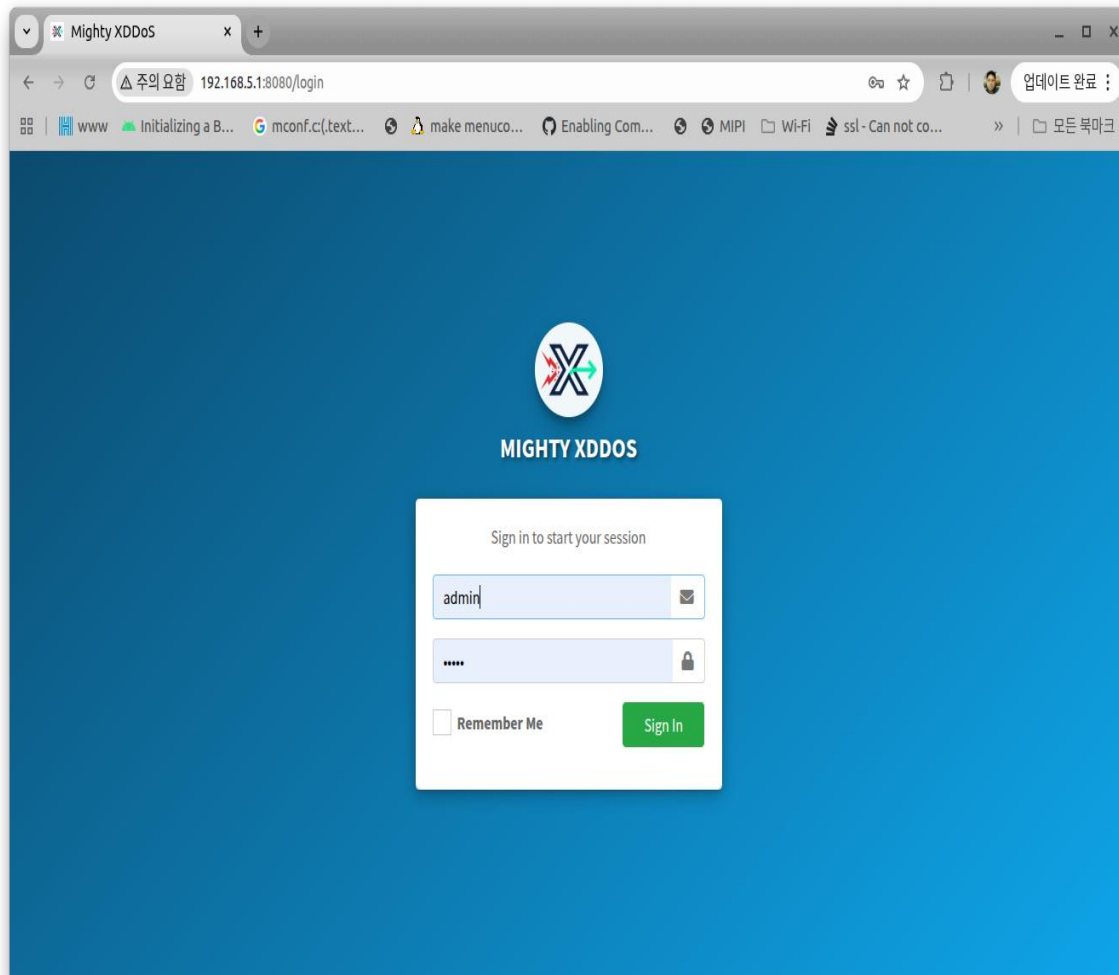


4. Mighty User Interface(1) – Mighty Engine(3)

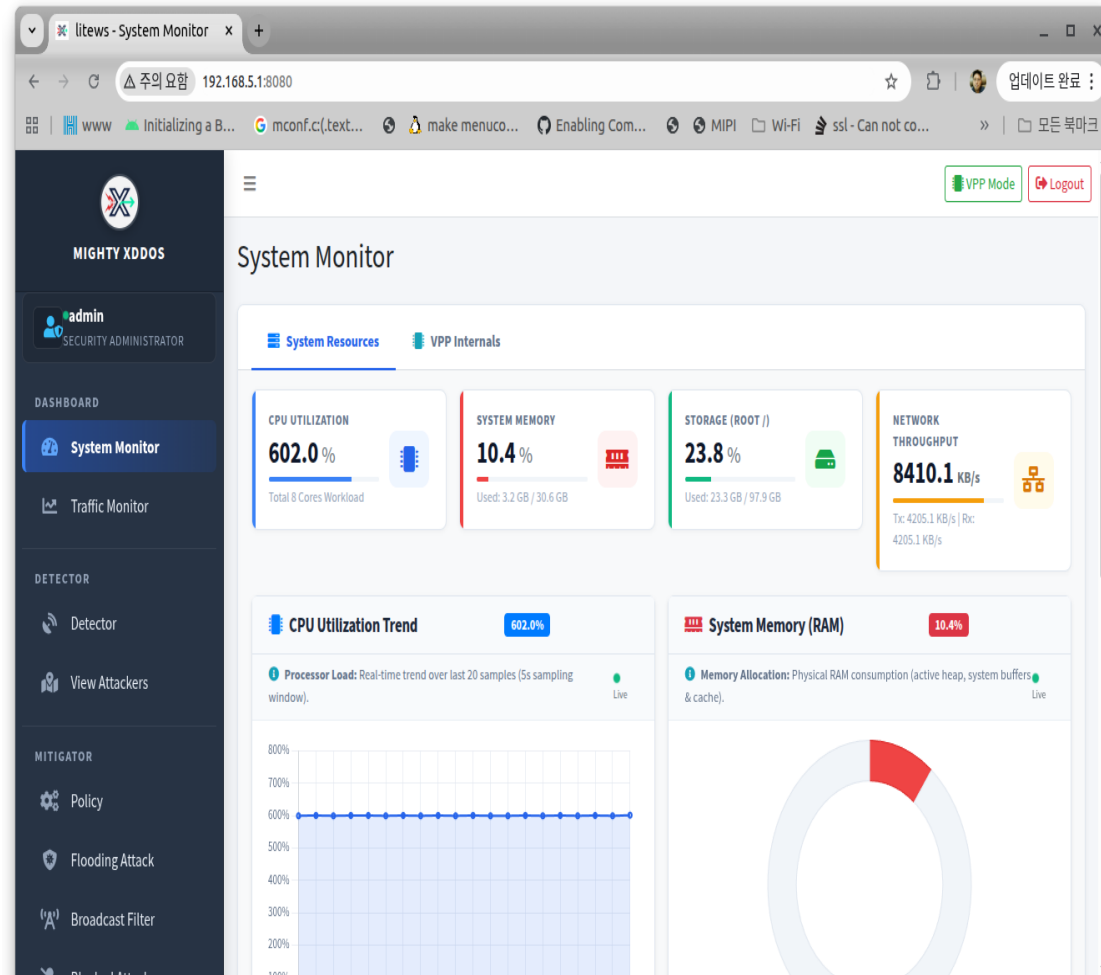


4. Mighty User Interface(2) – WebUI(1)

Login page

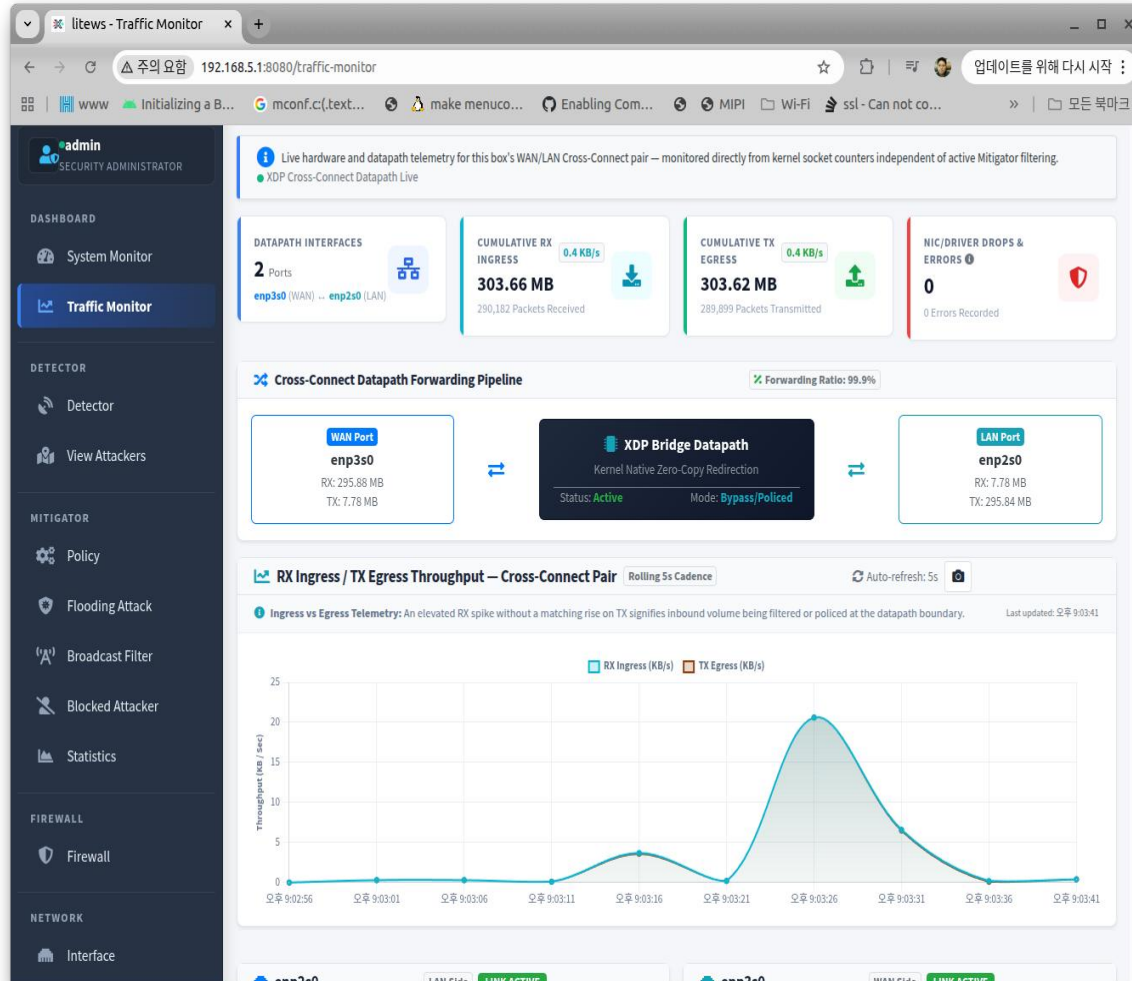


DashBoard – System Monitor Page

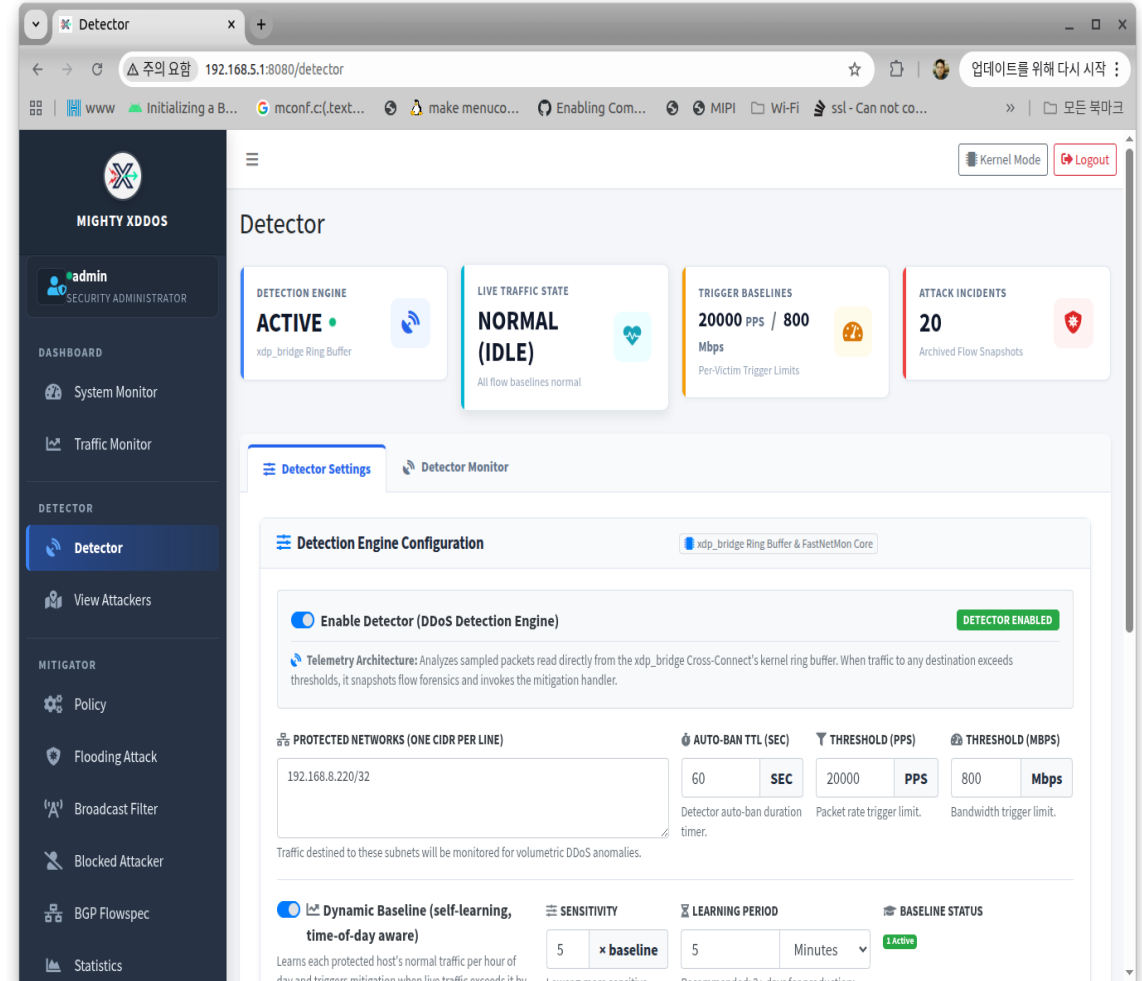


4. Mighty User Interface(2) – WebUI(2)

DashBoard – Traffic Monitor Page



Detector Page#1



4. Mighty User Interface(2) – WebUI(3)

Detector Page#2

The screenshot shows the 'Detector' page of the Mighty XDDOS web interface. The left sidebar contains a navigation menu with sections: MIGHTY XDDOS, admin (SECURITY ADMINISTRATOR), DASHBOARD (System Monitor, Traffic Monitor), DETECTOR (Detector, View Attackers), and MITIGATOR (Policy, Flooding Attack, Broadcast Filter, Blocked Attacker, BGP Flowspec, Statistics). The main content area is titled 'Detector' and features four summary cards: 'DETECTION ENGINE ACTIVE' (xdp_bridge Ring Buffer), 'LIVE TRAFFIC STATE TRAFFIC DETECTED' (High flow anomaly detected!), 'TRIGGER BASELINES 20000 pps / 800 Mbps' (Per-Victim Trigger Limits), and 'ATTACK INCIDENTS 20' (Archived Flow Snapshots). Below these is a 'Detector Monitor (Live Telemetry)' section with a 'TRAFFIC ACTIVE' status and a 'Real-time Stream' button. It displays a 'FastNetMon Engine Stream' with live datapath telemetry from /tmp/fastnetmon.dat. The stream shows incoming traffic from 192.168.8.220 at 74475 pps and 23 mbps, which is marked as '*banned'. Other traffic (outgoing, internal, other) shows 0 pps and 0 mbps. A terminal window at the bottom shows the FastNetMon 1.2.10 interface with the same traffic statistics and a screen update time of 5.3e-05 sec.

Detector - View Attackers page

The screenshot shows the 'View Attackers' page of the Mighty XDDOS web interface. The left sidebar is identical to the Detector page. The main content area is titled 'View Attackers' and includes a note: 'Only attacks fastnetmon traced to a single, distinct source IP are plotted here. Most floods (especially spoofed-source ones) involve many/random source addresses, where showing any one of them as "the attacker" would be misleading rather than useful – those are intentionally left off the map. An empty map most of the time is expected, not a sign something's broken.' Below this is an 'Attacker Locations' section with a '1 plotted' indicator and a map of North America. A location pin is visible in the United States, with a tooltip showing the IP address 23.34.172.165, Location: Unknown, United States, Target victim: 192.168.8.220, Attack type: unknown, and Time: 09 09 26 11:19:21.

4. Mighty User Interface(2) – WebUI(4)

Mitigator - Policy page

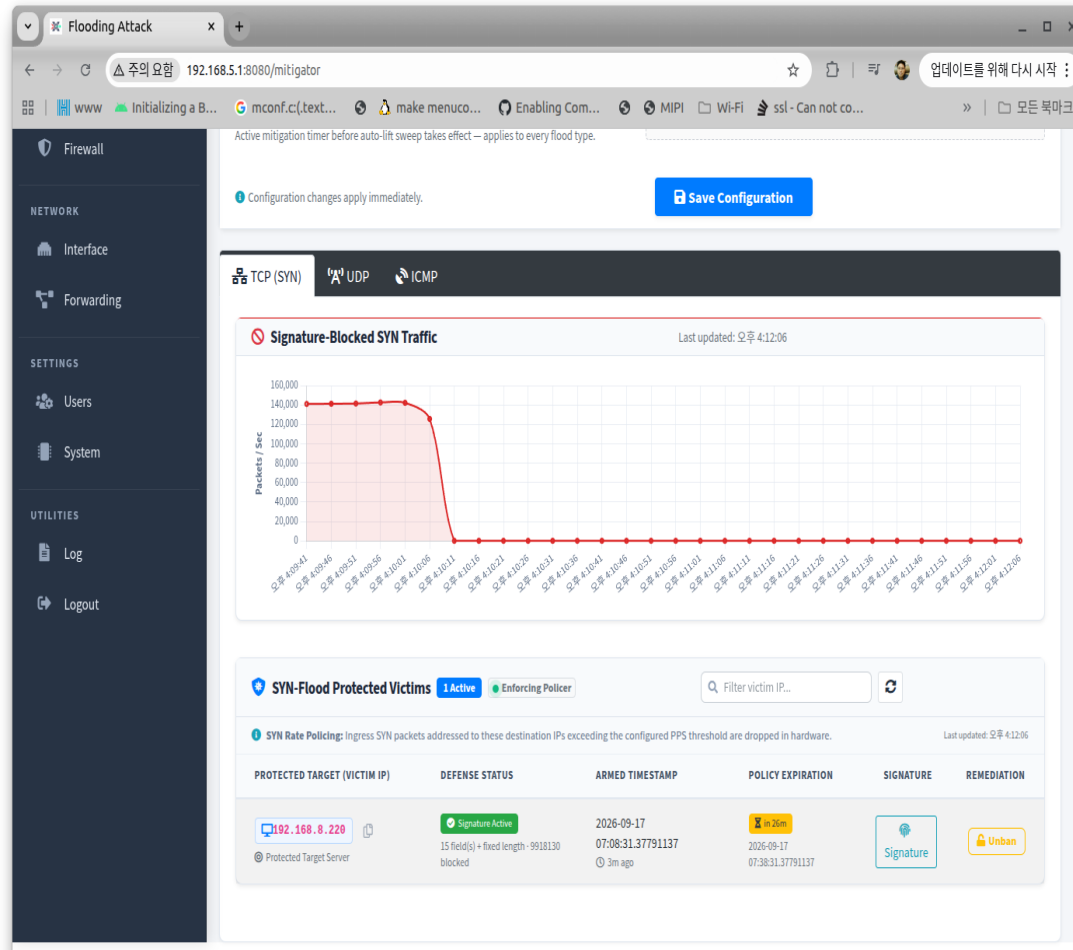
The screenshot shows the 'Policy' page of the Mitigator web interface. The browser address bar shows '192.168.5.1:8080/mitigator/policy'. The left sidebar contains a navigation menu with 'MIGHTY XDDOS' at the top, followed by 'admin SECURITY ADMINISTRATOR', 'DASHBOARD' (with 'System Monitor' and 'Traffic Monitor'), 'DETECTOR' (with 'Detector' and 'View Attackers'), and 'MITIGATOR' (with 'Policy' selected, and 'Flooding Attack', 'Broadcast Filter', 'Blocked Attacker', and 'Statistics'). The main content area is titled 'Policy' and features a section 'ACL / Mitigator Interface Binding Scope'. This section explains that it controls which bridge member interfaces get enforcement layer bound to them. It offers two options: 'All (LAN & WAN)' (Default) which binds to every bridge member interface, and 'WAN Only' which binds only to WAN-role interfaces. Below this, there is a 'WAN Service Port Policy' section with a toggle switch labeled 'Enable WAN Service Port Policy' which is currently turned on. A 'Save' button is visible. At the bottom, there is an 'ALLOWED PORTS' section with an 'Add Port Rule' button. The top right of the interface shows 'VPP Mode' and a 'Logout' button.

Mitigator - Blocked Attacker page

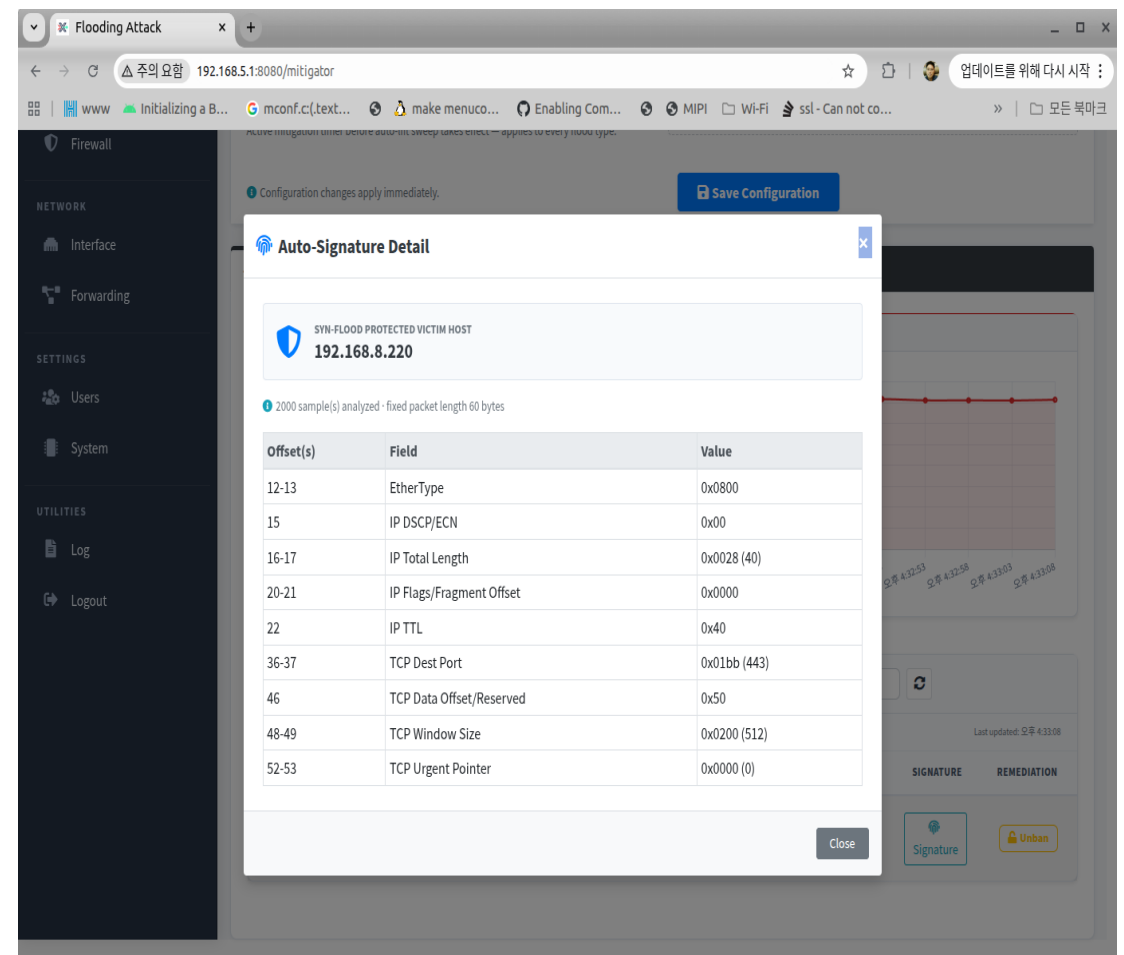
The screenshot shows the 'Blocked Attacker' page of the Mitigator web interface. The browser address bar shows '192.168.5.1:8080/mitigator/blocked-attacker'. The left sidebar is identical to the previous page, with 'Blocked Attacker' selected under the 'MITIGATOR' section. The main content area is titled 'Blocked Attacker' and displays four summary cards: 'MITIGATION ENGINE' (ACTIVE, L2 Bridge Hardware Policer), 'PROTECTED VICTIMS' (1, Flood-Type Rate Policed), 'BLOCKED ATTACKERS' (0, Hardware ACL Drop Enforced), and 'FLOOD DROP RATE' (—, Live Drop Share (Active Tab)). Below these cards, there is a section for 'Blocked Attacker Source IPs' showing '0 Blocked' and a search bar. A message states: 'ACL Enforcement: Ingress traffic matching these hostile origin addresses is discarded at the L2 Bridge before reaching internal services. Last updated: 오후 12:40:18'. A table with columns 'ATTACKER SOURCE IP', 'TARGETED VICTIM', 'BLOCKED AT', 'EXPIRATION / TTL', and 'REMEDIATION' is present but empty. A large green checkmark icon is displayed with the text 'No Hostile Sources Blocked' and a note: 'Hardware ACL filter table is clean. No malicious origin IPs are currently blacklisted.' The top right of the interface shows 'VPP Mode' and a 'Logout' button.

4. Mighty User Interface(2) – WebUI(5)

Mitigator - Flooding Attack/TCP(SYN) page

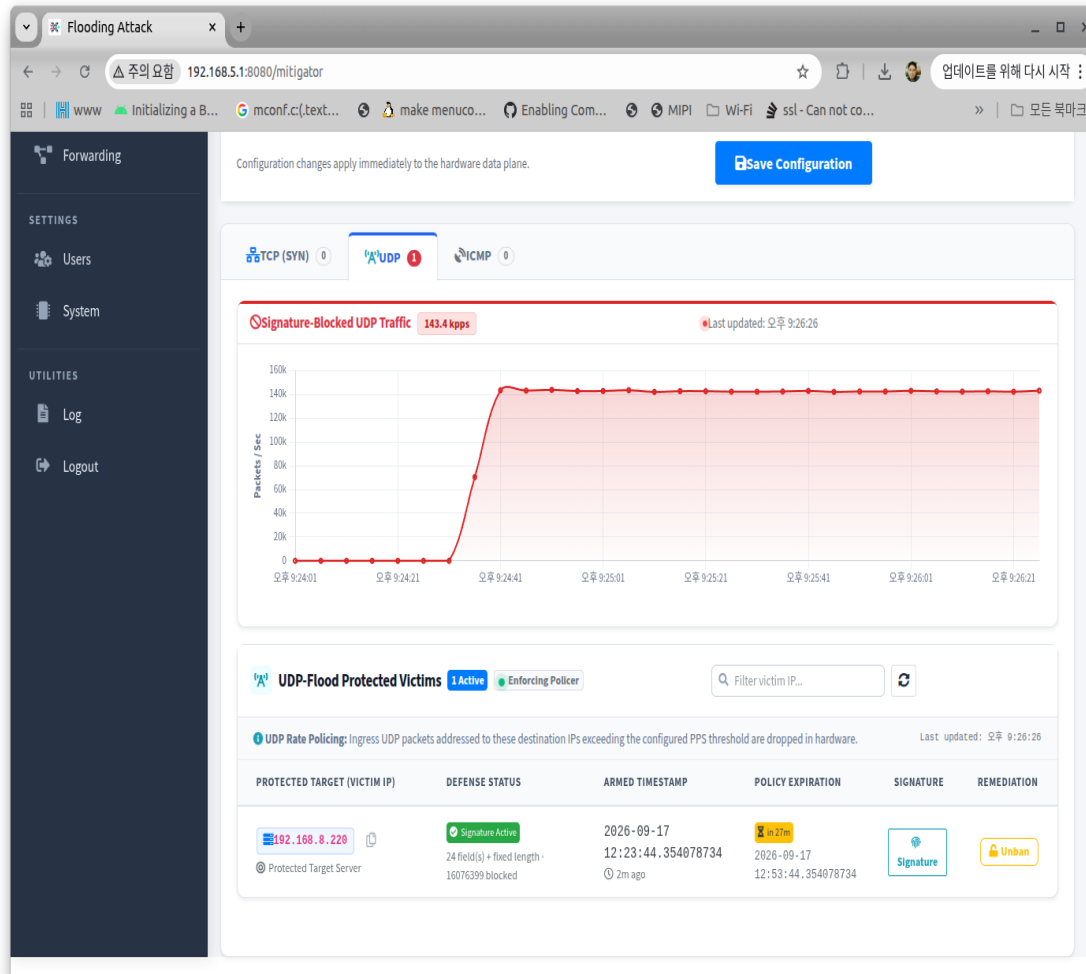


Mitigator - Flooding Attack/TCP(SYN) page - Signature

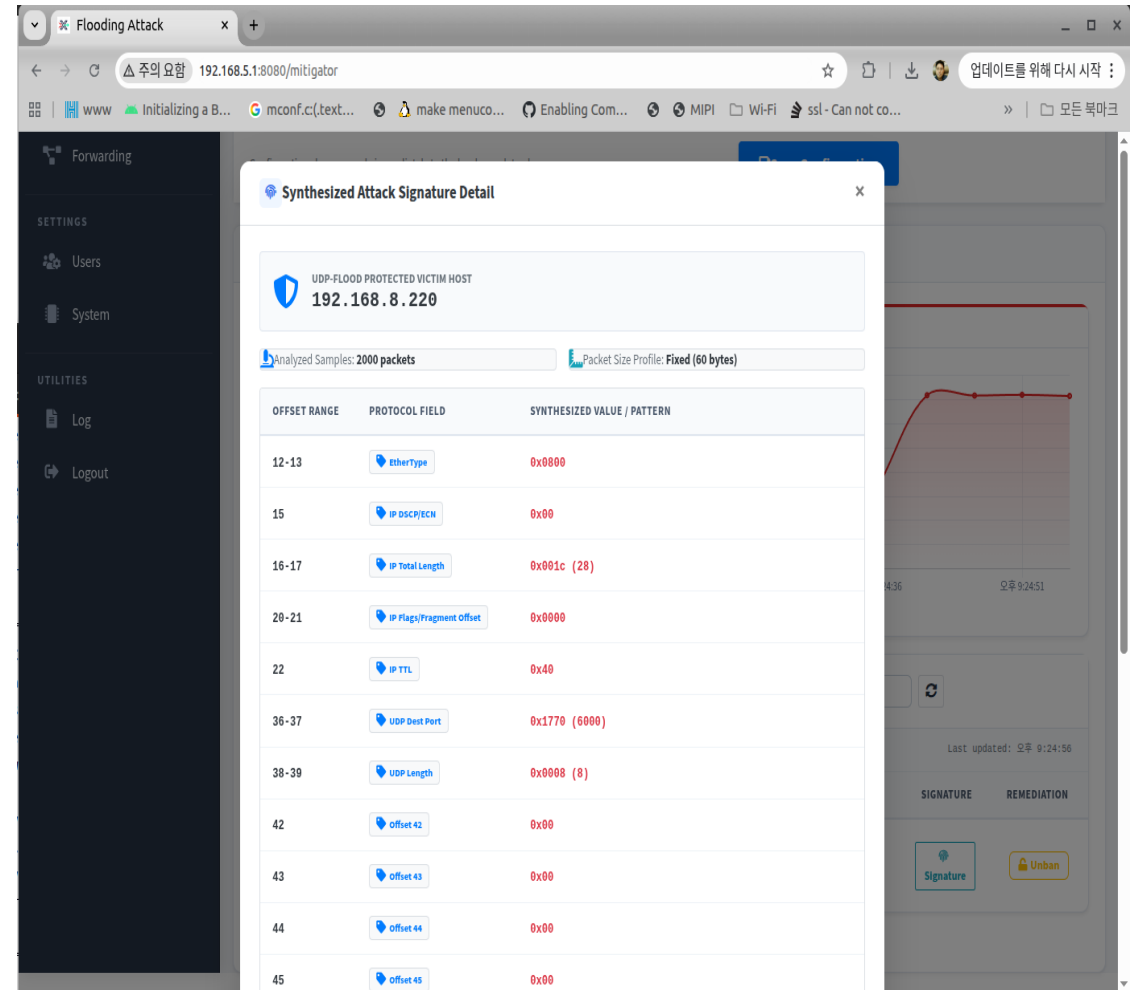


4. Mighty User Interface(2) – WebUI(6)

Mitigator - Flooding Attack/UDP page

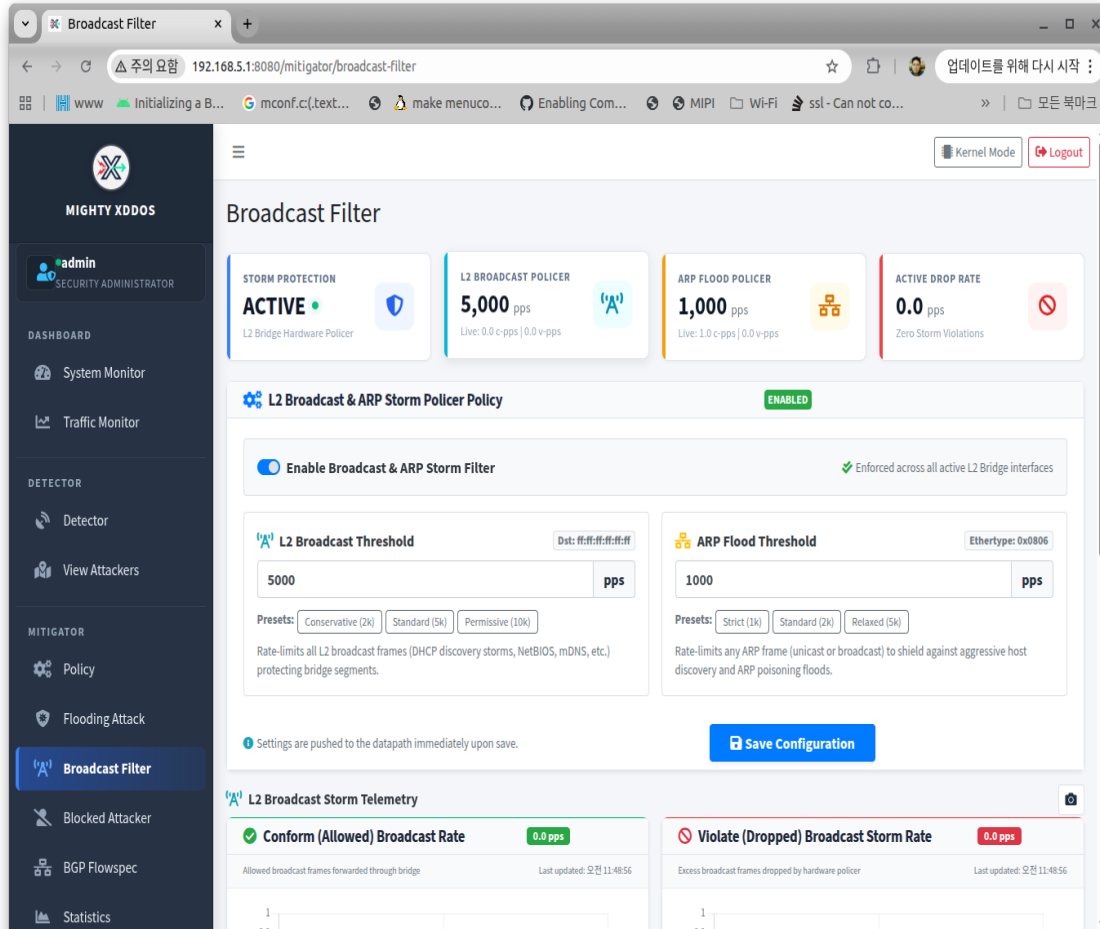


Mitigator - Flooding Attack/UDP - Signature page

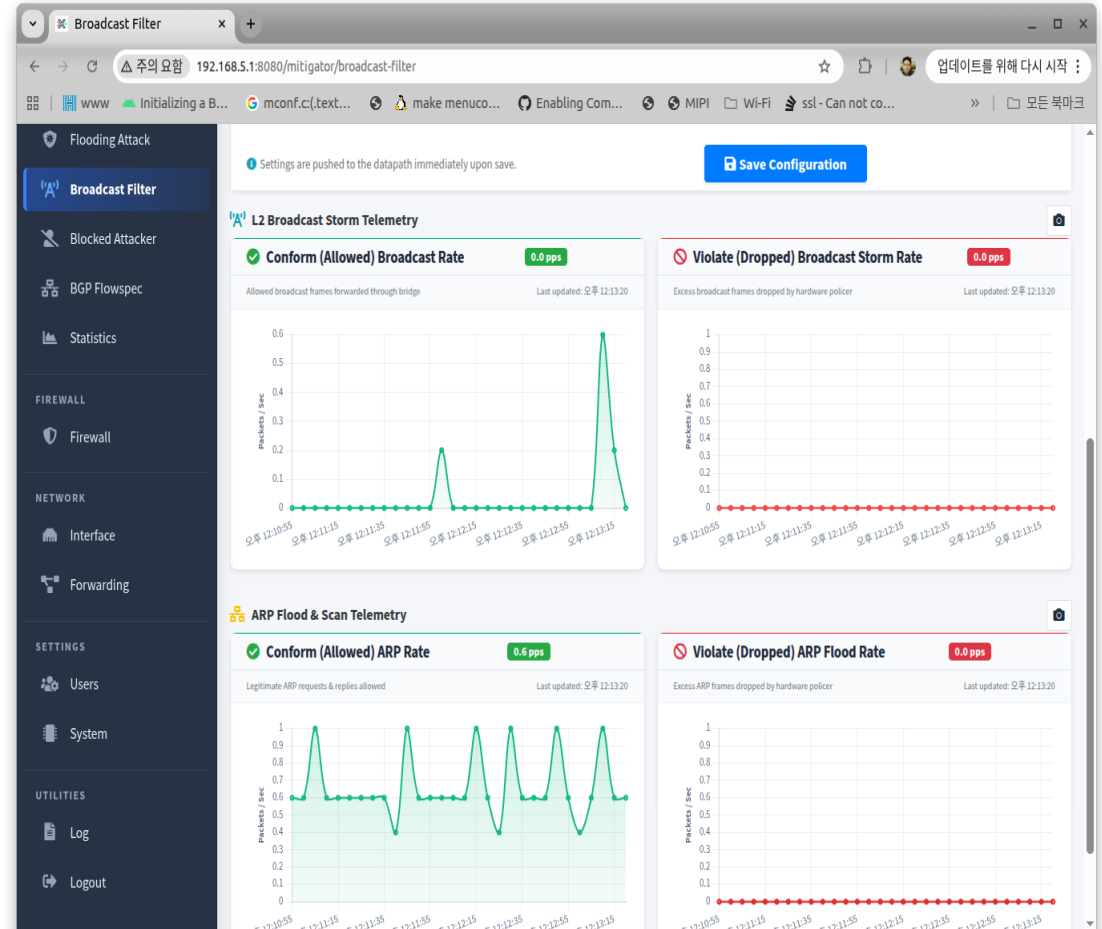


4. Mighty User Interface(2) – WebUI(7)

Mitigator - Broadcast Filter Page#1



Mitigator - Broadcast Filter Page#2



4. Mighty User Interface(2) – WebUI(8)

Mitigator - BGP Flowspec Page#1

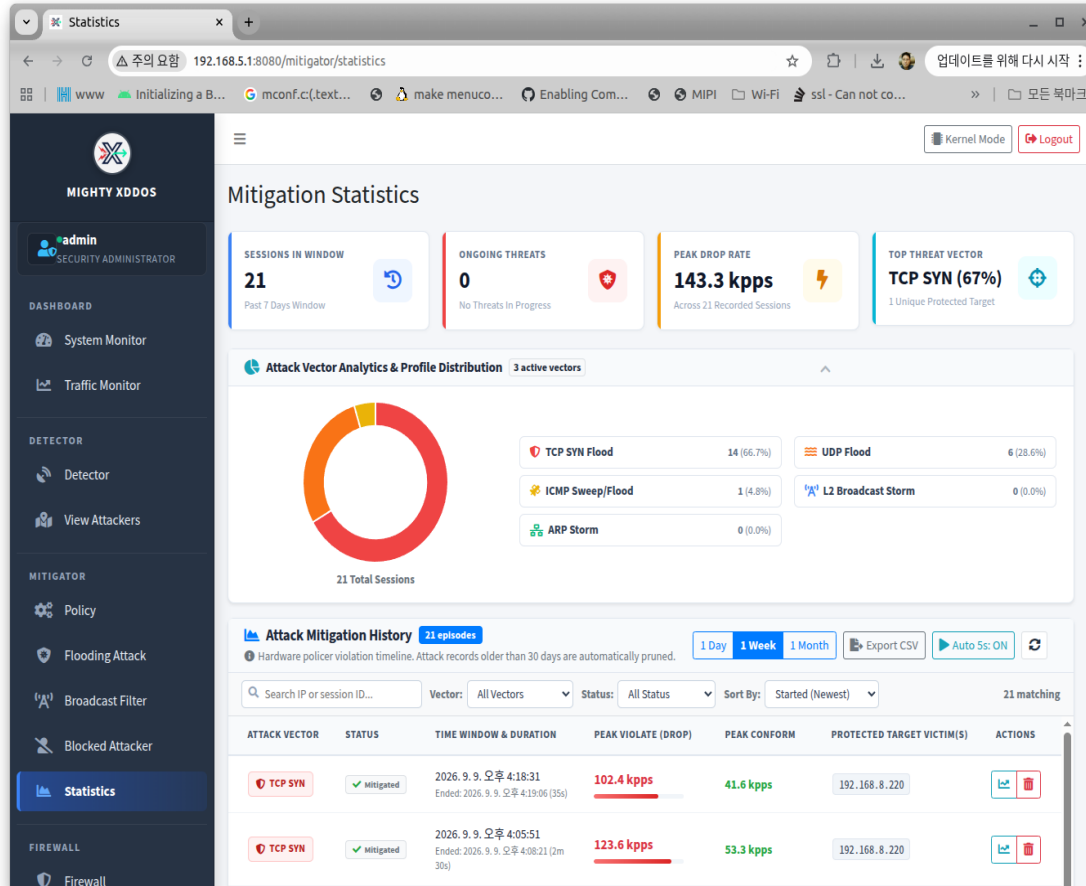
The screenshot shows the 'BGP Flowspec' configuration page in the Mitigator web interface. The left sidebar contains navigation links: admin, DASHBOARD, DETECTOR, MITIGATOR, and UTILITIES. The main content area displays a summary of the BGP Flowspec engine status, including 'BGP ENGINE SERVICE' (ENABLED), 'PEERING SESSIONS' (1/1), 'ACTIVE FLOWSPEC RULES' (0), and 'WAN LINK SATURATION' (Normal). Below this, the 'BGP Flowspec Engine Configuration' section shows the engine is active and provides details on the selective offloading architecture. The 'LOCAL AS NUMBER' is set to 65001, the 'ROUTER-ID (BGP ID)' is 192.168.8.114, and the 'LINK SATURATION THRESHOLD' is 5%. The 'MONITORED INTERFACES' section lists 'enp3s0'. The 'Upstream BGP Neighbors (Peering Routers)' section shows one peer configured with IP 192.168.8.139, Remote AS 65000, and a hold time of 90 seconds.

Mitigator - BGP Flowspec Page#2

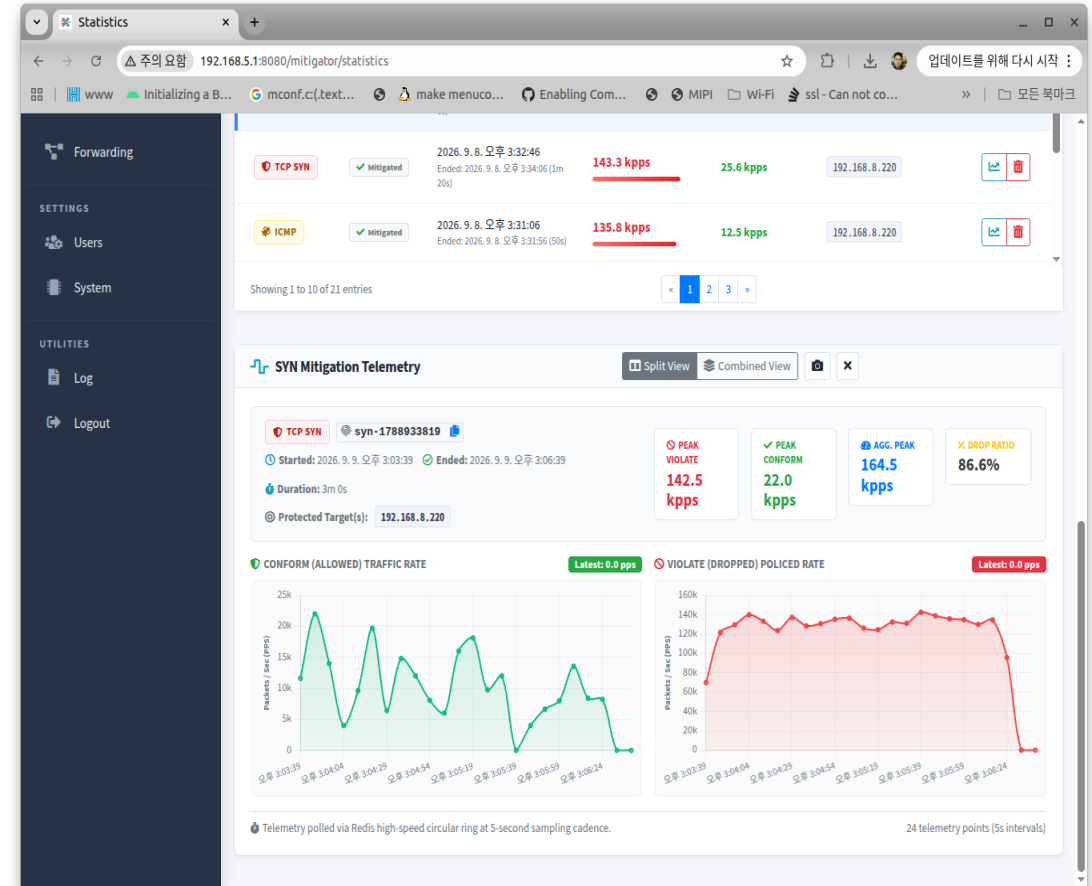
The screenshot shows the 'BGP Flowspec' configuration page in the Mitigator web interface, specifically the 'Upstream BGP Neighbors (Peering Routers)' section. The left sidebar is the same as in Page#1. The main content area shows the configuration for the upstream peer with IP 192.168.8.139, Remote AS 65000, and a hold time of 90 seconds. The 'BGP Peering Session Telemetry' section shows the session is established and operational. The 'Active Flowspec Rules (Upstream Advertised)' section shows one rule configured for flood type 'UDP Flood' with a match filter for destination IP 192.168.8.220 and port 6000. The 'BGP Flowspec (RFC 8955 / RFC 5575) Architecture Reference' section is also visible at the bottom.

4. Mighty User Interface(2) – WebUI(9)

Mitigator - Statistics Page#1



Mitigator - Statistics Page#2



4. Mighty User Interface(2) – WebUI(10)

Network - Interfaces Page

The screenshot shows the 'Network Interfaces' page. The left sidebar contains a navigation menu with 'MIGHTY XDDOS' at the top, followed by 'admin SECURITY ADMINISTRATOR', 'DASHBOARD', 'System Monitor', 'Traffic Monitor', 'DETECTOR', 'Detector', 'View Attackers', 'MITIGATOR', 'Policy', 'Flooding Attack', 'Broadcast Filter', 'Blocked Attacker', and 'Statistics'. The main content area has a top bar with 'VPP Mode' and 'Logout'. Below this, there are four summary cards: 'DATAPATH ENGINE VPP FASTPATH' (Hardware DPDK poll-mode driver), 'ETHERNET INTERFACES 6' (Configured Physical & Bond Ports), 'L2 BRIDGE GROUPS 1' (Active L2 cross-connect & domains), and 'VRRP REDUNDANCY STOPPED' (0 Virtual Router Instances). A tab bar shows 'Interfaces 6', 'L2 Bridge 1', and 'VRRP 0'. The 'Interfaces' tab is active, displaying a table of 'Ethernet & Bonding Interfaces' with buttons for 'Restart Interfaces' and 'Create Bonding'.

INTERFACE	STATUS	IP ADDRESS	NETMASK	ROLE	DPDK	LINK INFO	ACTION
enp2s0	↑ up	0.0.0.0	0.0.0.0	LAN	DPDK	✓ Link 1000Mb/s (Full)	Edit
enp3s0	↑ up	0.0.0.0	0.0.0.0	WAN	DPDK	✓ Link 1000Mb/s (Full)	Edit
enp1s0	↑ up	192.168.5.1	255.255.255.0	MGMT	—	✓ Link 2500Mb/s (Full) Port: TP Mac: 2500baseT/Full Auto-Neg: On	Edit
enp4s0	↓ down	—	—	—	—	No Link Port: TP Mac: 2500baseT/Full Auto-Neg: On	Edit

Network - Forwarding Page

The screenshot shows the 'Forwarding' page. The left sidebar is identical to the previous page. The main content area has a top bar with 'eBPF XDP' and 'L2 BRIDGE GROUPS 1'. Below this, there are three summary cards: 'eBPF XDP' (eBPF/XDP High-Speed Linux Datapath), 'L2 BRIDGE GROUPS 1' (Configured Cross-Connect Pair), and 'DYNAMIC ARP ENTRIES 6' (Live MAC Addresses Learned by the Bridge). A 'ROUTING TABLE' card shows '3 Active IPv4 Forwarding Database Records'. A tab bar shows 'L2 Bridge 1' and 'L3 Route 3'. The 'L2 Bridge' tab is active, displaying a 'Cross-Connect Status' section for 'bridge0' with a table of members and their XDP modes. Below this is a 'Learned MAC Table' section showing a table of learned MAC addresses.

NAME	TYPE	MEMBERS	XDP MODE
bridge0	↔ Cross-Connect	enp3s0 enp2s0	native

MAC ADDRESS	BRIDGE DOMAIN	INTERFACE	STATIC	BVI
94:83:c4:0e:ac:78	↔ bridge0	enp3s0	no	no
98:83:89:9a:9b:c6	↔ bridge0	enp3s0	no	no
94:83:c4:0e:ac:77	↔ bridge0	enp3s0	no	no
f0:a7:31:a0:f4:d6	↔ bridge0	enp2s0	no	no
66:84:04:a9:79:f8	↔ bridge0	enp3s0	no	no

4. Mighty User Interface(2) – WebUI(11)

Utilities - Log Page#1

System Logs

Log Monitor Archive Settings

Filter log content... All Levels 100 Lines Copy Download

Syslog

Syslog Monitor

Sep 17 11:52:59 mightyxddos mighty-ddos[2727]: **WARN:** /usr/local/mightysg/etc/mightymail.conf not found, skipping email

Sep 17 11:52:59 mightyxddos mighty-ddos[2728]: BAN ip=192.168.8.220 direction=incoming pps=27875 attack_type=syn_flood source_ip_count=1 - mitigation applied, email alert sent

Sep 17 11:53:50 mightyxddos mighty-ddos[2751]: UNBAN (Detector-internal, no mitigation change) ip=192.168.8.220 direction=incoming pps=27875

Sep 17 12:00:24 mightyxddos redis-server[1064]: 1064:M 17 Sep 2026 12:00:24.100 * 1 changes in 900 seconds. Saving...

Sep 17 12:00:24 mightyxddos redis-server[1064]: 1064:M 17 Sep 2026 12:00:24.100 * Background saving started by pid 2861

Sep 17 12:00:24 mightyxddos redis-server[2861]: 2861:C 17 Sep 2026 12:00:24.107 * DB saved on disk

Sep 17 12:00:24 mightyxddos redis-server[2861]: 2861:C 17 Sep 2026 12:00:24.108 * Fork CoW for RDB: current 0 MB, peak 0 MB, average 0 MB

Sep 17 12:00:24 mightyxddos redis-server[1064]: 1064:M 17 Sep 2026 12:00:24.201 * Background saving terminated with success

Sep 17 12:00:39 mightyxddos mighty-ddos[2894]: **WARN:** /usr/local/mightysg/etc/mightymail.conf not found, skipping email

Sep 17 12:00:39 mightyxddos mighty-ddos[2896]: BAN ip=192.168.8.220 direction=incoming pps=26502 attack_type=syn_flood source_ip_count=51 - mitigation applied, email alert sent

Sep 17 12:02:34 mightyxddos mighty-ddos[2933]: UNBAN (Detector-internal, no mitigation change) ip=192.168.8.220 direction=incoming pps=26502

Sep 17 12:14:30 mightyxddos systemd[1]: Starting motd-news.service - Message of the Day...

Sep 17 12:14:30 mightyxddos systemd[1]: motd-news.service: Deactivated successfully.

Sep 17 12:14:30 mightyxddos systemd[1]: Finished motd-news.service - Message of the Day.

Sep 17 12:15:17 mightyxddos dbus-daemon[1050]: [system] Activating via systemd: service name='org.freedesktop.timedate1' unit='dbus-org.freedesktop.timedate1.service' requested by ':1.12' (uid=0 pid=3302 comm='timedactl status')

Utilities - Log Page#2

System Logs

Log Monitor Archive Settings

Log Disk Usage & Retention Policy

Total System Log Disk Usage (under /var/log)

1892.24 MB / 1,024.00 MB (100.0%)

This indicates the cumulative size of all system, service, and custom log files.

Rotation Interval: Daily Max Retention Copies: 7

Max File Size Limit (MB): 0

Force rotate immediately if any log exceeds this size, regardless of interval. Set to 0 to disable size limit check.

Compress Archived Logs (.gz) [checked]

Enable HDD Space Alert Email [unchecked]

HDD Warning Threshold (%): 85

Send warning email when disk space usage exceeds this percentage.

Current disk usage: 25.9%

Force Rotate Now Save Settings

4. Mighty User Interface(3) – CLI(1)

```
(mighty) 192.168.5.1 — Konsole
파일(F) 편집(E) 보기(V) 책갈피(B) 플러그인 설정(S) 도움말(H)
복사(C) 붙여넣기(P) 찾기(F)

chyi@earth:~$
chyi@earth:~$
chyi@earth:~$ ssh mighty@192.168.5.1
mighty@192.168.5.1's password:
Welcome to MightyXDDoS Security Box!

mightyxddos# ?

Available commands:
show          Display information
enable        Enter privileged mode
help          Show available commands (same as ?)
exit          Close shell

mightyxddos#
mightyxddos# show ?

Possible completions:
interface     Show ethernet interfaces
vrrp          Show VRRP settings and daemon status
route         Show routing table
firewall      Show firewall rules
bridge        Show L2 bridge/xconnect group(s)
detector      Show DDoS detection engine status
baseline      Show Dynamic Baseline learned per-host traffic profile
signature     Show Auto-Signature Synthesis progress for currently-armed victims
mitigator     Show DDoS Mitigator settings
policy        Show Mitigator Policy settings
bgp           Show BGP Flowspec peering/rule status
mac-security  Show MAC Flooding Protection settings
system        Show system settings

mightyxddos# show Display information
```

```
(mighty) 192.168.5.1 — Konsole
파일(F) 편집(E) 보기(V) 책갈피(B) 플러그인 설정(S) 도움말(H)
복사(C) 붙여넣기(P) 찾기(F)

firewall      Show firewall rules
bridge        Show L2 bridge/xconnect group(s)
detector      Show DDoS detection engine status
baseline      Show Dynamic Baseline learned per-host traffic profile
signature     Show Auto-Signature Synthesis progress for currently-armed victims
mitigator     Show DDoS Mitigator settings
policy        Show Mitigator Policy settings
bgp           Show BGP Flowspec peering/rule status
mac-security  Show MAC Flooding Protection settings
system        Show system settings

mightyxddos# enable
mightyxddos(en)# ?

Available commands:
show          Display information
interface     Network interface configuration
vrrp          VRRP cluster configuration
route         Routing table management
firewall      Firewall rule management
bridge        L2 Cross-Connect/Bridge Domain group management
detector      DDoS detection engine control (start/stop/restart)
mitigate      Unban DDoS-mitigated victims/sources
system        System operations
write         Commit staged changes to the system
clear         Discard staged changes
exit          Return to view mode

mightyxddos(en)#
show          Display information
interface     Network interface configuration
vrrp          VRRP cluster configuration
route         Routing table management
firewall      Firewall rule management
bridge        L2 Cross-Connect/Bridge Domain group management
```

4. Mighty User Interface(3) – CLI(2)

```
(mighty) 192.168.5.1 — Konsole
파일(F) 편집(E) 보기(V) 책갈피(B) 플러그인 설정(S) 도움말(H)
복사(C) 붙여넣기(P) 찾기(F)

clear          Discard staged changes
exit           Return to view mode
mightyxddos(en)# show interface
Interface Status IP Address Link State
-----
enp1s0 UP 192.168.5.1/24 up (Speed: 2500Mb/s, Duplex: Full)
enp2s0 UP - up (Speed: 1000Mb/s, Duplex: Full)
enp3s0 UP 192.168.8.114/24 up (Speed: 1000Mb/s, Duplex: Full)
enp4s0 DOWN - down
enp5s0 DOWN - down
enp6s0 DOWN - down

Interface enp1s0:
Link Detection : up (Speed: 2500Mb/s, Duplex: Full)
Supported Ports : TP
Max Link Mode : 2500baseT/Full
Auto-Negotiation : supported (currently on)

Interface enp2s0:
Link Detection : up (Speed: 1000Mb/s, Duplex: Full)
Supported Ports : TP
Max Link Mode : 2500baseT/Full
Auto-Negotiation : supported (currently on)

Interface enp3s0:
Link Detection : up (Speed: 1000Mb/s, Duplex: Full)
Supported Ports : TP
Max Link Mode : 2500baseT/Full
Auto-Negotiation : supported (currently on)

Interface enp4s0:
Link Detection : down
Supported Ports : TP
```

```
(mighty) 192.168.5.1 — Konsole
파일(F) 편집(E) 보기(V) 책갈피(B) 플러그인 설정(S) 도움말(H)
복사(C) 붙여넣기(P) 찾기(F)

Max Link Mode : 2500baseT/Full
Auto-Negotiation : supported (currently on)

Interface enp6s0:
Link Detection : down
Supported Ports : TP
Max Link Mode : 2500baseT/Full
Auto-Negotiation : supported (currently on)

mightyxddos(en)# show bridge
Name : bridge0
Type : xconnect
Members : enp3s0, enp2s0
Running : on
Mode : native
Learn : on
UU-Flood : on
Local IP : 192.168.8.114/255.255.255.0
Detector Sample Rate: 1-in-75

mightyxddos(en)# show route
Codes: C - connected, S - static, D - default
Type Destination Gateway Interface Protocol
-----
[D] 0.0.0.0/0 192.168.8.1 enp3s0 -
[C] 192.168.5.0/24 0.0.0.0 enp1s0 kernel
[C] 192.168.8.0/24 0.0.0.0 enp3s0 kernel

mightyxddos(en)#
show Display information
interface Network interface configuration
vrrp VRRP cluster configuration
route Routing table management
firewall Firewall rule management
bridge L2 Cross-Connect/Bridge Domain group management
```

4. Mighty User Interface(3) – CLI(3)

```
(mighty) 192.168.5.1 — Konsole
파일(F) 편집(E) 보기(V) 책갈피(B) 플러그인 설정(S) 도움말(H)
복사(C) 붙여넣기(P) 찾기(F)

exit          Return to view mode
mightyxddos(en)# show detector
Status       : Running (pid 11006)
Enabled      : Yes
Ban TTL      : 60 sec
Threshold    : 20000 pps / 800 Mbps

Protected Networks:
Subnet
-----
192.168.8.220/32
mightyxddos(en)#
mightyxddos(en)# show mitigator
Enabled      : Yes
Ban Duration : 30 min

Flood Protection (xdp_bridge SYN policer — Live):
Flood Type  Threshold  Conform Pkts  Violate Pkts  Protected Victims
-----
SYN          20000 pps  0             0             0
UDP          2000 pps  0             0             0
ICMP         10000 pps  0             0             0

Source-IP Block (xdp_bridge ddos_blocklist_map — Live):
No source IPs currently blocked in xdp_bridge.

Flooding Attack Protected Victims:
No victims currently protected.

Blocked Attacker Source IPs:
No source IPs currently blocked.

Broadcast Filter:
```

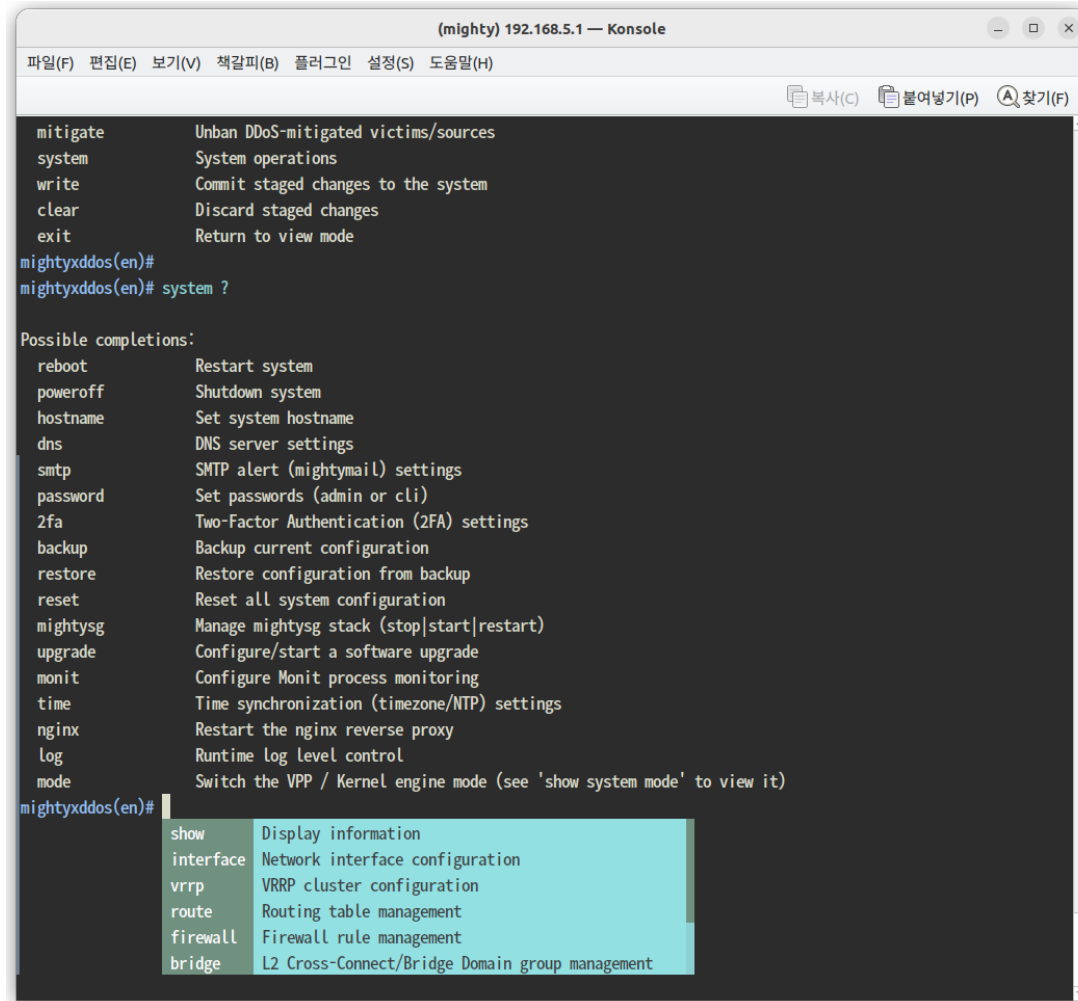
```
(mighty) 192.168.5.1 — Konsole
파일(F) 편집(E) 보기(V) 책갈피(B) 플러그인 설정(S) 도움말(H)
복사(C) 붙여넣기(P) 찾기(F)

mightyxddos(en)#
mightyxddos(en)# show baseline
Dynamic Baseline : on
Sensitivity       : 5.0x learned baseline
Learning Period   : 5 minute(s)
                  (shorter than the 3 day(s) recommended for production —
                  for real deployments, restore this to at least that before relying
                  on Dynamic Baseline for automatic bans.)
Protected Networks : 192.168.8.220/32

Host      Status  First Seen      SYN pps  UDP pps  ICMP pps
-----
192.168.8.220 Active  2026-09-15 15:06:05  0.0      0.1      0.0
mightyxddos(en)#
mightyxddos(en)#
mightyxddos(en)#
mightyxddos(en)# show signature
Flood Type  Victim      Samples  Status
-----
SYN          192.168.8.220  2000     Active

Signature detail: SYN / 192.168.8.220
Fixed packet length: 60 bytes
Offset(s)  Field      Value
-----
12-13      EtherType  0x0800
15          IP DSCP/ECN  0x00
16-17      IP Total Length  0x0028 (40)
20-21      IP Flags/Fragment Offset  0x0000
22          IP TTL      0x40
36-37      TCP Dest Port  0x01bb (443)
46          TCP Data Offset/Reserved  0x50
```

4. Mighty User Interface(3) – CLI(4)



The screenshot shows a terminal window titled "(mighty) 192.168.5.1 — Konsole". The window contains a list of commands and their descriptions, followed by a prompt for the user to enter a command. The commands are listed in a table-like format:

mitigate	Unban DDoS-mitigated victims/sources
system	System operations
write	Commit staged changes to the system
clear	Discard staged changes
exit	Return to view mode

mightyxddos(en)#
mightyxddos(en)# system ?

Possible completions:

reboot	Restart system
poweroff	Shutdown system
hostname	Set system hostname
dns	DNS server settings
smtp	SMTP alert (mightyml) settings
password	Set passwords (admin or cli)
2fa	Two-Factor Authentication (2FA) settings
backup	Backup current configuration
restore	Restore configuration from backup
reset	Reset all system configuration
mightysg	Manage mightysg stack (stop start restart)
upgrade	Configure/start a software upgrade
monit	Configure Monit process monitoring
time	Time synchronization (timezone/NTP) settings
nginx	Restart the nginx reverse proxy
log	Runtime log level control
mode	Switch the VPP / Kernel engine mode (see 'show system mode' to view it)

mightyxddos(en)#

show	Display information
interface	Network interface configuration
vrrp	VRRP cluster configuration
route	Routing table management
firewall	Firewall rule management
bridge	L2 Cross-Connect/Bridge Domain group management

References

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